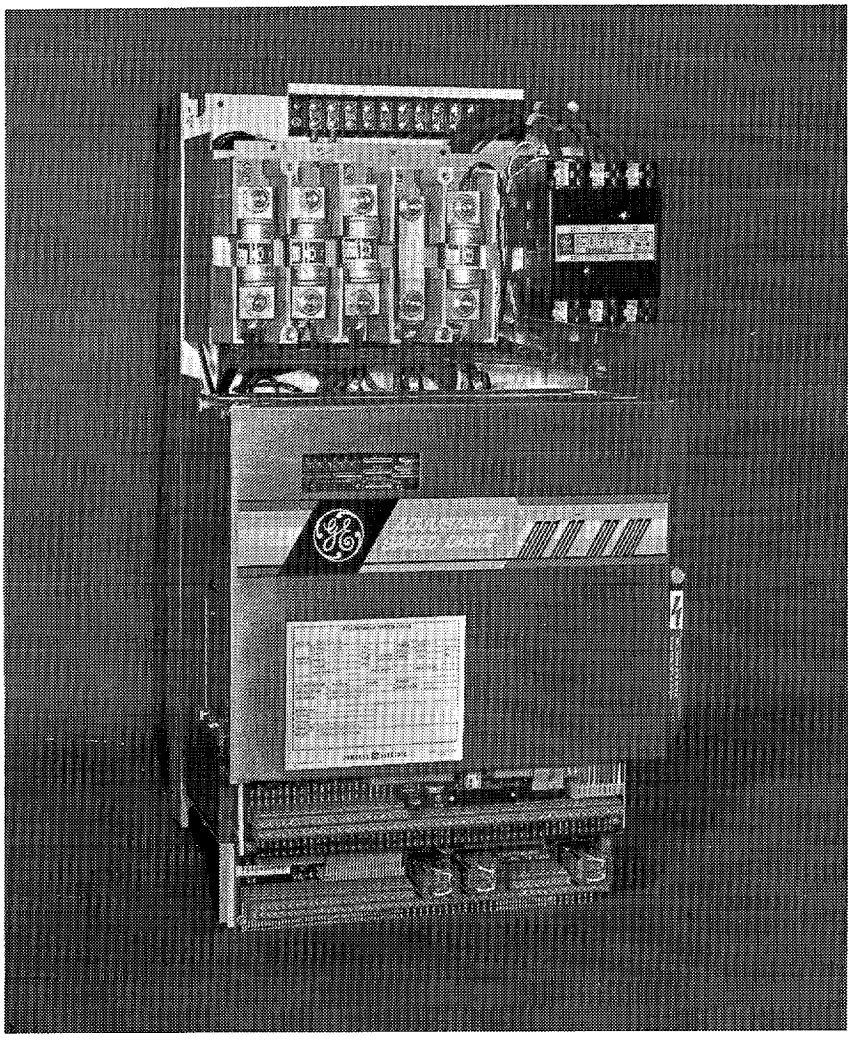
Preliminary GEK-85774

GE Drives

INSTRUCTIONS



DC-300™ E Adjustable Speed Drives

TM TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

TABLE OF CONTENTS

TITLE	PAGE
WARNINGS	2
RECEIVING AND STORAGE	3
INSTALLATION	3
MAIN CONTROLLER ELEMENTS AND THEIR FUNCTIONS	4
ELECTRICAL CONNECTIONS -- POWER AND CONTROL	21
START-UP PROCEDURE	37
ADJUSTMENTS	41
HARDWARE -- JUMPERS AND POT ADJUSTMENTS	41
HARDWARE -- TEST POINTS	52
"KEY" PARAMETER ADJUSTMENTS	54
PARAMETER MAP	55
TROUBLESHOOTING	62
BUGBUSTER™ TROUBLESHOOTING CHART	63
SPECIFIC FAULT TROUBLESHOOTING	71
SPECIFIC ERROR TROUBLESHOOTING	86
TROUBLESHOOTING STABILITY PROBLEMS	89
WAVEFORMS	92
USE OF OPTIONAL PROGRAMMER MODULE	96
OPERATE MODE	97
PARAMETER MODE	98
DIAGNOSTICS MODE	99
DIAGNOSTIC MODE FOR DRIVE SETUP	103
PARTS REPLACEMENT	105
SPARE AND RENEWAL PARTS	108
WARRANTY PARTS AND SERVICE	111
RECOMMENDED POWER STUD WIRING AND TERMINALS	112
GLOSSARY OF TERMS	113

WARNINGS, CAUTIONS, AND NOTES AS USED ON EQUIPMENT

WARNING

DENOTES OPERATING PROCEDURES AND PRACTICES THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CORRECTLY FOLLOWED.

WHEN A WARNING LABEL IS PROVIDED IT WILL BE RED IN COLOR WITH BLACK OR WHITE LETTERING.

CAUTION

DENOTES OPERATING PROCEDURES AND PRACTICES THAT, IF NOT STRICTLY OBSERVED, MAY RESULT IN DAMAGE TO, OR DESTRUCTION OF THE EQUIPMENT.

WHEN A CAUTION LABEL IS PROVIDED IT WILL BE AMBER IN COLOR WITH BLACK LETTERING.

NOTE

NOTES MERELY CALL ATTENTION TO INFORMATION THAT IS ESPECIALLY SIGNIFICANT IN UNDERSTANDING AND OPERATING THE EQUIPMENT.

WHEN A NOTE LABEL IS PROVIDED IT WILL BE WHITE IN COLOR WITH BLACK LETTERING.

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment.

WARNINGS

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY. LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. CIRCUIT BREAKERS, IF SUPPLIED AS PART OF THE TOTAL SYSTEM, MAY NOT DISCONNECT ALL POWER TO THE EQUIPMENT. SEE SYSTEM ELEMENTARY DIAGRAMS. WHETHER THE AC VOLTAGE SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGE TO GROUND WILL BE PRESENT AT MANY POINTS.

WARNINGS

WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT GREAT CAUTION MUST BE USED. THE OSCILLOSCOPE'S CHASSIS SHOULD BE GROUNDED AND A DIFFERENTIAL AMPLIFIER INPUT SHOULD BE USED. CARE SHOULD BE USED IN THE SELECTION OF PROBES AND LEADS AND IN THE ADJUSTMENT OF THE OSCILLOSCOPE SO THAT ACCURATE READINGS MAY BE MADE. SEE OSCILLOSCOPE MANUFACTURER'S INSTRUCTION BOOK FOR PROPER OPERATION AND ADJUSTMENTS TO THE OSCILLOSCOPE.

SOME CONTROLLERS ARE FURNISHED WITH PARTIAL ENCLOSURES OPEN AT THE TOP AND BOTTOM. THESE ARE INTENDED ONLY FOR MOUNTING IN ANOTHER ENCLOSURE OR IN A CONTROL ROOM HAVING ACCESS BY QUALIFIED PERSONNEL ONLY.

EXPLOSIONS OR FIRES MIGHT RESULT FROM MOUNTING DRIVE CONTROLLERS IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE FLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. DRIVE CONTROLLERS SHOULD BE INSTALLED AWAY FROM HAZARDOUS AREAS, EVEN IF USED WITH DC MOTORS SUITABLE FOR USE IN THESE LOCATIONS.

ALL MOTOR BASES AND EQUIPMENT ENCLOSURE HOUSINGS SHOULD BE CONNECTED TO THE FACTORY OR FACILITY EARTH GROUNDING SYSTEMS.

CAUTIONS

DO NOT REMOVE CONNECTIONS OR PRINTED CIRCUIT CARDS FROM THE EQUIPMENT WHILE POWER IS APPLIED. THIS CAN DAMAGE THE EQUIPMENT.

DO NOT REMOVE INPUT POWER FROM THE DRIVE UNTIL IT HAS FULLY EXECUTED A STOP SEQUENCE AS THIS CAN DAMAGE THE DRIVE SYSTEM.

INSTALLATION WIRING MUST BE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE AND BE CONSISTENT WITH ALL LOCAL CODES. SECONDARIES OF 115 VOLT CONTROL TRANSFORMERS TYPICALLY HAVE ONE SIDE FUSED AND THE OTHER GROUNDED OR AVAILABLE FOR GROUNDING BY THE USER.

CAUTIONS

THE ELEMENTARY DIAGRAMS SUPPLIED WITH THE DRIVE CONTROLLER IDENTIFY SOME SIGNAL CONNECTIONS AS "LOW LEVEL ANALOG AND OR DIGITAL SIGNALS" AND INDICATE SPECIAL WIRING PRACTICES. THESE SIGNALS SHOULD BE ISOLATED FROM ALL OTHER WIRING (PARTICULARLY ANY POWER WIRES) AND RUN IN A SEPARATE CONDUIT OR WIRE RUN TO PREVENT POSSIBLE MISOPERATION OF THE CONTROLLER.

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HI POT WITHOUT CONSULTING **GE DRIVES.**

DO NOT CONNECT ANY EXTERNAL CIRCUITS OTHER THAN THOSE SHOWN ON THE ELEMENTARY DIAGRAM. CONNECTION OF DEVICES SUCH AS AMMETERS ON THE SHUNT OR VOLTMETERS ON THE TACHOMETER MAY DEGRADE THE PERFORMANCE OF THE DRIVE SYSTEM.

DO NOT USE POWER FACTOR CORRECTION CAPACITORS WITH THIS EQUIPMENT WITHOUT CONSULTING **GE DRIVES.** DAMAGE MAY RESULT FROM HIGH VOLTAGES GENERATED WHEN CAPACITORS ARE SWITCHED.

NOTES

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLESHOOTING THE EQUIPMENT. FOLLOW THE STARTUP PROCEDURES STEP BY STEP.

READ AND HEED ALL **WARNING, CAUTION** AND **NOTE** LABELS POSTED ON THE EQUIPMENT.

RECEIVING AND STORAGE

RECEIVING

The controller should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVES, ERIE, PA.** When seeking assistance please use serial number, requisition number, model number and drive code to identify the equipment **(Telephone 814-875-3851).**

STORAGE

If this controller is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures from -20°C (-4°F) to 55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage environment exceeds a 15°C (27°F) drop in temperature at 50% humidity over a 4-hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat). Higher humidities with smaller temperature changes can also cause condensation.

INSTALLATION

LOCATION

DC-SCR controllers are suitable for most areas where industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C (32°F) to 40°C (104°F) and relative humidities up to 90%. It should be recognized, however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20°C (68°F) and 30°C (87°F).

Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

WIRING LEVELS

1) Wiring should be run in separate conduits or wireways for signal, control and power wiring levels.

   a) *Signal* – Low level analog and digital signals
      – Speed and/or position signals
      – Power supplies and logic signals
   (See system schematics and notes to identify signal level wires)
   b) Control – AC or DC control circuits
   c) Power – Field leads, armature leads
      – 115 VAC circuits

2) Signal wiring and power wiring may cross at right angles with a minimum one-inch separation. Avoid parallel runs between signal level wires and power or control wires. If signal wires must be run in parallel to control or power wires, a minimum of a four-inch separation must be maintained between the wires.

GROUNDING

1) Drive common (PCOMX) should be grounded at **only one** point. If the reference is supplied by numerical control or process instrument with a grounded common, the drive common should not be grounded separately.

2) If an isolation transformer is used and must be grounded, it is recommended that a high resistance ground be used.

3) Shields for shielded and twisted, shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

4) Proper case grounding that conforms to CSA standards should utilize wire sizes and crimp-type terminals per the following table:

RATING OF CIRCUIT (AMPS) Not to Exceed:	COPPER GROUND WIRE AWG	AMP TERMINAL NUMBER	AMP TOOL NUMBER
20	14	34120	49900
60	10	33457	49900
100	8	33458	69062
200	6	33465	69062
300	4	31811	69062
500	2	320383	69062
800	1/0	36917	69120
1000	2/0	36923	69120
1200	3/0	36925	69120
1600	4/0	321878	69120

COMMONS

Connection of reference, meter or any other external, commons can only be made at 3TB–PCOMX, unless otherwise indicated by the system elementaries.

SUPPRESSION

Control system relays, solenoids or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volt capacitor can be used.

TIGHTEN ALL CONNECTIONS

1) Incoming AC line connections

2) Power connections to motor armature and motor field

3) 1TB, 2TB, 3TB, 4TB and MATB if furnished

ADDITIONAL CONSIDERATIONS

Attention should be given to the National Electrical Code and any applicable local codes when installing any drives. Wire size and insulation type, conduit sizing, enclosures, etc., should be determined per these codes.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dust, dirt or foreign matter.
2. Vibration or shock.
3. Moisture or vapors.
4. Rapid temperature excursions or high ambient temperatures.
5. Caustic fumes.
6. Power line fluctuations.
7. Electromagnetic interference or **"noise"** introduced by:
   a. Radio frequency signals, typically from portable transmitters used in the vicinity of the equipment or its wiring.
   b. Stray high voltage or high frequency signals as might be provided by arc welders, or by the operation of unsuppressed relay, contactor, or brake coils as a part of, or in the vicinity of, drive control circuits.

MOUNTING

Totally enclosed controllers should be positioned to permit heat radiation from all surfaces.

A wall-mounted enclosure may be placed side by side with another enclosure. Clearance at least to the width of the enclosure doors should be available in front so that the door may be fully opened for easy access.

Wall-mounted controllers may be mounted on any firm, reasonably flat, vertical surface.

MAIN CONTROLLER ELEMENTS AND THEIR FUNCTIONS

GENERAL DESCRIPTION (See Figures 1 thru 4)

The **DC-300 E** adjustable speed DC drive is a digitally controlled programmable drive utilizing microprocessors to provide digital regulation. Drive parameters are stored in EEROM memory, and therefore, are accurate and drift free.

The **DC-300 E** controller provides over 80 diagnostic messages which can be decoded via a LED diagnostic readout or an optional Programmer module with alpha-numeric readout. Diagnostic messages can also be read via an optional coded

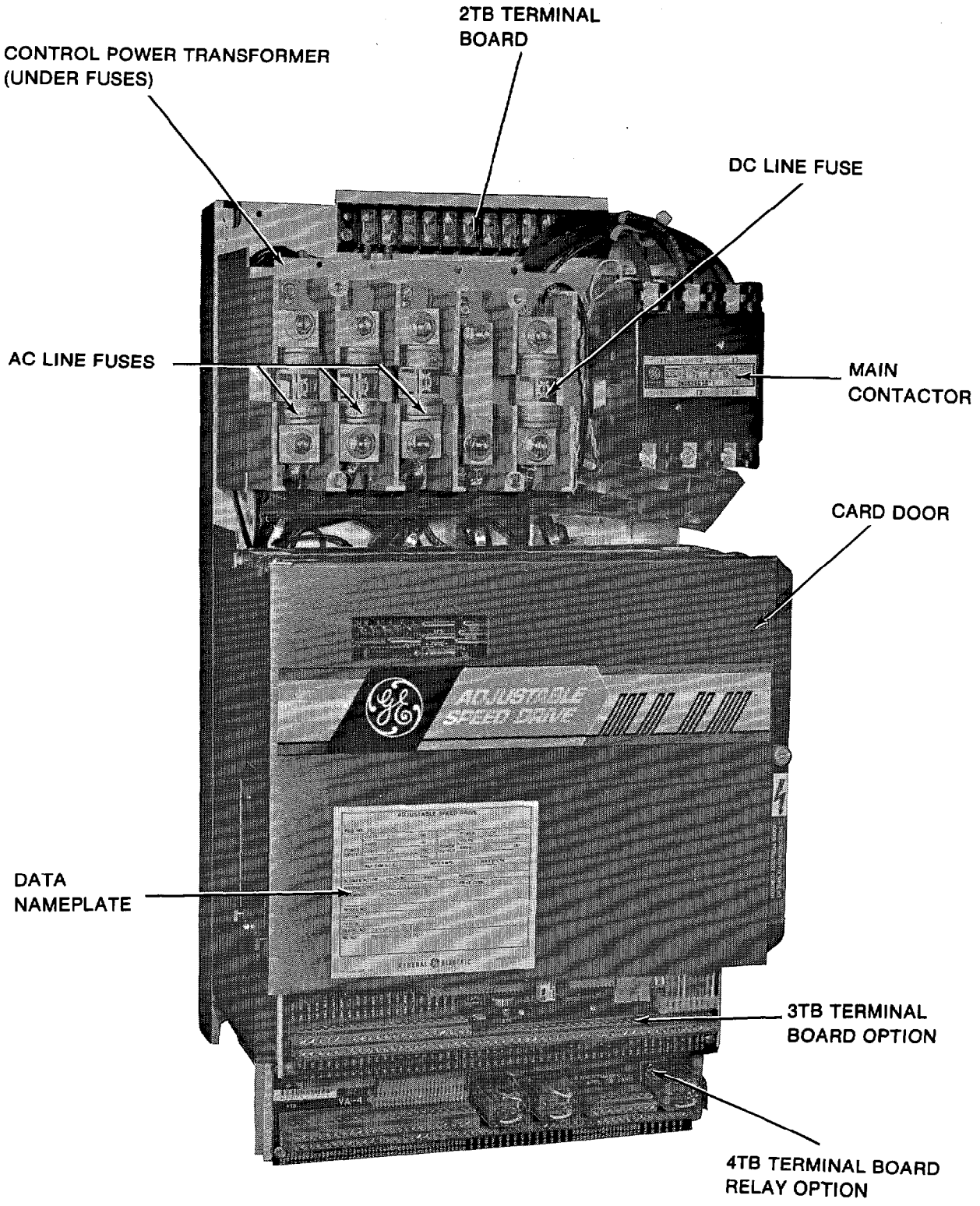


FIGURE 1
DC-300 E DRIVE CONTROLLER, WALL MOUNTED
OPEN CHASSIS CONSTRUCTION

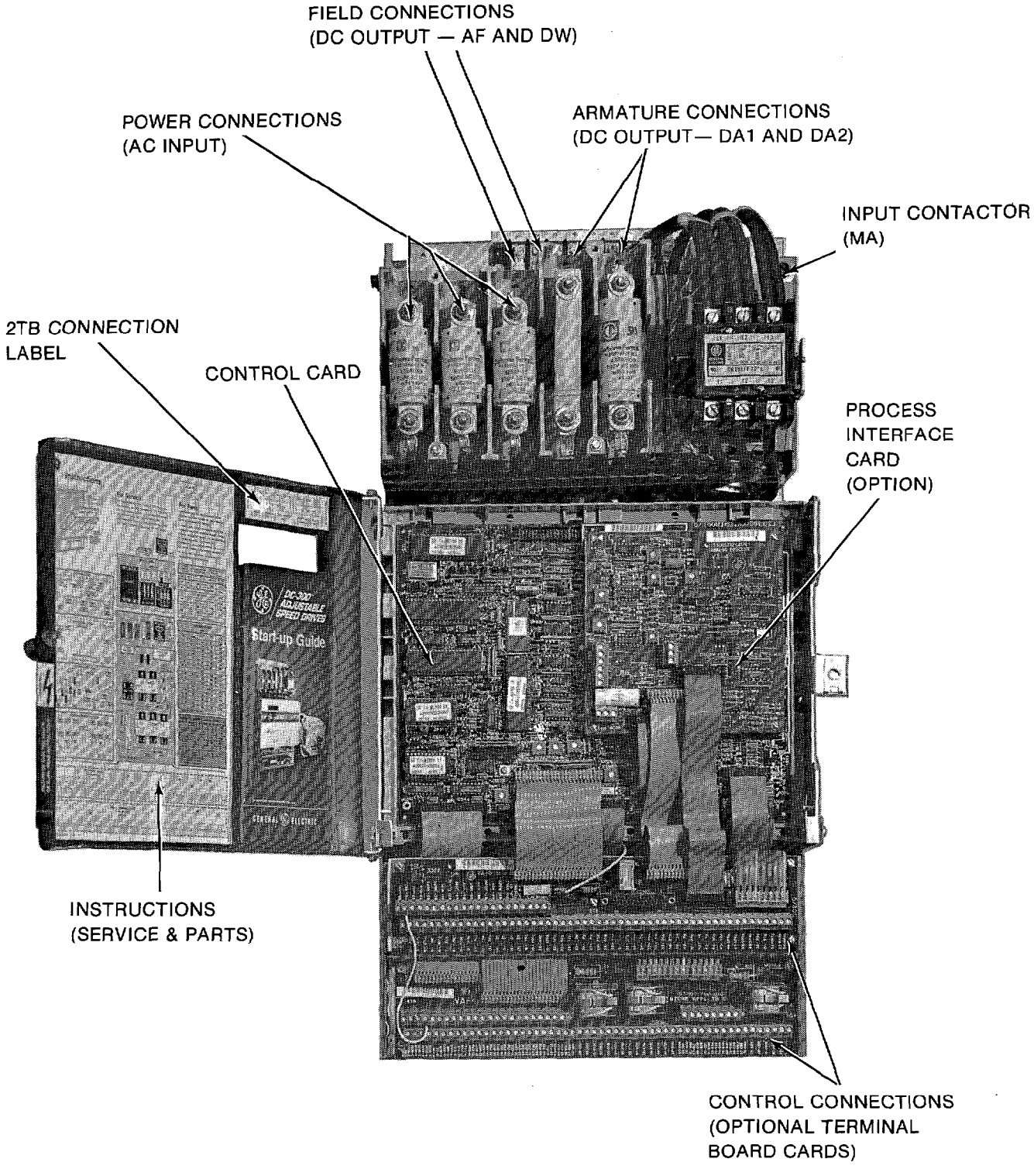


FIGURE 2
DC-300 E DRIVE CONTROLLER WITH CARD DOOR
IN OPEN POSITION

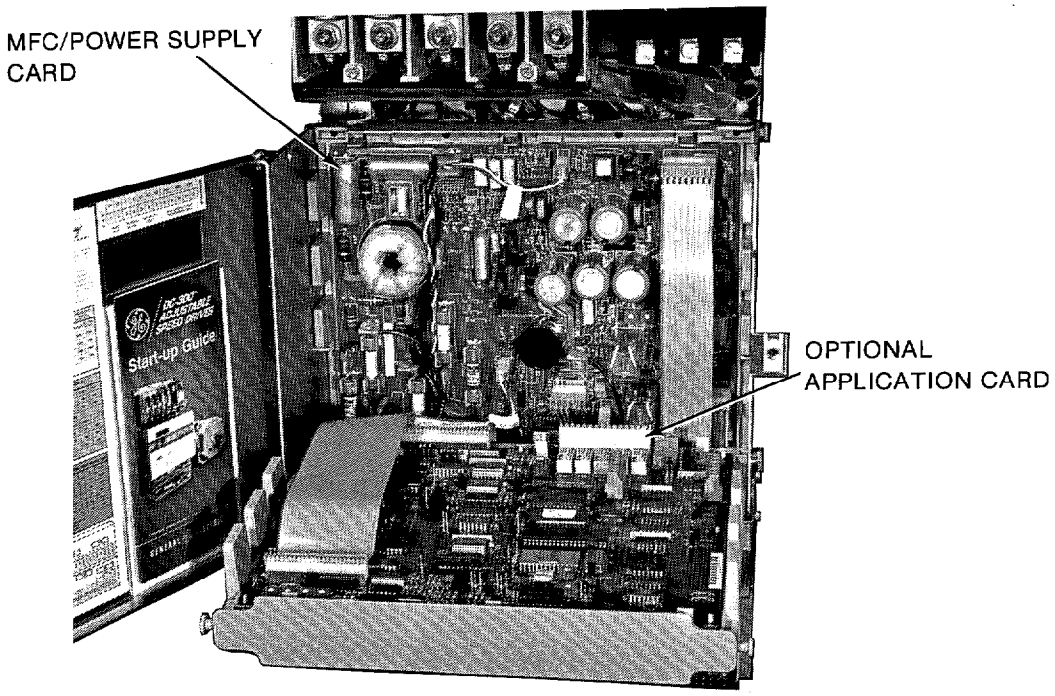


FIGURE 3
DC-300 E CONTROLLER WITH
FRONT CARD CARRIER LOWERED

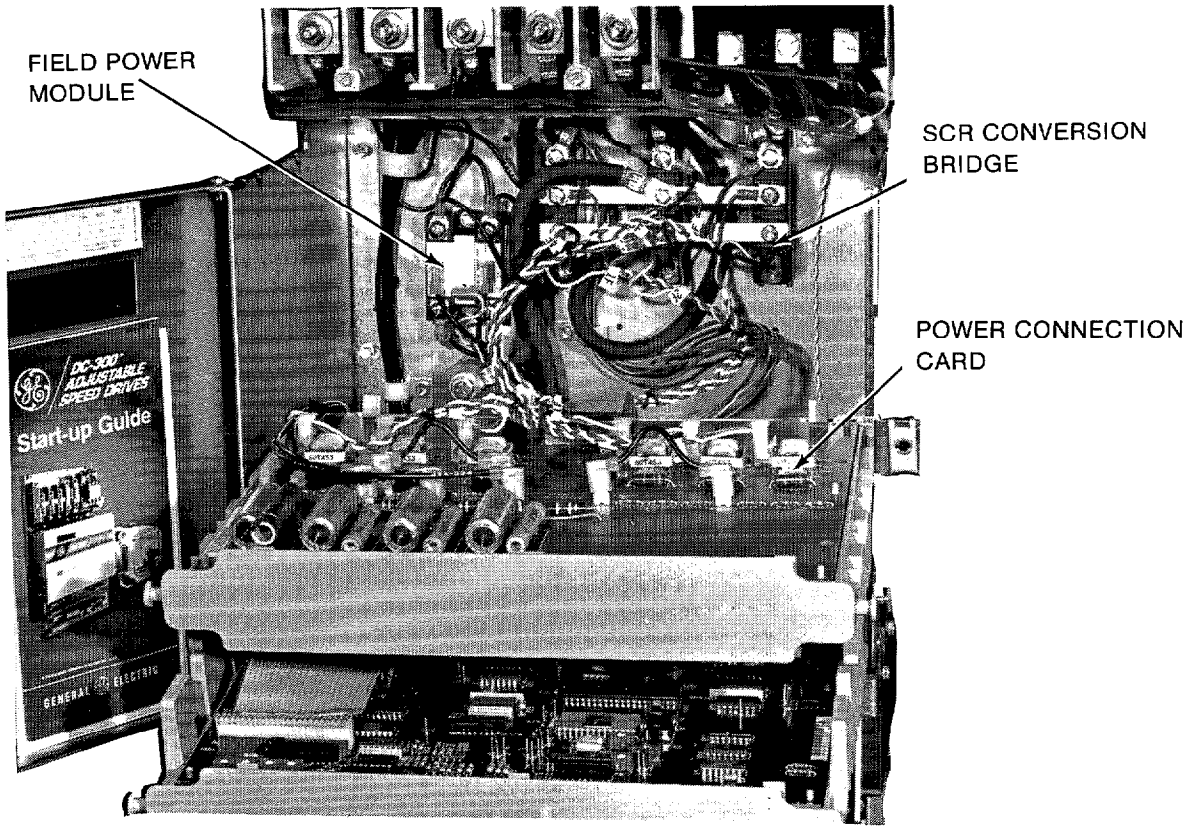


FIGURE 4
DC-300 E CONTROLLER WITH
REAR CARD CARRIER LOWERED

8-wire output or over the optional RS422/232C communication link. Drive operating parameters stored in the EEROM may also be examined and changed via the RS422/232C link. In addition, the controller provides diagnostics which can provide a power-up self check down to the printed circuit board level.

The **DC-300 E** controller requires only four printed circuit cards (Control card, Process Interface card, Power Supply card and Power Connection card) for operation and offers optional cards and terminal boards for additional functions. Optional cards include the Application card, Programmer card, and Terminal Board cards (3TB and 4TB). The Programmer can be purchased in a Local or Hand-held version which plugs into the Control card for inspection and adjustment of parameters when tuning up the drive, as well as diagnostic function.

The Analog Process Interface card provides an analog speed regulator outer loop with a digital current regulator inner loop on the Control card. Individually timed acceleration and deceleration, adjustable over rates of 0 (current limit acceleration) to 999.9 seconds, are available through the optional "S"-Curve circuit.

Output metering signals for armature current, armature voltage, field current, speed, torque and horsepower are also available from the **DC-300 E** controller. In addition, all signals can be brought out as an analog signal via one of three optional digital-to-analog converters or digitally via the optional RS422/232C serial link. (These signals and others can be programmed to activate optional programmable signal level detectors.)

Power is fed through the AC fuses, current transformers and MA contactor, and enters the power conversion SCR modules where it is converted to adjustable DC voltage. DC current is fed through a shunt and a DC link fuse to the DC motor armature.

Analog voltage, frequency or digital inputs can be used as the references.

POWER CIRCUITS (See Figures 5 and 6)

The motor field control (MFC) may be provided by an internal 10 Amp maximum field supply (Figure 5A) or an optional external field supply rated to 24 AMPS (Figure 5B).

For the 10 Amp MFC, the Power Supply card contains the field supply fuses (CFU1, CFU2 and CFU3), metal oxide varistors (MOV), field current feedback transformer (T3), and line reactor (L4). The field power module is mounted on the main controller heatsink.

The 24 Amp external MFC provides a separate heatsink with field supply fuses (CFU1-1, CFU2-1 and CFU3-1), line reactor (L4), field power module and MFC Suppression card containing the MOVs and field current feedback transformer (T1).

NOTE: There is a variation in fuse and transformer designations between the two motor field control versions.

The operation of the MFC may be best understood by reading the following while referring to the appropriate figure. Field power is supplied through fuses CFU1, 2 and 3 (CFU1-1, 2-1 and 3-1 on the external MFC). MOVs protect from line spikes and transients.

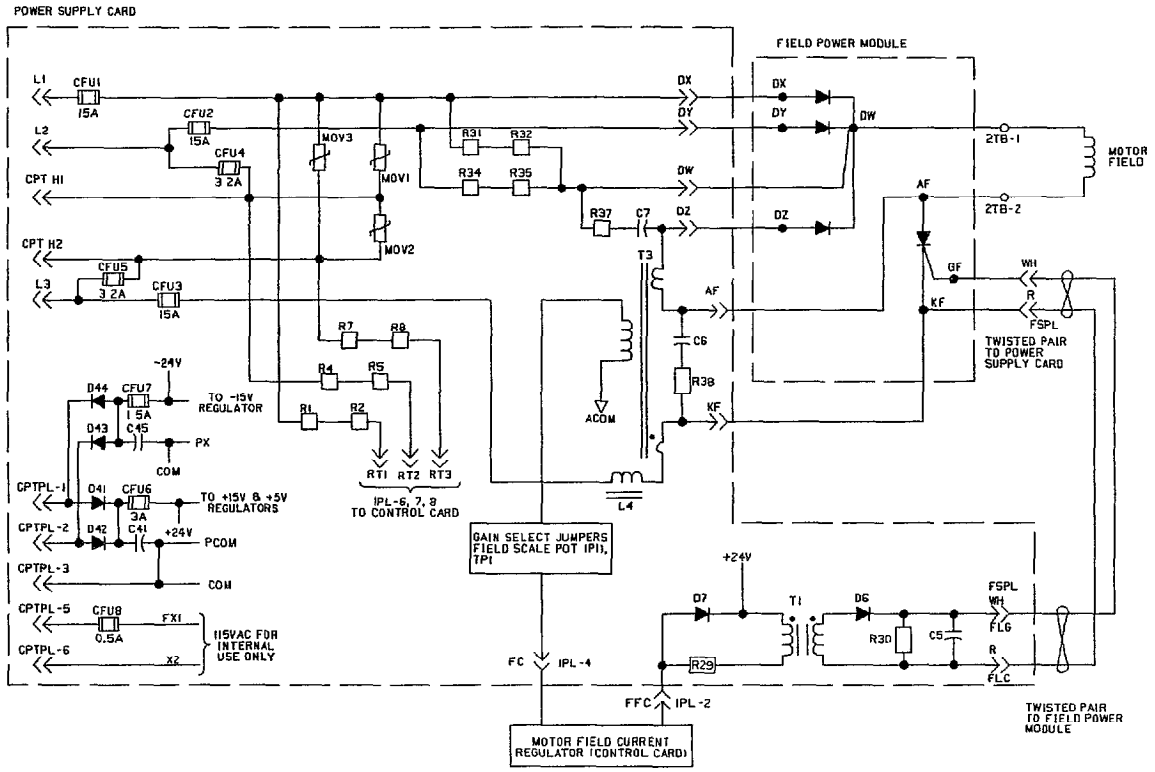
The power to the motor field control is "two-thirds" wave rectified via the field power module. The SCR in the field power module controls field current as follows: When AC line L1 is high with respect to L2 and L3 the SCR is fired and current passes through the diode between DX and DW to the motor field. The circuit is completed through the SCR and returns through the line reactor to AC line L3. When L2 becomes high, current flows through the diode between DY and DW to the field and back through the SCR, line reactor and L3. When L3 becomes high, the SCR is reversed biased and the energy stored in the field winding circulates current through the diode between DZ and DW (acting as a free wheeling diode) until L1 becomes high again and the SCR is fired.

Field current is sensed by the current feedback transformer (T1 or T3 depending on the MFC rating) and fed into the field gain select circuit on the Power Supply card (through FCPL with external MFC) and to the Control card.

The Power Supply card provides input voltage to the control power transformer through CFU4 and 5 via terminals CPT H1 and CPT H2. RT1, 2 and 3 are the AC line sync signals to the Control card.

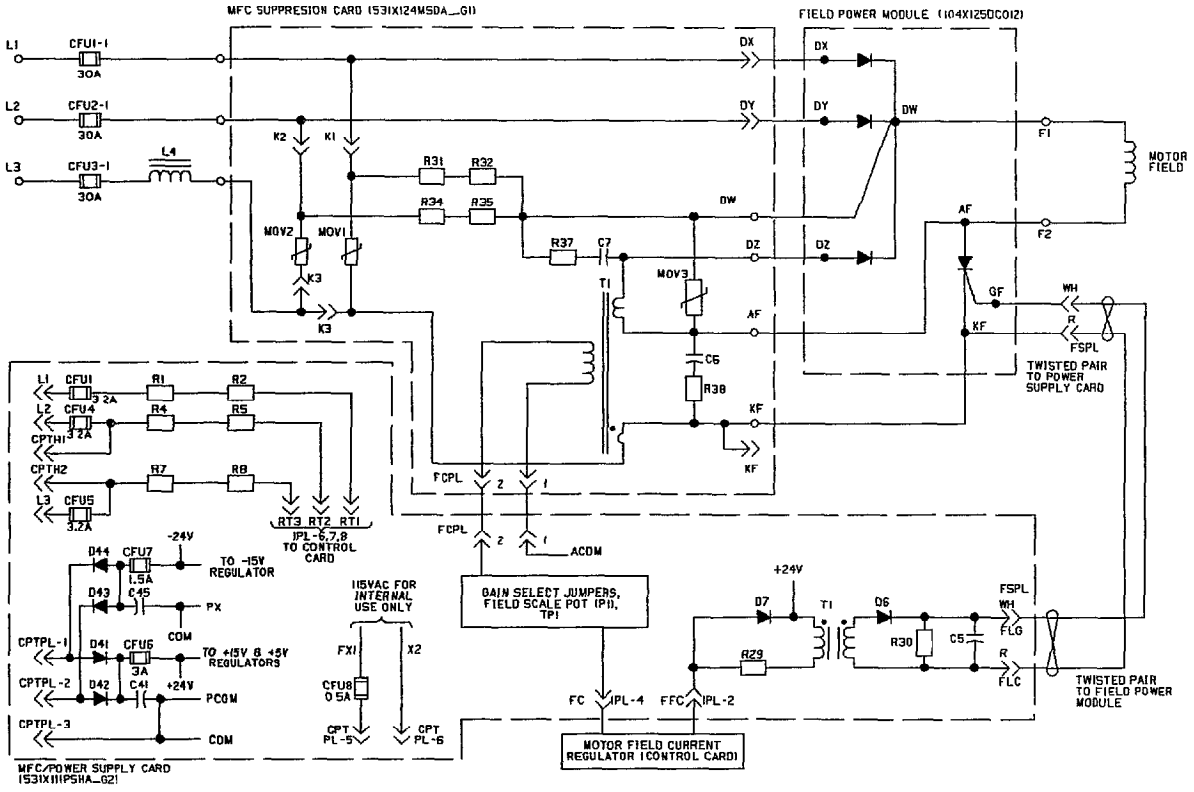
Fuses CFU6, CFU7 and CFU8 are located on the Power Supply card. The internal DC power supplies are protected by CFU6 and CFU7. The 115V AC internal supply is protected by CFU8. These circuits are fed from the control power transformer through plug CPTPL.

The regenerative SCR Power Conversion Bridge is shown in Figure 6. Three-phase power is delivered to the conversion bridge through AC line fuses (FU1-3) and the MA contactor. Two line current transformers (CT1 and CT3) provide a current feedback signal for commutation failure indication and reactors provide protection against AC line transients.



S MISC.ELEC 445 (R9/28/87)

FIGURE 5A
MOTOR FIELD CONTROL (10 AMP)



MISC ELEC GEK-85766 SH1 (R9/28/87)

FIGURE 5B
MOTOR FIELD CONTROL (24 AMP)

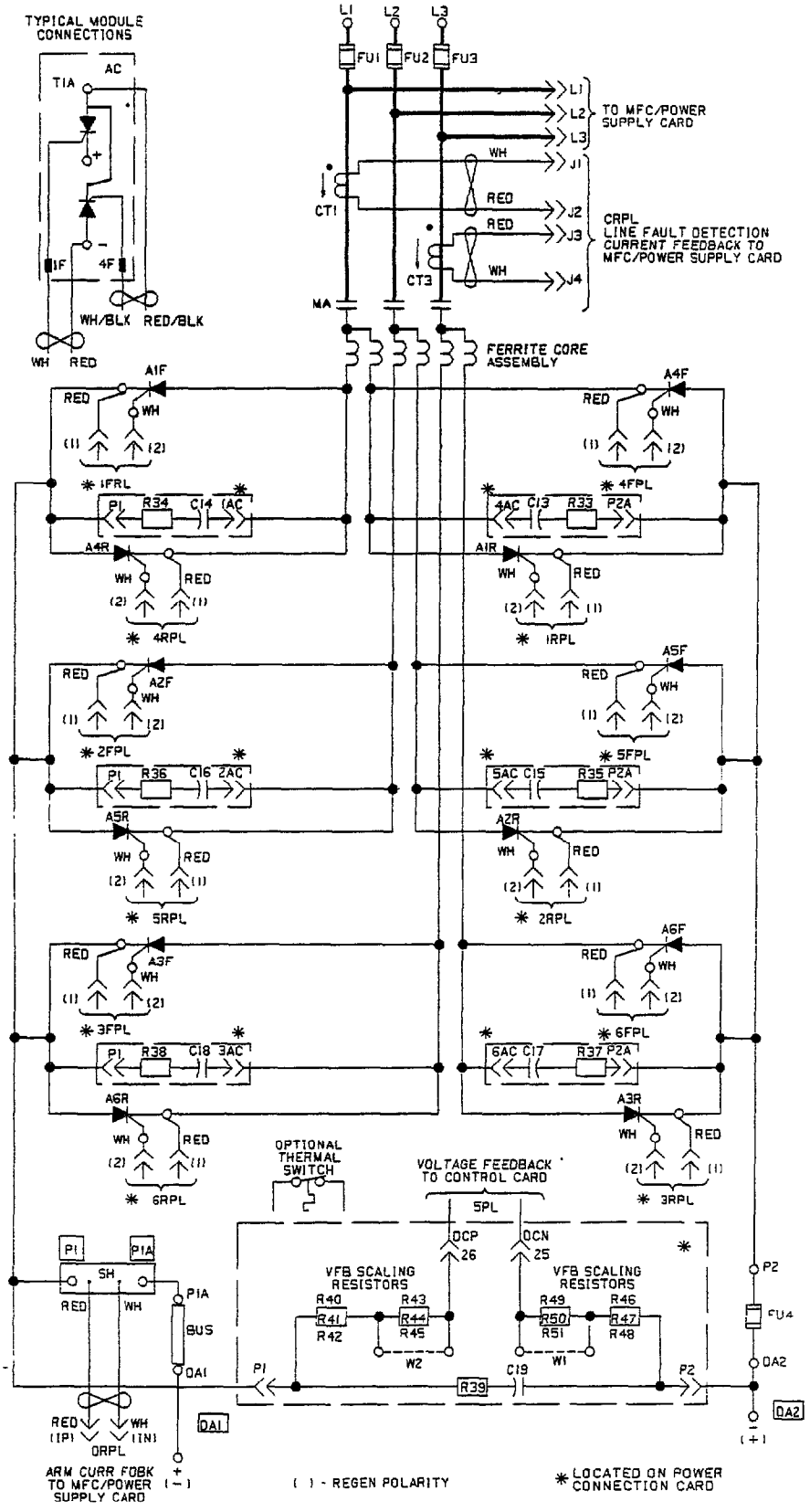


FIGURE 6
REGENERATIVE SCR POWER CONVERSION BRIDGE

RC snubber networks, located on the Power Connection card, provide protection for the SCRs and are connected across a forward/reverse SCR pair on regenerative drives. The SCR firing pulses are generated through pulse transformers located on the Power Connection card in the firing order 1-6-2-4-3-5 for both forward or reverse.

DC power is supplied to the armature through the shunt and DA1 and returns through DA2 which is fused. An RC network across P1 and DA2 aids latching of the SCRs. The voltage feedback signal is in parallel with the snubber and current feedback is derived from the shunt. An optional thermal switch mounted on the heatsink is also available.

AC POWER REQUIREMENTS

The **DC-300 E** drive is available in either 230 VAC +10%, -5% or 460 VAC +10%, -5%, three-phase power configurations. Nominal line frequency is 50/60 Hz with tolerance of ±2 Hz. Other input voltage configurations are also available on special order.

AC LINE FUSES (FU1-3)

AC line fuses provide short circuit protection for the SCRs, and the wiring inside the controller. Additional overcurrent protection is provided by both instantaneous and timed overcurrent functions.

FERRITE CORE ASSEMBLY (REACTORS)

The ferrite core assembly is provided on the AC power lines between the AC line fuses and the SCRs to protect the SCRs from possible misoperation due to transient currents and voltages during regeneration.

MAIN CONTACTOR (MA)

The MA contactor provides a controllable disconnect between the AC power line and the power conversion bridge. The contactor picks up when running mode is selected and no fault exists in the drive. The MA contactor drops out under the following conditions:

1. When "STOP" is commanded and the motor slows to near zero speed on **regenerative drives.**

2. When a fault condition occurs.

DC LINE FUSE (FU4)

A DC line fuse is provided in the DC output line to the motor armature and protects the SCRs and motor from overcurrent sourced from the motor.

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuses (CFU4 and 5) on the Power Supply card. The secondary winding provides two isolated voltages:

1. 115 VAC to operate the coil of the main (MA) contactor, RUN (K1) Relay and controller cooling fan if required.

2. 38 VAC center tapped for the Power Supply card to generate the DC control voltages necessary for controller operation.

SHUNT (SH)

The shunt provides the armature current feedback signal to the control circuit. A nominal 100mV output signal is generated at the current rating stamped on the shunt.

CURRENT TRANSFORMER ASM (CT)

The current transformer assembly is used to provide AC line current feedback signals that are used for circulating current fault detection.

CONNECTION DIAGRAM (See Figure 7)

The connection diagram shown in Figure 7 graphically describes the internal connections for the plugs and cables. The diagram aids control hook up as well as troubleshooting when connection integrity of one of the cables is in question. The functions of the connectors are described in the "Electrical Connection" section.

CONTROL CARD (See Figure 9)

This card provides the intelligence for the **DC-300 E** drive controller. It contains three microprocessors and performs most of the regulating functions, SCR burst firing, diagnostic and protective functions for the drive along with many customer and motor interfacing functions. The functions performed on the card are implemented in hardware and/or software.

The Control card (Figure 9) contains six Diagnostic LEDs to display the status of the drive. The LEDs which are lit are interpreted as a binary number indicative of the fault. See the troubleshooting section for listing of Fault codes. Note that during proper controller operation, the LEDs will blink sequentially right to left when the drive is stopped and sequentially in pairs when the drive is running.

REGULATORS

A special Analog Process Interface card is provided on the **DC-300 E** drive. Analog DC tachometer input and scaling, a differential reference amplifier and an analog speed regulator

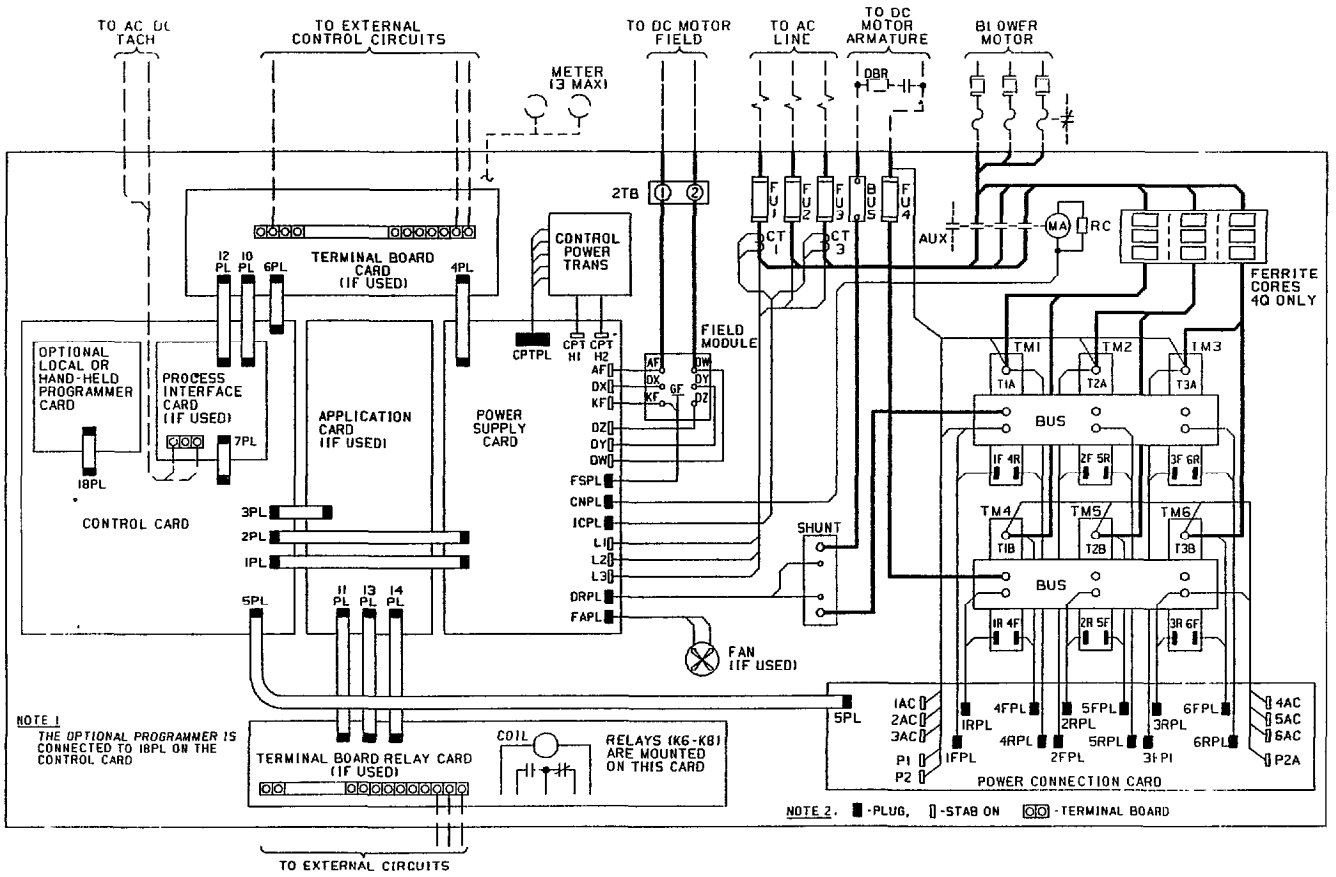


FIGURE 7
CONNECTION DIAGRAM – REGENERATIVE CONTROLLER

are provided on the Analog Process Interface card for the high response drive. The output of the analog speed regulator is fed through plug 7PL-4 ASFB to provide the current reference input to the digital current regulator located on the Control card. Increased current regulator gain and resolution further enhance the performance of the **DC-300 E** drive.

MFC/POWER SUPPLY CARD (See Figures 10 and 11)

This power supply card provides the necessary +5 VDC, ±15 VDC, and ±24 VDC power used throughout the controller. The armature current feedback isolation circuit as well as three relays, "RUN" (K1), "MAX" (K2) and "FAULT" (K3) are also on this card. Additionally, the motor field control is provided by this card. The version G1 card is used with motors requiring field current up to 10 amps and contains the pulse transformer, current transformer, choke, MOVs and fuses for the motor field circuitry as previously described. The version G2 card is used in conjunction with the external field control assembly for higher field current control.

The External Motor Field Control Assembly (Figure 12) is used with the version G2 MFC/Power Supply card for controllers requiring up to 24 amps for the motor field. The assembly contains the fuses (CFU1-1, CFU2-1 and CFU3-1), power module, current feedback transformer, choke and MOVs, which provide power module protection.

POWER CONNECTION CARD (See Figure 13)

This card contains SCR and DC output resistor-capacitor (RC) snubbers and the gate pulse transformers for the SCRs. The DC voltage feedback isolation resistors are also mounted on this card.

OPTIONAL APPLICATION CARD (See Figure 14)

This optional card contains a combination of analog and digital interfaces between the Control card and external equipment. These interfaces allow the core drive to be expanded to handle optional functions. The card, pictured in Figure 14, contains the following optional functions:

- Eight Analog Inputs
- Eight Digital Inputs
- Five Programmable Signal-Level-Detector Relays (SLDs)
- Seven-Bit Digital Fault Status Output
- Three Programmable Status Bit Outputs
- Two Analog Outputs
- Two General Purpose Amplifiers (Version APM card only)

ANALOG PROCESS INTERFACE CARD (See Figure 15)

This card contains a mixture of analog and digital interfaces between the Control card and the equipment. The optional functions on this card are:

- Analog DC tachometer interface.
- Analog speed meter driver (analog tachs only).
- Digital voltmeter probe.
- RS232C and RS422 serial link interface.
- Sharp, low frequency cutoff, filter in speed loop.
- Armature voltage rate circuit.

OPTIONAL TERMINAL BOARD CARDS (3TB and 4TB) (See Figures 16 and 17))

These optional cards provide convenient termination points for control interface to the drive. Connections to the Control, Power Supply and Process Interface cards are made thru 3TB, which also contains an optional signal-level-detector (SLD5). Connections to the optional Application card are made thru 4TB. In addition, three 115 VAC relays are available on this card.

OPTIONAL PROGRAMMER MODULE

The Programmer is available as either a "Local" Programmer, mounted on the Control card, or as a "Hand-held" Programmer, which is plugged into 18PL on the Control card. The Programmer contains a 10-digit fluorescent alphanumeric display and keypad for communicating with the drive.

The Programmer has three modes:

- **Operate** mode provides all necessary functions to run the drive and displays the drive status including fault messages.
- **Parameter** mode allows examination and changes, if necessary, of the system operating parameters.
- **Diagnostic** mode allows the user to monitor drive values while in operation, self test the controller, and access diagnostic STATIC mode. Diagnostic mode also provides a number of drive tests. Refer to "Use of Optional Programmer Module" section for further information.

SERVICE AND PARTS INSTRUCTION LABELS (See Figure 8)

Each drive contains, inside the door, a multi-color label similar in appearance to that shown in Figure 8. This label provides information on connection point and component locations essential in connecting or troubleshooting the drive. The label contains the following information in color-coded format:

- Card location
- Thyristor module locations
- Fuse information and nomenclature
- Plug locations
- Incoming control connections at 3TB and 4TB
- Power connections
- Factory "hot line" telephone number **(1-814-455-3219)**

For Service

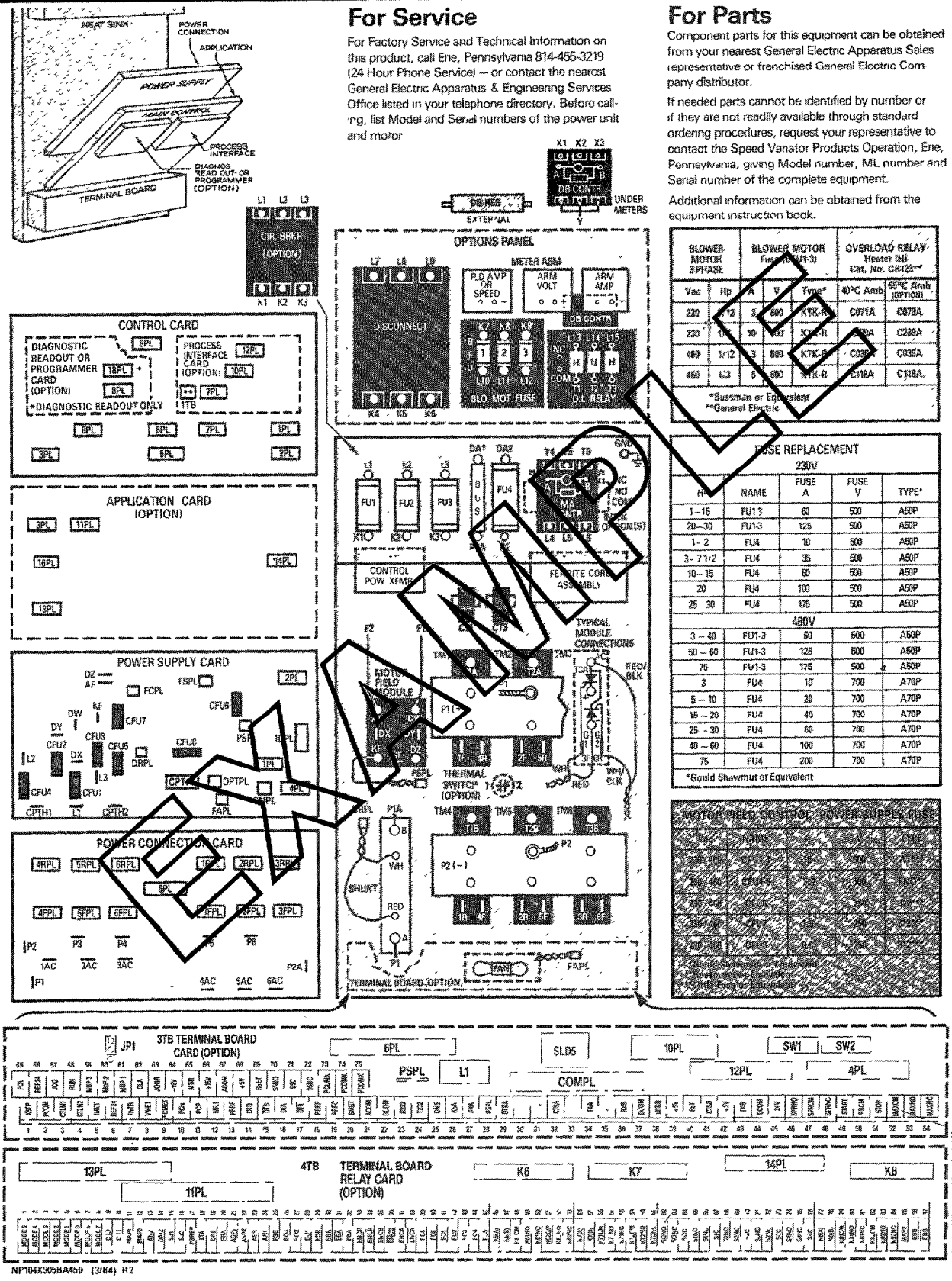
For Factory Service and Technical Information on this product, call Erie, Pennsylvania 814-455-3219 (24 Hour Phone Service) — or contact the nearest General Electric Apparatus & Engineering Services Office listed in your telephone directory. Before calling, list Model and Serial numbers of the power unit and motor

For Parts

Component parts for this equipment can be obtained from your nearest General Electric Apparatus Sales representative or franchised General Electric Company distributor.

If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures, request your representative to contact the Speed Variator Products Operation, Erie, Pennsylvania, giving Model number, ML number and Serial number of the complete equipment.

Additional information can be obtained from the equipment instruction book.



BLOWER MOTOR 3PHASE		BLOWER MOTOR Fuses (FU1-3)		OVERLOAD RELAY Heater (HI) Cat. No. CR123**	
Vac	Hp	A	V	Type*	40°C Amb / 55°C Amb (OPTION)
230	1/2	3	600	KTK-R	C071A / C078A
230	1	10	600	KTK-R	C239A
460	1/2	3	600	KTK-R	C035A
460	1/3	3	600	KTK-R	C118A / C118A

*Bussman or Equivalent
**General Electric

FUSE REPLACEMENT 230V				
HP	NAME	FUSE A	FUSE V	TYPE*
1–15	FU1-3	60	500	A50P
20–30	FU1-3	125	500	A50P
1- 2	FU4	10	500	A50P
3- 7 1/2	FU4	35	500	A50P
10–15	FU4	60	500	A50P
20	FU4	100	500	A50P
25 30	FU4	175	500	A50P
460V				
3 – 40	FU1-3	60	500	A50P
50 – 60	FU1-3	125	500	A50P
75	FU1-3	175	500	A50P
3	FU4	10	700	A70P
5 – 10	FU4	20	700	A70P
15 – 20	FU4	40	700	A70P
25 - 30	FU4	60	700	A70P
40 – 60	FU4	100	700	A70P
75	FU4	200	700	A70P

*Gould Shawmut or Equivalent

EXAMPLE

FIGURE 8
TYPICAL INFORMATION LABEL

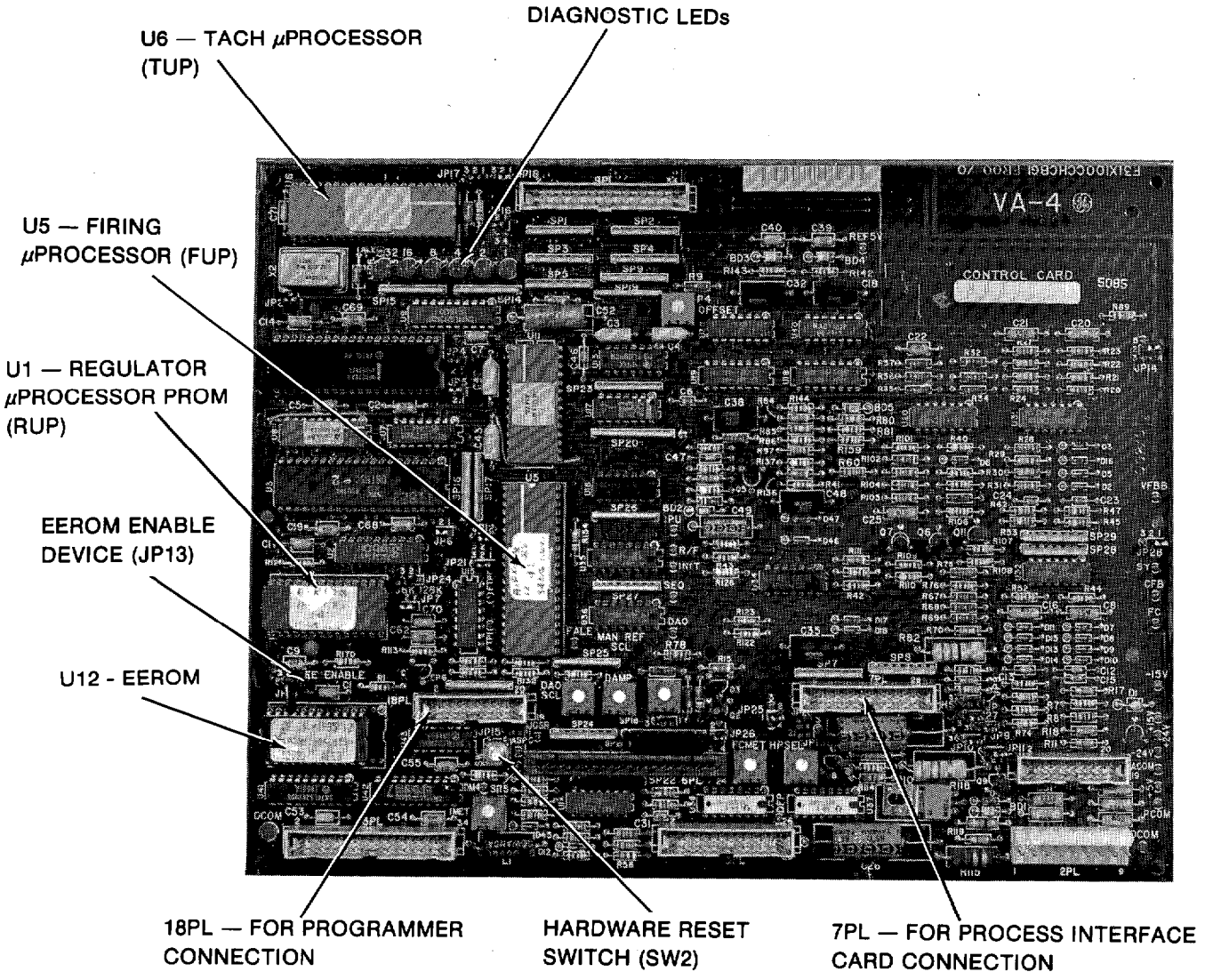


FIGURE 9
CONTROL CARD

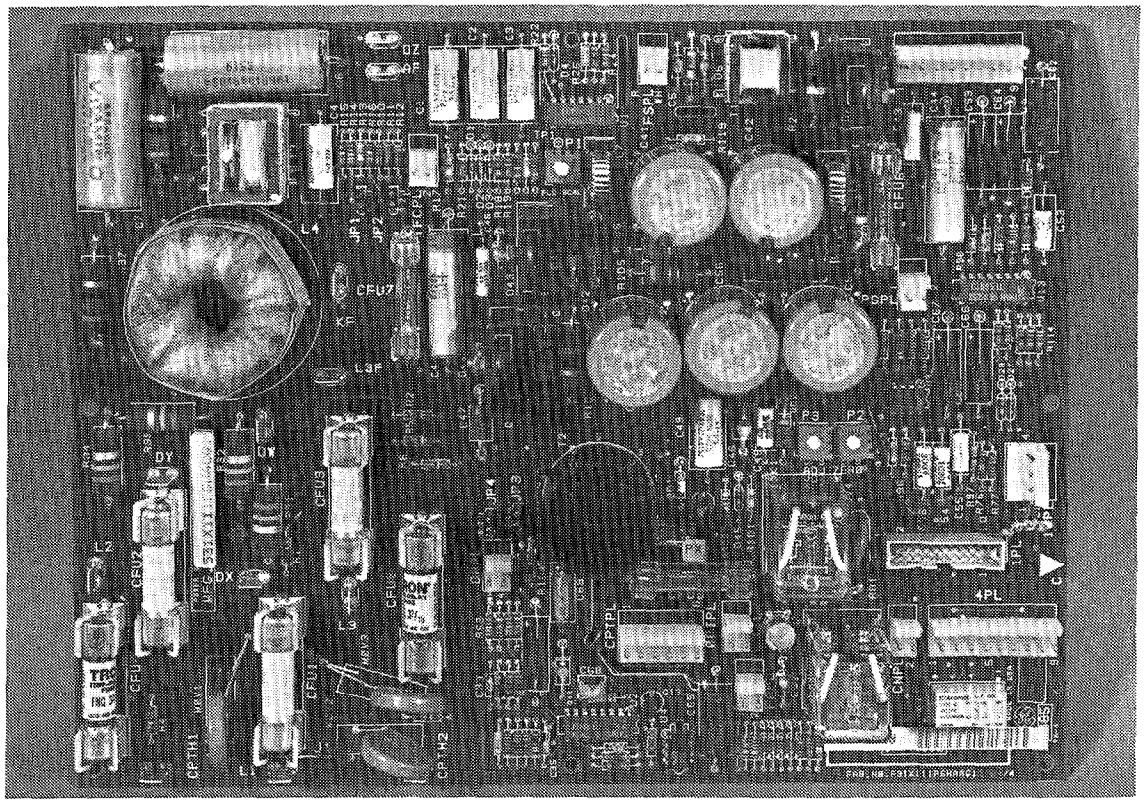


FIGURE 10
MFC/POWER SUPPLY CARD (VERSION G1)

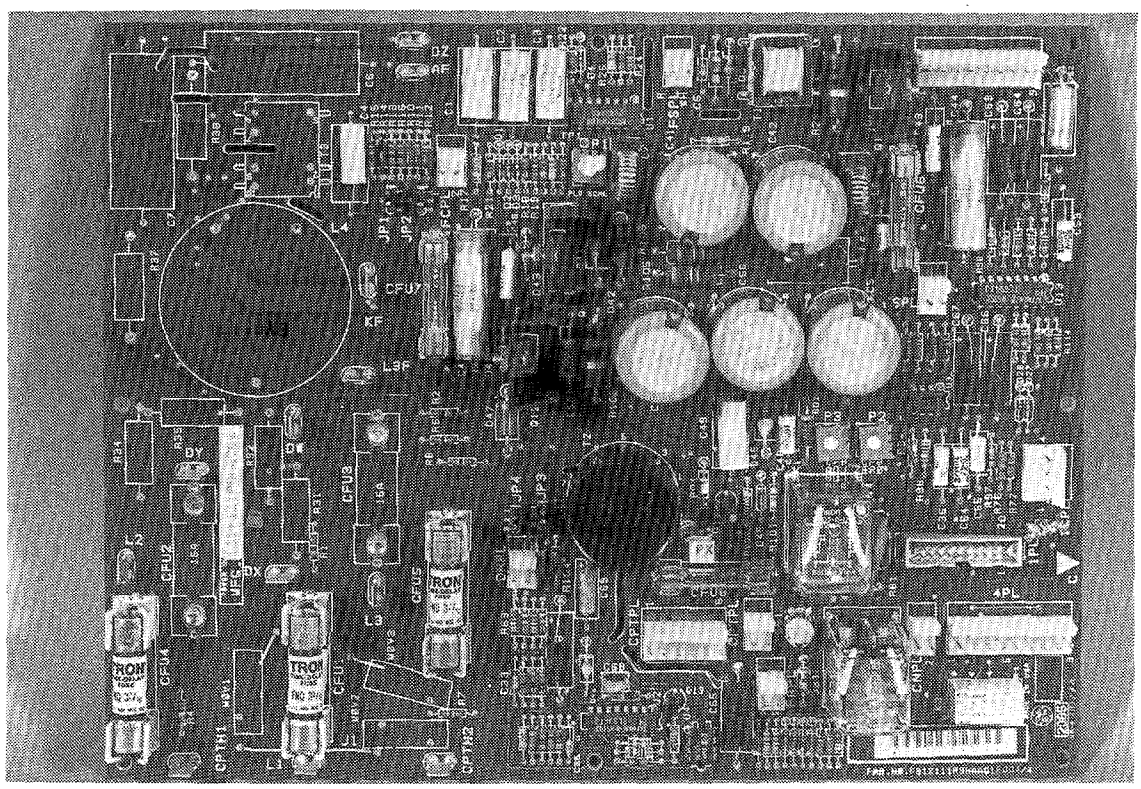


FIGURE 11
MFC/POWER SUPPLY CARD
(VERSION G2 REQUIRES EXTERNAL MFC ASSEMBLY)

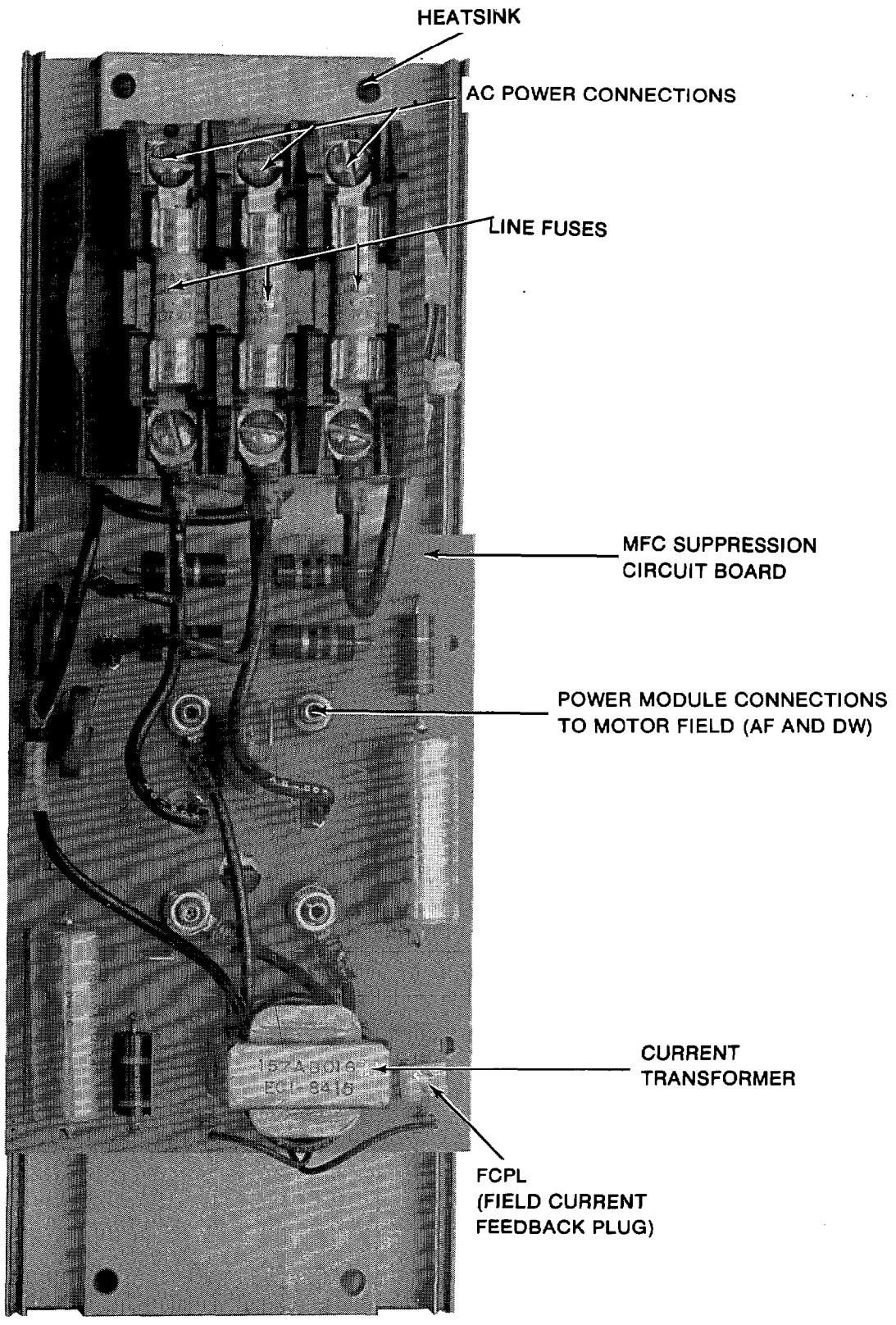


FIGURE 12
EXTERNAL MOTOR FIELD CONTROL ASSEMBLY

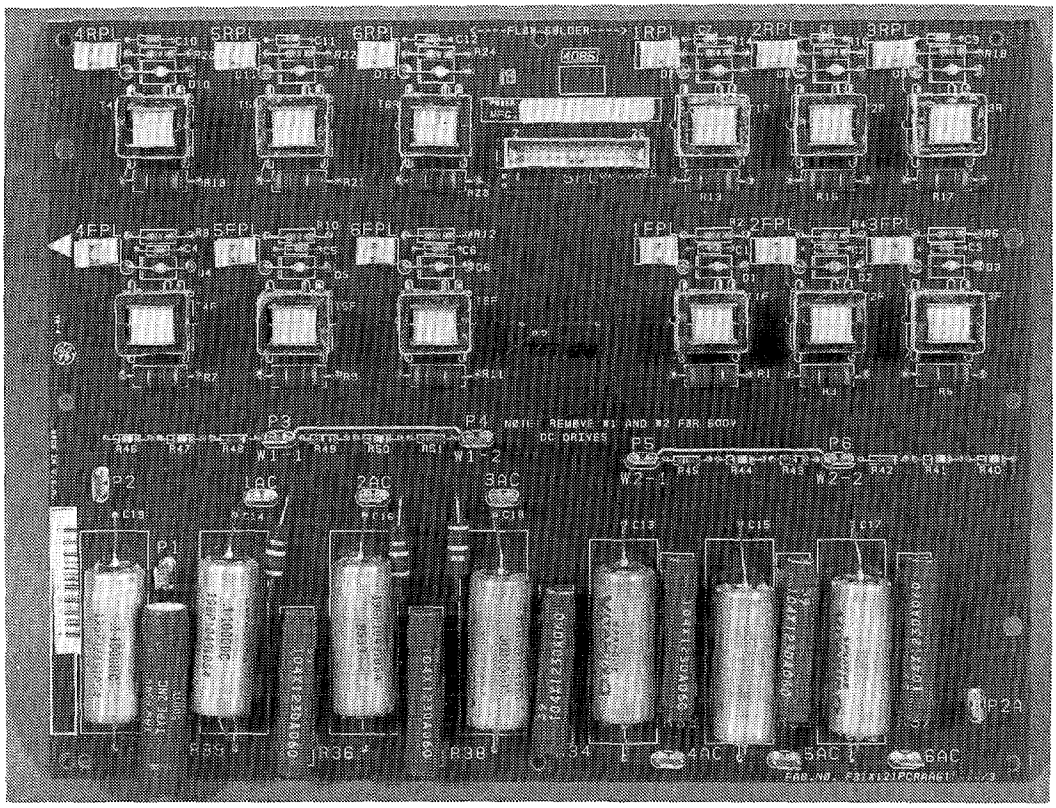


FIGURE 13
POWER CONNECTION CARD

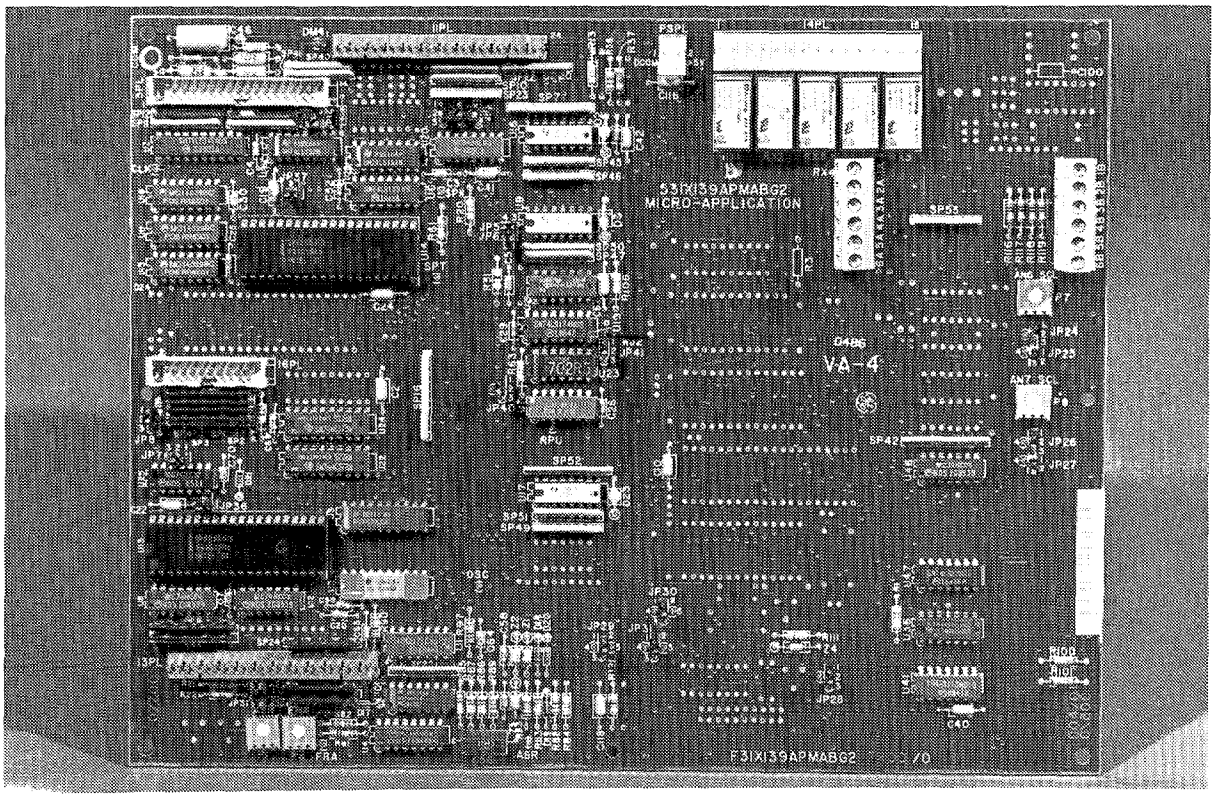


FIGURE 14
OPTIONAL APPLICATION CARD

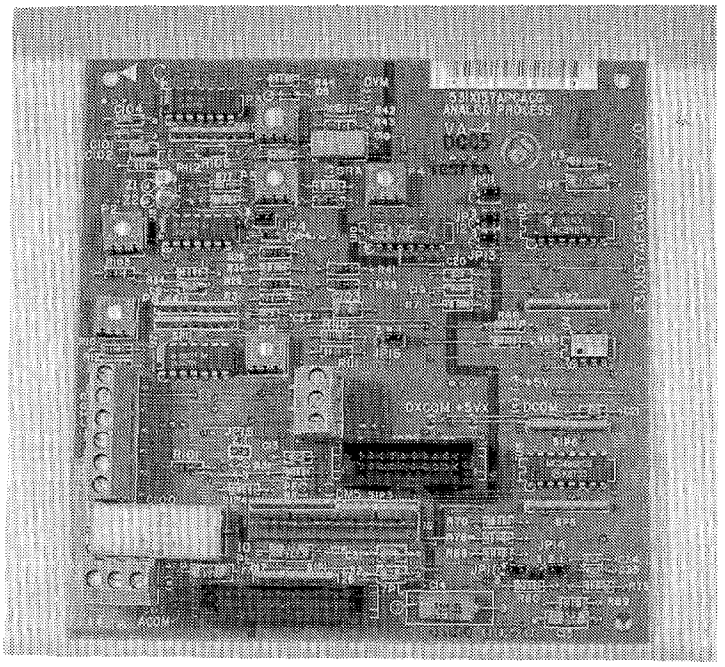


FIGURE 15
ANALOG PROCESS INTERFACE CARD

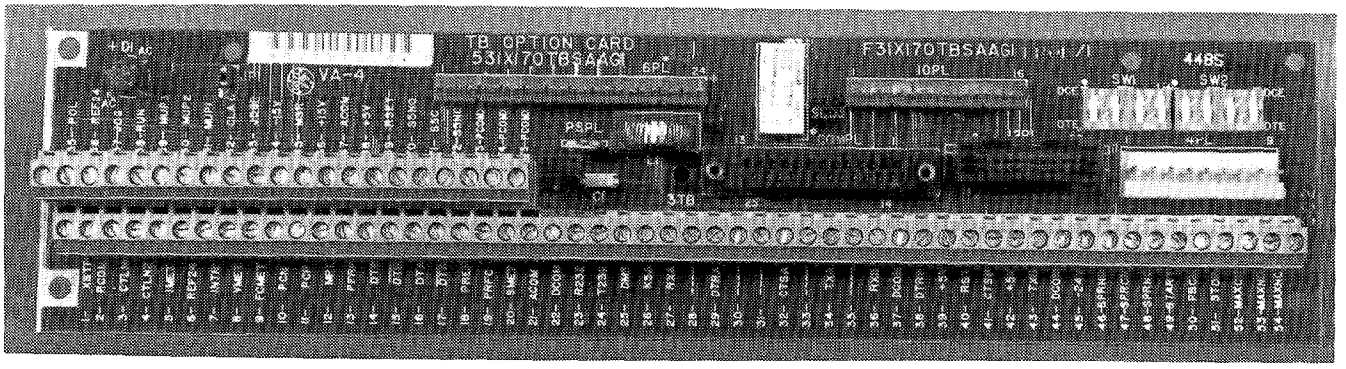


FIGURE 16
OPTIONAL TERMINAL BOARD (3TB)

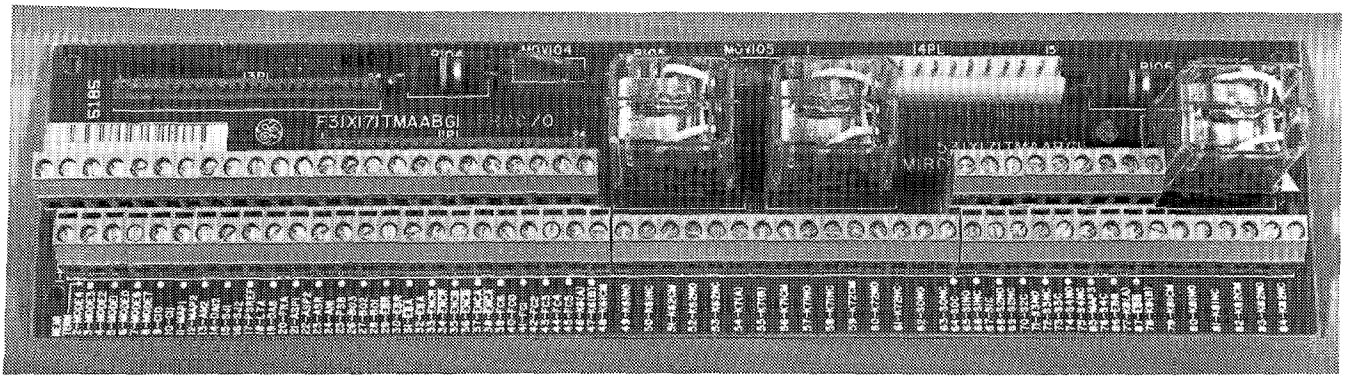


FIGURE 17
OPTIONAL TERMINAL/RELAY CARD (4TB)

CONNECTOR 1PL – BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

PIN	NOMENCLATURE	DESCRIPTION
1	DCOM	Digital Common.
2	FFC	Field firing control--drives field gate pulse transformer.
3	DCOM	
4	FC	Field current feedback from PSC.
5	DCOM	
6	RT1	AC phase 1 input from PSC.
7	RT2	AC phase 2 input from PSC.
8	RT3	AC phase 3 input from PSC (RT1, RT2, & RT3 are resistance isolated).
9	FRC	Fault relay control - drives the 24-volt fault relay on PSC.
10	-24V	Unregulated -24V return path for FRC and RP.
11	RP	MA contactor pilot relay driver (MAX - K2 on PSC).
12	RUN	Drive run signal from RUN (K1) relay contact on PSC.
13	REF24	Supplies +/-24V to RUN (K1) relay on PSC depending on setting of jumper JP20.
14	W5V	Watchdog reference--provides 5V reference from PSC to undervoltage detector on Control card.
15	J1	Input from line phase 1 CT.
16	J2	Input from line phase 1 CT.
17	J3	Input from line phase 3 CT.
18	J4	Input from line phase 3 CT.
19	DCOM	
20	CFB	Armature current feedback input from PSC.

CONNECTOR 2PL – BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

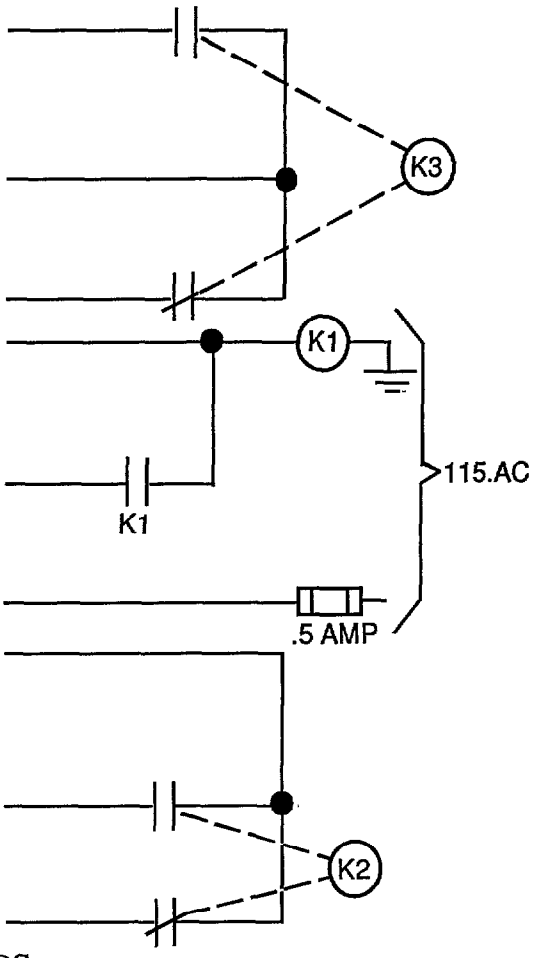
PIN	NOMENCLATURE	DESCRIPTION
1	ACOM	Analog signal return for +/-15V.
2	-15V	-15V regulated analog power supply input.
3	+15V	+15V regulated power supply input.
4	+24V	+24V unregulated raw power supply input.
5	-24V	-24V unregulated raw power supply input.
6	+5V	+5V regulated digital power supply input.
7	DCOM	Digital signal return for +5V.
8	+5V	
9	PCOM	Raw power supply return for +/-24V.

CONNECTOR 3PL – BETWEEN CONTROL CARD AND OPTIONAL APPLICATION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	D0	U4 data bus bit 0 (least significant bit).
2	D1	U4 data bus bit 1.
3	DCOM	Digital common.
4	D2	U4 data bus bit 2.
5	D3	U4 data bus bit 3.
6	D4	U4 data bus bit 4.
7	DCOM	
8	D5	U4 data bus bit 5.
9	D6	U4 data bus bit 6.
10	D7	U4 data bus bit 7.
11	DCOM	
12	A8	U4 address bus bit 8.
13	A9	U4 address bus bit 9.
14	A10	U4 address bus bit 10.
15	DCOM	
16	A11	U4 address bus bit 11.
17	SEL	Signal from U3. Used for Micro-application card synchronization.
18	ALE	Address latch from U4.
19	DCOM	
20	/RD	Read control line from U4.
21	/WR	Write control line from U4.
22	A15	U4 address bus bit 15, used for Application card I/O select.
23	RST	Card reset signal for Application card.
24	+5V	+5V regulated digital supply to Application card.
25	+5V	
26	+5V	
27	PCOM	+/-24V power supply return.
28	+24V	+24V non-regulated raw power supply to Application card.
29	-24V	-24V non-regulated power supply to Application card.
30	DM4	Provides "dummy" connection to Application card.
31	+15V	+15V regulated analog supply to Application card.
32	-15V	-15V regulated analog supply to Application card.
33	ACOM	Analog signal return for +/-15V.
34	OPTA	Input to A-D converter from Application card multiplexer.

CONNECTOR 4PL – BETWEEN POWER SUPPLY CARD (PSC) AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	46	SPRNO	Fault Relay Normally Open Contact
2	47	SPRCM	Fault Relay Common Connection
3	48	SPRNC	Fault Relay Normally Closed Contact
4	49	START	Connection Point for Start PB to "RUN" relay (K1)
5	50	PBCM	Common Point for Start and Stop Pushbuttons
6	51	STOP	Connection for Stop Pushbutton
7	52	MAXCM	MA Auxiliary Relay Common Connection
8	53	MAXNO	MA Auxiliary Relay Normally Open Contact
9	54	MAXNC	MA Auxiliary Relay Normally Closed Contact



Note: Contact rating K2 relay is 1.2 amp at 115 volts AC or 28 volts DC.
Contact rating K3 relay is 1 amp at 115 volts AC or 28 volts DC.

CONNECTOR 5PL – BETWEEN CONTROL CARD AND POWER CONNECTION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	A6F	Drives cell 6F gate pulse transformer.
2	DFP	Delayed firing power for A6F and A5F.
3	A5F	Drives cell 5F gate pulse transformer.
4	PCOM	Shield wire.
5	A4F	Drives cell 4F gate pulse transformer.
6	DFP	Delayed firing power for A4F and A3F.
7	A3F	Drives cell 3F gate pulse transformer.
8	PCOM	Shield wire.
9	A2F	Drives cell 2F gate pulse transformer.
10	DFP	Delayed firing power for A2F and A1F.
11	A1F	Drives cell 1F gate pulse transformer.
12	PCOM	Shield wire.
13	A1R	Drives cell 1R gate pulse transformer.
14	DFP	Delayed firing power for A1R and A2R.
15	A2R	Drives cell 2R gate pulse transformer.
16	PCOM	Shield wire.
17	A3R	Drives cell 3R gate pulse transformer.
18	DFP	Delayed firing power for A3R and A4R.
19	A4R	Drives cell 4R gate pulse transformer.
20	PCOM	Shield wire.
21	A5R	Drives cell 5R gate pulse transformer.
22	DFP	Delayed firing power for A5R and A6R.
23	A6R	Drives cell 6R gate pulse transformer.
24	PCOM	Shield wire.
25	DCN	Resistance isolated armature voltage.
26	DCP	Resistance isolated armature voltage.

CONNECTOR 6PL – BETWEEN CONTROL CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	69	RSET	A hard reset of drive is initiated when RSET is connected to +15V. Must be open to run. RSET stops firing of all SCRs, removes delayed firing power, drops out the MA contactor and fault relay, and holds all microprocessors in a reset state until RSET is opened. The drive will not automatically restart if reset while running. Do not apply RSET while running.
2	68	+5V	Not for use external to drive.
3	67	ACOM	Not for use external to drive.
4	66	+15V	Signal voltage for System Reference pot, RSET circuit and others.
5	65	MSR	Manual system reference input to drive, ±15V max. Positive for forward motoring with POL at zero or open.
6	64	-15V	Signal voltage for System Reference pot and others.
7	63	JOGR	Jog reference input (can be selected to bypass linear time), ±14V = Top speed. Programmable to 7V = top speed for fast update rate. Refer to Custom Software Description.
8	62	CLA	Current limit adjust--adds to software setting of current limit. +10V increases current limit by 100% of rated, -10V decreases it by 100%.
9	61	MUP1	Multiuse point 1 - connects to connection test point MUP1, which may be jumpered to another connection point. Example: DA0 may be jumpered to MUP1 for output on 3TB.
10	60	MUP2	Multiuse point 2 - connects to connection test point MUP2, which may be jumpered to another connection point.
11	59	MUP3	Multiuse point 3 - connects to connection test point MUP3, which may be jumpered to another connection point.
12	58	RUN	Alternate run command input point in place of "RUN" relay. Do not pull high if run relay is also being used. Drive runs when connected to 3TB56 (6PL-14), REF24. With JP19 and JP20 on the Control card in the 1-2 and 3-4 positions respectively, input is active (controller runs) when RUN is pulled down to -24V DC. Input is inactive (controller stops) when RUN is open or driven to +24V DC. With JP19 and JP20 in the 1-3 and 2-4 positions respectively, input is active (controller runs) when RUN is driven to +24V DC. Input is inactive (controller stops) when RUN is open or pulled down to -24V DC.
13	57	JOG	Jog command input point. Drive jogs when connected to 3TB56 (6PL-14), REF24. Polarity controlled same as RUN above.
14	56	REF24	Selectable + or -24V supply for RUN, JOG, POL or XTSP input. Polarity set by JP19 and JP20 on Control card.

CONNECTOR 6PL – BETWEEN CONTROL CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
15	55	POL	Control input which reverses the direction of the drive for any applied reference; e.g. activating POL low is equivalent to reversing the polarity of the input at MSR. Activated/deactivated same as RUN above. Programmable for multiple functions. Refer to Custom Software Description and elementaries if used.
16	1	XSTP	Must be activated to enable drive. Activated/deactivated same as RUN above. Causes a normal, coast, quick or trip stop when open (see ADR004-STPJP in the Custom Software Description).
17	2	PCOM	Not for use external to the drive.
18	3	CTLN1	Control on -- must be connected to CTLN2 to allow the drive to run. Opening this connection unconditionally drops MA contactor after a 50 millisec attempt by microprocessor to precondition the drive.
19	4	CTLN2	See CTLN1.
20	5	IMET	Analog load meter output proportional to armature current, adjustable for 1 to 8V DC output at rated load current. See P2 and JP15 on the Control card (Table 1) for details. Load not to exceed 1000 ohms/volt.
21	6	REF24	Selectable ±24V supply for RUN, JOG, POL or XSTP input. Polarity set by JP19 and JP20 on the Control card.
22	7	INTR	Programmable 24V drive from Control card to the optional signal level detector SLD5 on 3TB terminal board. Controls contacts at 3TB70, 71 and 72. See JP1 on the 3TB Terminal Board card (Table 1) for details.
23	8	VMET	Analog armature voltmeter output. This output is bipolar, normally 4 volts at rated armature voltage output. (See P8 on the Control card, Table 1, for details.) Maximum load is 10 milliamps.
24	9	FCMET	Analog field current meter output, 1 to 5 VDC output. See P5 on the Control card (Table 1) for details. Load not to exceed 1000 ohms/volt.
–	26	K5A	Optional SLD5 control input from REF24 for external control of SLD5. Controls contacts at 3TB70, 71, and 72. See JP1 on the 3TB Terminal Board card (Table 1) for details.
–	70	S5NO	Optional SLD5 normally open contact.
–	71	S5C	Optional SLD5 common contact.
–	72	S5NC	Optional SLD5 normally closed contact.

CONNECTOR 7PL – BETWEEN CONTROL CARD AND ANALOG PROCESS INTERFACE CARD

PIN	NOMENCLATURE	DESCRIPTION
1	ACOM	Analog signal return.
2	DVM	Output from digital voltmeter.
3	SFB	Output from analog tach circuit.
4	ASFB	Amplified signal from analog regulator circuit, used as armature current reference.
5	ACOM	
6	-15V	-15V supply to analog circuits on Process Interface (P.I.) card.
7	+15V	+15V supply to analog circuits on P.I. card.
8	+5V	+5V supply to logic circuits on P.I. card.
9	+5V	
10	TRS	Regulator precondition signal.
11	DCOM	Digital common.
12	DIR	Not used.
13	TPN	Not used.
14	PRF	Not used.
15	DCOM	
16	DCOM	
17	RXD	Input from either the RS422 or RS232C serial link receiver from the P.I. card.
18	TXD	Output to the RS422 and RS232C serial link drivers from the P.I. card.
19	/DTR	Data terminal ready--output indicating the P.I. card is ready to receive a byte on the serial link. Low when ready, high when not.
20	/CTS	Clear to send--input from serial link signalling when the receiver on the other end of the link is ready to receive a byte. When low, drive may transmit. When high, drive must not send anything out on TXD.
21	RST	Reset line from Control card to P.I. card, active high.
22	+5V	
23	+5V	
24	DCOM	
25	DCOM	
26	-24V	-24V raw power to P.I. card.

CONNECTOR 10PL – BETWEEN ANALOG PROCESS INTERFACE CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	10	PCN	Input for external reference.
2	11	PCP	Common (ground) input for external reference.
3	12	MRI	Not used.
4	13	PFRF	Optional reference interface output (0-10V).
5	14	DTB	Not used.
6	15	/DTB	Not used.
7	16	DTA	Not used.
8	17	/DTA	Not used.
9	18	PREF	Optional reference, (0-10 VDC) non-adjustable.
10	19	PRFC	Not used.
11	20	SMET	Analog speed meter output, 10V DC. See P2 on the P.I. card (Table 1).
12	21	ACOM	Common for optional RS232C communications.
13	22	DCOM	Not for use external drive.
14	23	R232	Optional RS232C receive line.
15	24	T232	Optional RS232C transmit line.
16	25	DM5	Spare connection tied to undedicated point DM5.

NOTE: /or $\overline{}$ = NOT: Example /C or $\overline{C}$ = Not C

TERMINAL BOARD 1TB LOCATED ON ANALOG PROCESS INTERFACE CARD

PIN	NOMENCLATURE	DESCRIPTION
1	TKN	Negative tachometer feedback signal. Maximum input is 300V DC.
3	TKP	Positive tachometer feedback signal.

CONNECTOR 11PL – BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	30	EBR	No connection (used with Micro-application card).
2	29	/EBR	No connection (used with Micro-application card).
3	85	MAP3	Micro-application multiuse point 3 - connects to connection point MAP3, which may be jumpered to another connection point.
4	25	PSB	Strobe for optional 16 bit parallel input on 16PL. See Application card JP7 (Table 1). Inputs must be set 50 micro-sec before strobe, and strobe held for 100 micro-sec.
5	26	B03	Optional programmable 24V bit output #3 (5V TTL on version APG card).
6	27	B02	Optional programmable 24V bit output #2 (5V TTL on version APG card).
7	28	B01	Optional programmable 24V bit output #1 (5V TTL on version APG card).
8	32	EBA	No connection (used with Microapplication card).
9	31	/EBA	No connection (used with Microapplication card).
10	86	EBB	No connection (used with Microapplication card).
11	87	/EBB	No connection (used with Microapplication card).
12	33	ENCR	Not used.
13	34	/ENCR	Not used.
14	35	ENCB	Not used.
15	36	/ENCB	Not used.
16	37	ENCA	Not used.
17	38	/ENCA	Not used.
18	39	FC6	Bit 6 of optional fault code 24V output (5V TTL on version APG card).
19	40	FC0	Bit 0 of optional fault code 24V output (5V TTL on version APG card).
20	41	FC1	Bit 1 of optional fault code 24V output (5V TTL on version APG card).
21	42	FC2	Bit 2 of optional fault code 24V output (5V TTL on version APG card).
22	43	FC3	Bit 3 of optional fault code 24V output (5V TTL on version APG card).
23	44	FC4	Bit 4 of optional fault code 24V output (5V TTL on version APG card).
24	45	FC5	Bit 5 of optional fault code 24V output (5V TTL on version APG card).

CONNECTOR 12PL – BETWEEN PROCESS INTERFACE CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	–	–	No connection.
2	–	–	No connection.
3	–	–	No connection.
4	–	–	No connection.
5	–	–	No connection.
6	–	–	No connection.
7	32	CTSA	Optional inverted RS422 clear-to-send input from serial link signaling when the Control card may transmit a byte, active high. (EIA standard nomenclature is CTSA.)
8	29	DTRA	Optional inverted RS422 data-terminal-ready output from Control card signaling when it is ready to receive a byte on the serial link, active high. (EIA standard nomenclature is DTRA.)
9	34	TXA	Optional inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXA.)
10	27	RXA	Optional inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXA.)
11	36	RXB	Optional non-inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXB.)
12	43	TXB	Optional non-inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXB.)
13	38	DTRB	Optional non-inverted RS422 data-terminal-ready output from Control card. (EIA standard nomenclature is DTRB.)
14	41	CTSB	Optional non-inverted RS422 clear-to-send input from serial link. (EIA standard nomenclature is CTSB.)
15	40	RST	Reset signal from Control card, active high; not for use external to drive.
16	39	+5V	Not for use external to drive.
17	42	+5V	Not for use external to drive.
18	37	DCOM	Not for use external to drive.
19	44	DCOM	Not for use external to drive.
20	45	-24V	Not for use external to drive.

CONNECTOR 13PL – BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

(**NOTE:** Refer to the Custom Instruction Book, software jumper description, for definitions of I/O through 13PL for particular application.)

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	6	MODE 0	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
2	5	MODE 1	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
3	4	MODE 2	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
4	3	MODE 3	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
5	2	MODE 4	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
6	1	MODE 5	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
7	7	MODE 6	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
8	8	MODE 7	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
9	9	CI0	No connection (used with Micro-application card).
10	10	CI1	No connection (used with Micro-application card).
11	11	MAP1	Micro-application multiuse point 1 - connects to connection point MAP1, which may be jumpered to another connection point.
12	12	MAP2	Micro-application multiuse point 2 - connects to connection point MAP2, which may be jumpered to another connection point.
13	13	AN2	No connection (used with Micro-application card).
14	14	DA12	Optional analog output from 12-bit D-A converter. Range = -10V to +10V, 1 MA, with 450 ohms output impedance. Update rate is 30 to 40 times/second.
15	15	SJ1	Optional analog input to summing junction #1. Operating range = ±10V; maximum input = ±15V.
16	16	SJ2	Optional analog input to summing junction #2. Operating range = ±10V; maximum input = ±15V.
17	17	PSREF	Optional analog position or process reference input (±10V).
18	18	LTA	Optional analog linear time adjust. Positive voltages increase timing, negative voltages decrease timing. ±10 volts maximum. 10 volts is equal to 66 second adjust.
19	19	DA8	Optional analog output from 8-bit D-A converter. See DA12 for specifications. Normal update rate of 180 times/sec. Reduced to 30 to 40 times/sec if fast update selected for DA0.

CONNECTOR 13PL – BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB (CONTINUED)

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
20	20	FRA	Optional analog field reference adjust. Positive voltages up to +10V increase field current. Negative voltages down to -10V decrease field current. See Application card JP32 and P2 (Table 1).
21	21	ASP1	Optional spare analog input #1 (programmable).
22	22	ASP2	Optional spare analog input #2 (programmable).
23	23	ASR	Optional auto speed reference input to drive, ±15V maximum, positive for forward cell firing while motoring. ASR is the active speed reference in the auto mode when analog auto reference is selected.
24	24	AN1	No connection (used with Micro-application card).

NOTE: Functions of MODE 0 - MODE 7 are determined by Addresses 000-015 (ADR000 - ADR015) in the software description.

CONNECTOR 14PL – BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	62	S0NO	Optional SLD #0 normally open contact.
2	63	S0NC	Optional SLD #0 normally closed contact.
3	64	S0C	Optional SLD #0 common contact.
4	65	S1NO	Optional SLD #1 normally open contact.
5	66	S1NC	Optional SLD #1 normally closed contact.
6	67	S1C	Optional SLD #1 common contact.
7	68	S2NO	Optional SLD #2 normally open contact.
8	69	S2NC	Optional SLD #2 normally closed contact.
9	70	S2C	Optional SLD #2 common contact.
10	71	S3NO	Optional SLD #3 normally open contact.
11	72	S3NC	Optional SLD #3 normally closed contact.
12	73	S3C	Optional SLD #3 common contact.
13	74	S4NO	Optional SLD #4 normally open contact.
14	75	S4NC	Optional SLD #4 normally closed contact.
15	76	S4C	Optional SLD #4 common contact.

CONNECTOR 16PL – LOCATED ON OPTIONAL APPLICATION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	DCOM	Digital common.
2	+5V	Not used.
3	PI0	Optional parallel word input least significant binary bit input. For all inputs refer to Table 1, JP8 description for voltage level select. Input must be set for 50 micro-sec prior to strobe.
4	PI1	Optional parallel binary bit input 1.
5	PI2	Optional parallel binary bit input 2.
6	PI3	Optional parallel binary bit input 3 (or PI0–PI3 = BCD units digit).
7	PI4	Optional parallel binary bit input 4.
8	PI5	Optional parallel binary bit input 5.
9	PI6	Optional parallel binary bit input 6.
10	PI7	Optional parallel binary bit input 7 (or PI4–PI7 = BCD tens digit).
11	PI8	Optional parallel binary bit input 8.
12	PI9	Optional parallel binary bit input 9.
13	PI10	Optional parallel binary bit input 10.
14	PI11	Optional parallel binary bit input 11 (or PI8–PI11 = BCD hundreds digit).
15	PI12	Optional parallel binary bit input 12.
16	PI13	Optional parallel binary bit input 13.
17	PI14	Optional parallel binary bit input 14 (or PI12–PI14 = BCD thousands digit).
18	PI15	Optional parallel sign bit input.
19	DCOM	Digital common.
20	DCOM	Digital common.

CONNECTOR 18PL – BETWEEN CONTROL CARD AND OPTIONAL PROGRAMMER CARD

PIN	NOMENCLATURE	DESCRIPTION
1		No connection.
2		No connection.
3		No connection.
4		No connection.
5		No connection.
6		No connection.
7		No connection.
8		No connection.
9		No connection.
10		No connection.
11	RXT	Input from the Programmer card serial link driver to Control card.
12	TXT	Output from the Control card to the Programmer card serial link receiver.
13	/DTRD	Data terminal ready–output from Control card indicating that it is ready to receive a byte from the Programmer card. Signal goes low when ready, high when not.
14	/CTSD	Clear to send–input from Programmer card signalling when it is ready to receive a byte. Signal low when ready, high when not.
15	RST	Reset line from Control card to the Programmer card, active high.
16	+5V	+5V regulated power supply to the Programmer card.
17	+5V	
18	DCOM	Signal and power return for the Programmer card.
19	DCOM	
20	-24V	-24V unregulated power supply to Programmer card.

ADDITIONAL CONNECTORS

CONNECTOR 1CPL – BETWEEN POWER SUPPLY CARD AND AC LINE CTs (CT1 AND CT3)

PIN	NOMENCLATURE	DESCRIPTION
1	J1	Input from line phase 1 CT.
2	J2	Input from line phase 1 CT.
3	J3	Input from line phase 3 CT.
4	J4	Input from line phase 3 CT.

CONNECTOR CNPL – BETWEEN POWER SUPPLY CARD AND MA CONTACTOR

PIN	NOMENCLATURE	DESCRIPTION
1	MAP	MA contactor control output.
2	X2	MA contactor control output.

CONNECTOR CPTPL – BETWEEN POWER SUPPLY CARD AND CONTROL POWER TRANSFORMER

PIN	NOMENCLATURE	DESCRIPTION
1	–	38 VAC, center tapped for controller power supplies.
2	–	38 VAC, center tapped for controller power supplies.
3	–	Center tap point for 38 VAC.
4	–	Not used.
5	–	115 VAC for controller cooling fan (if supplied).
6	–	115 VAC for controller cooling fan (if supplied).

CONNECTOR DRPL – BETWEEN POWER SUPPLY CARD AND SHUNT

PIN	NOMENCLATURE	DESCRIPTION
RED	IP	Armature current feedback from shunt, 0-100mV. Positive when motoring with the forward power circuit, Red wire.
WH	IN	Armature current feedback from shunt. Negative when motoring with the forward power circuit, White wire.

CONNECTOR FAPL – BETWEEN POWER SUPPLY CARD AND CONTROLLER COOLING FAN (IF USED)

PIN	NOMENCLATURE	DESCRIPTION
1	X1	115 VAC for controller cooling fan.
2	X2	115 VAC for controller cooling fan.

CONNECTOR FCPL – BETWEEN POWER SUPPLY CARD AND EXTERNAL MOTOR FIELD CONTROL ASSEMBLY

PIN	NOMENCLATURE	DESCRIPTION
1	ACOM	Analog common.
2	–	External motor field control feedback. Used only with external motor field control assembly.

CONNECTOR FSPL – BETWEEN POWER SUPPLY CARD AND FIELD POWER MODULE

PIN	NOMENCLATURE	DESCRIPTION
1	FLC	Gate pulse transformer output to field power module SCR, Red wire. Not used with external motor field control assembly.
2	FLG	Gate pulse transformer output to field power module SCR, White wire.

CONNECTOR OPTPL – LOCATED ON POWER SUPPLY CARD

PIN	NOMENCLATURE	DESCRIPTION
1	FX1	Fused 115 VAC from Control Power Transformer.
2	X2	115 VAC return from Control Power Transformer.

CONNECTOR PSPL – LOCATED ON POWER SUPPLY CARD

PIN	NOMENCLATURE	DESCRIPTION
1	+5V	+5V regulated digital supply (not for use external to drive).
2	DCOM	Digital signal return for +5V (not for use external to drive).

CONNECTOR COMPL – OPTIONAL R232C SERIAL COMMUNICATIONS CONNECTOR ON OPTIONAL 3TB

PIN	NOMENCLATURE	DESCRIPTION
1	GND	Ground.
2	RXA or TXA	For Data Communications Equipment (DCE), dumb terminal mode, pin 2 is the RS232C receive line (RXA).
		For Data Terminal Equipment (DTE), modem or PDS system, pin 2 is the RS232C transmit line (TXA).
3	TXA or RXA	For DCE, pin 3 is the TXA line.
		For DTE, pin 3 is the RXA line.
4	–	Not used.
5	DTRA or CTSA	For DCE, pin 5 is the data-terminal-ready (DTRA) line.
		For DTE, pin 5 is the clear-to-send (CTSA) line.
6	DSR	Data set ready, no connection in DCE or DTE mode.
7	DCOM	Pin 7 is the communication system common.
8 thru 19	–	Not used.
20	CTSA or DTRA	For DCE, pin 20 is the clear-to-send (CTSA) line.
		For DTE, pin 20 is the data-terminal-ready (DTRA) line.
21 thru 25	–	Not used.

NOTE: Refer to 3TB SW1 and SW2 description in Table 1, for setting DCE and DTE modes.

STARTUP PROCEDURE

This startup procedure is meant as a general guide for the initial checkout and startup of a **DC-300 E** drive. It references some general hardware and software checks relative to most basic drive applications. Refer to the sections on "Hardware Jumpers and Pot Adjustments" and "Key Parameter Adjustments" and the accompanying "Parameter Map" (Figure 23) for more complete information on adjustments available on the **DC-300 E** drive. Refer to the Custom Software Description Elementaries and system instructions in the Custom Instruction Book for information on wiring and adjustments specific to your drive application.

NOTE: Prior to beginning actual drive startup, read this entire manual and be aware of all pertinent information. If an optional Programmer was supplied with the drive, review the "Use of Optional Programmer Module" and "Diagnostic Mode for Drive Setup" sections. Special diagnostic functions are available which will greatly aid drive startup.

BEFORE APPLYING POWER

(Review warning and caution notices at the beginning of this instruction book before proceeding.)

Every **DC-300 E** DC SCR drive controller has been factory tested and is ready to operate provided the power and control connections have been properly made and the following steps are performed:

A. Verify that all incoming wiring is in agreement with the elementary drawings supplied with the drive, and conforms to approved wiring practices (wire size, separation of power from control, etc.).

B. Verify that incoming 3-phase voltage is in accordance with nameplate requirements (+10%, -5%), unless otherwise specified.

C. Confirm that the following jumpers (JP) and switch positions (SW) are in accordance with the jumper selection chart supplied with the drive in the drive door pocket and Custom Software Description. (Refer to the card outlines, Figures 18 thru 22, for the location of jumpers.)

1. Control Card:

JP1 - Reference Voltage Range For Maximum Reference to MSR (if used)	Position of JP1
8 - 10 V	1-2
5 - 8 V	2-3

JP15 - Loadmeter Select For Percent Load Output	Positon of JP15
Absoluted (Zero Left Scale)	2-3
Bipolar (Zero Center Scale)	1-2

2. Power Supply Card:

JP1 & JP2 – Field Current Range (version G1 Power Supply Card, Figure 10.)

For Motor Max Field Amps	Position of JP1	JP2	Gain Factor (GF)
0.64 - 1.6A	2-3	2-3	0.4
1.6 - 4.0A	1-2	2-3	1.0
4.0 -10.0A	1-2	1-2	2.5

JP1 & JP2 – Field Current Range (version G2 Power Supply Card, Figure 11.)

For Motor Max Field Amps	Position of JP1	JP2	Gain Factor (GF)
1.5 - 6.0A	2-3	2-3	1.5
6.0 - 12.0A	1-2	2-3	3.0
12.0 - 24.0A	1-2	1-2	6.0

NOTE: Gain Factor required in "Apply AC Power" section of startup procedure, Item G.

JP3 and JP4 – Armature Current Feedback Signal

For Operating Mode	Position of JP3	JP4
Normal	1-2	1-2

3. Process Interface Card:

Refer to the Custom Instruction Book for tach range and analog regulator setup (selection of R_{fb}, R_{rg}, C_{rg}, etc.).

D. Verify that all connections are tight.

E. Verify that the DC motor has been properly installed and that the shaft is free to rotate.

APPLY AC POWER

A. Check for proper direction of rotation of all three-phase blower motors on both the power unit(s) and the DC motor(s).

B. Verify that the Diagnostic LEDs are blinking one at a time from right to left. If the optional Programmer is supplied, confirm that it displays the word OPERATE.

NOTE: If at any time the LEDs remain locked, indicating a fault message, the Programmer displays a fault code, or either display is blank, refer to the "Troubleshooting" section for corrective procedures.

C. Check the field for proper voltage. Since the drive may be in field economy until a "START" is called for, check the setting in ADR050-FLDEC in the Custom Software Description to determine the proper field voltage.

If an optional programmer is supplied with the drive, proceed to the following step. If no Programmer is available, skip to step 0.

D. Place the EEROM write device on the Control card in the "ENABLE" position to allow programming.

NOTE: The EEROM write protect device is JP13 on the Control card. To enable programming move JP13 to the 2-3 position.

CAUTION

DO NOT PERFORM A "HARD RESET" OR TURN POWER "OFF" OR "ON" WITH THE EEROM WRITE DEVICE IN THE "ENABLE" POSITION.

E. Access the Parameter mode by pressing [SET] [DRV#] [7] [7] and [ENTER] on the keypad (where the number or function is encased in brackets [ ] indicates the Programmer key to be pressed).

The word PARAMETER should be displayed. The **BLACK** keypad functions are now usable. To access a parameter address, press all three numbers on the keypad corresponding to the three-digit number of the desired address. The name of the address and its present value will be displayed.

To access a new parameter, press [CLEAR/MODE] and the three digits corresponding to the next parameter desired.

F. In order to allow programming of new parameter values, first verify that the parameter address #003, DGNJP (ADR003-DGNJP) contains an odd number (1, 3, 9, etc.) by pressing [0] [0] [3]. If this address contains an **even** number, add a "1" to this number and enter the new value.

Example: ADR003 contains the number 8. Add 1 to obtain 9. Press [9] and [ENTER].

Press [CLEAR/MODE] and the three digits for the next desired parameter.

NOTE: Confirm that the drive parameters agree with the "as shipped" parameter list inside the drive door or in the Custom Software Description and System Start Up notes (where provided) prior to making any adjustments. Document any changes.

G. Adjust maximum field current (ADR076-FDLMX) to the motor nameplate value per the following formula (must be in Parameter mode):

1. $\text{MAX FLD SETTING} = \dfrac{\text{MAX N.P. FLD. CURRENT} \times 2048}{\text{GAIN FACTOR} \times 5}$

NOTE: Obtain the gain factor (GF) from item C2, under "BEFORE APPLYING POWER".

2. Example: Maximum Nameplate Field Current = 4.5A, G1 Power Supply Card, GF = 2.5

   a. $\text{SETTING} = \dfrac{4.5 \times 2048}{2.5 \times 5} = 737$

   b. Call up ADR076 - FLDMX by pressing:
      [0] [7] [6]

   c. Enter setting by pressing:
      [7] [3] [7] [ENTER]

   d. Press [CLEAR/MODE]

H. Similarly adjust minimum field by calling up ADR077-FLDMN. (Minimum field is defined at 60% of the lowest running field current.)

NOTE: Field current is at the field economy setting ADR050-FLDEC until a "START" is called for (START, RUN, JOG, etc.).

1. MIN FIELD SETTING:

$$\text{MIN FLD SETTING} = \frac{\text{MIN N.P. FLD. CURRENT} \times 0.6 \times 2048}{\text{GAIN FACTOR} \times 5}$$

2. Example: Minimum Nameplate Field Current = 2.0 amps; other parameters same as above:

   a. SETTING = $\frac{2.0 \times 0.6 \times 2048}{2.5 \times 5} = 196.6$

   b. Press [1] [9] [7] [ENTER]

   c. Press [CLEAR/MODE]

I. Some drives may have an optional "S-curve" capability. Refer to the Custom Software Description for setup information.

J. Call up ADR070-RPMSF and enter the desired motor top speed in RPM (ADR068-REFSF should contain the same value).

Example: Motor top speed is to be 1500 RPM. Enter 1500 in both ADR068 and ADR070.

K. Armature Voltage Settings

1. With the motor armature at rated voltage and test point VFB equal to 4V, the values for ADR079-CROSS or ADR034-ADVLM can be obtained from the following:

   A setting of 1638 in ADR079-CROSS is equal to rated armature volts.

   $$\text{DESIRED PARAMETER VALUE} = \frac{\text{Required Volts} \times 1638}{\text{Rated Volts}}$$

CAUTION

ADR034-ADVLM IS SET AT APPROXIMATELY 105% OF RATED ARMATURE VOLTAGE FOR REGENERATIVE DRIVES. DAMAGE MAY OCCUR TO FUSES AND SCRS DURING REGENERATION FROM TOP SPEED IF ADVLM IS SET TOO HIGH.

L. Current Feedback

1. The current feedback circuit is factory adjusted to give 1 VDC at CFB for the rated current of the shunt in the armature circuit.

2. All current parameter adjustments can be set in percent (100 = 100% shunt rated current).

3. Check P2-CFB ZERO on the Power Supply card and adjust so that test point "CFB" is as close to zero volts as possible with no current flowing in armature circuit (see Table 1 for CFB adjustments).

4. If low current limits are required, check setting of ADR040-ILMMN. No current limits from any source may be lower than this setting. In other words, set this slightly lower than the lowest current limit commanded from other sources in or external to the drive.

5. Parameter ADR129-IRATD may be set to 100 if it is desired to read % amps instead of amps on the Programmer, Diagnostic Test 5.

M. Exit the PARAMETER mode by pressing [CLEAR/MODE] [CLEAR/MODE] [CLEAR/MODE]. "OPERATE" should appear. The **RED** keypad functions will now control the drive.

NOTE: The speed reference referred to in the following steps may be preprogrammed in the drive software. Refer to the Custom Software Description and elementaries to determine the reference source.

N. Input Reference: The reference input should be connected to PCN (3TB10) and PCP (3TB11) with the ground or common side connected to PCP.

NOTE: For proper control on stops, a run interlock must be added in series with the reference lead connection to PCN – unless the reference is automatically set to zero on a stop command.

An alternate reference input is available at 3TB18-PREF which is a fixed input with 10V equal to 100% (±10%) speed.

O. Set the speed reference (speed pot, N/C reference, etc.) at its minimum setting. Start the drive (press "START" on an operator station or call for a "RUN" from the control). Verify that the MA contactor picks up and the field current is of the correct value with an amp probe or the field current panel meter if supplied.

P. Ensure that all "STOP" circuits are functional, dropping out MA. "STOP" from an operator station (115 VAC START/STOP circuit - 3TB49, 50 & 51) or a "RUN" input at 3TB58 will not function if the motor is started from the Programmer keypad.

Q. Restart the drive and slowly increase the speed reference until the motor starts to rotate. If the motor does not rotate, or rotates in the wrong direction, check the Diagnostic LEDs or the Programmer display. If DC tach feedback is provided, a tach polarity fault may be indicated. Refer to the fault codes in the Troubleshooting section for corrective action.

NOTE: If an optional Programmer is included in the drive, an on-board digital voltmeter (DVM) and electronic wire check are available to verify wiring and make the following adjustments. Refer to the "Use of Optional Programmer Module" section for instructions.

R. If MSR is connected to PFRF: With the drive **stopped**, increase the speed reference to 100% speed. Adjust P1-REF SCALE on the Control card, to obtain 4 VDC at test point SRS (Manual Speed Reference only. Jumper JP1 on the Control card selects the manual speed reference input voltage range). Measure with volt-ohmmeter (or the DVM diagnostic test as mentioned above) at the appropriate test point and PCOMX. Set the speed reference to 0% speed.

S. Start the drive and gradually increase speed reference to maximum. Check for 4 VDC at test points SRS and SFB on the Control and Process Interface cards. Adjust P8-SFB SCALE if necessary to get 4.0 V at SFB at top speed.

WARNING

EXERCISE CAUTION AS HIGH VOLTAGE MAY BE PRESENT.

T. Confirm good motor commutation (absence of sparking between brushes and commutator) under load.

U. Slowly decrease the speed. Verify stability through the full speed range. If the drive is **NOT** stable, refer to the Stability Adjustment procedure in the Troubleshooting section and Start Up Notes, where provided for the system.

V. After the drive is stable throughout the full speed range, confirm that the armature current does not exceed the nameplate rating.

W. The above list of adjustments should result in a properly running drive with desired readouts at any diagnostic test on the Programmer.

Before continuing, it is recommended that the following be done:

- The list of "as shipped" parameter settings provided with the controller be checked against the actual parameters in the drive.

- All incoming signals be checked for proper value and correct connection. Refer to the "Use of Optional Programmer Module" section Test 20 - Electronic Wire Check.

X. Similarly, other parameters as sold and shipped with the drive can be accessed and changed as necessary. (A complete list of pertinent parameters is contained in the Custom Software Description shipped with the drive.) A partial listing of "KEY" Parameters, their descriptions, and a Parameter Map follows.

NOTE: Final responsibility for parameter settings resides with the customer. Document all changes to parameter settings in the space provided in the Custom Software Description.

Y. Further accidental reprogramming should be prevented by subtracting a 1 from the odd value presently in ADR003-DGNJP (refer to Paragraph "F" this section), and entering the new value; then place the EEROM write device in the "PROTECT" position.

Z. Exit the parameter mode by pressing [CLEAR/MODE] [CLEAR/MODE] [CLEAR/MODE]. OPERATE should appear.

The drive is now ready for operation.

ADJUSTMENTS

The following section details adjustments available in hardware and software. The hardware portion describes the location and function of all jumpers and potentiometers on the DC-300 E controller cards. The software portion describes key software "parameter" adjustments available and provides a "Parameter Map" that relates the adjustments to electrical hardware equivalents.

HARDWARE JUMPER AND POT ADJUSTMENTS

Most of the jumper selections and pot adjustments have been factory set. When one of the cards has to be changed, make sure the replacement card switches, jumpers and pots are set to be the same as on the replaced card. For a full description of the jumper and pot adjustments, see Table 1.

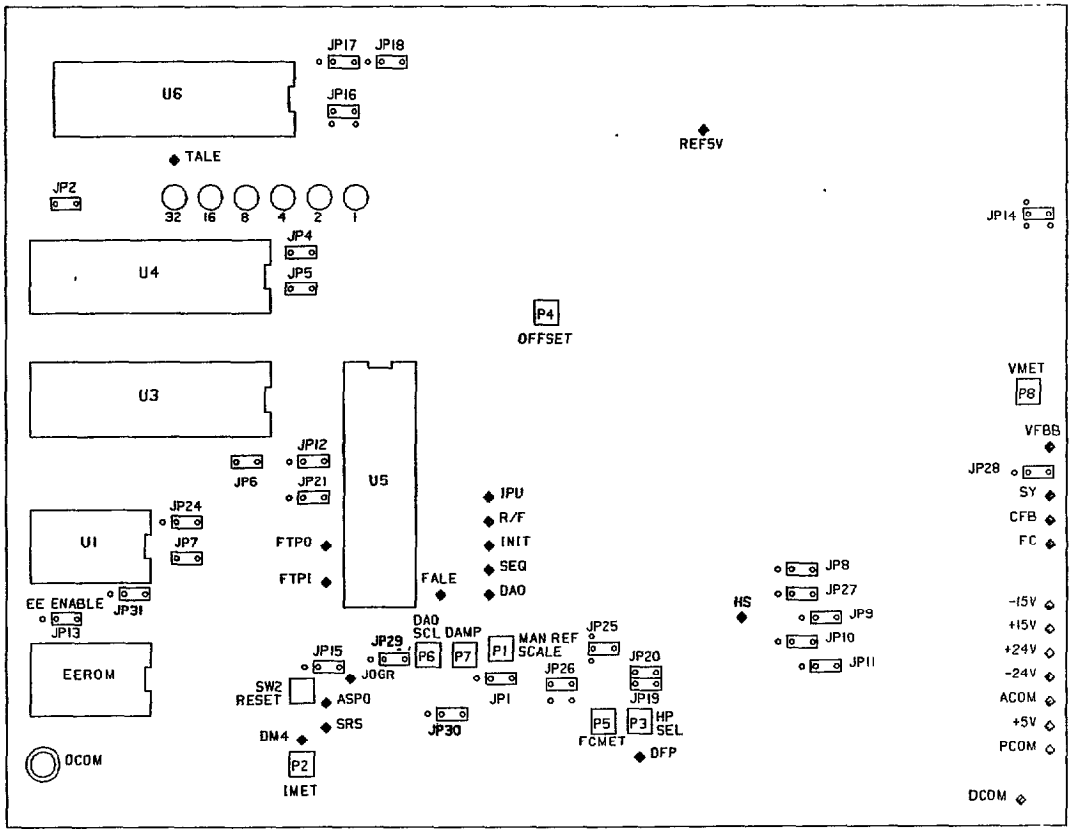
NOTE: If the Control card is replaced, the EEROM (U12) from the original card should be put into the replacement card. If the same failure symptoms still exist, the new EEROM, shipped with the replacement card, can be reinstalled but must be reprogrammed per the customer software jumper and adjustment values supplied in the Custom Software Description or parameter list in the drive door pocket.

Hardware jumpers and pots are located on the Control card, Power Supply card, Process Interface card, optional Application card, and optional Programmer card. Figures 18 thru 22 show the positions of pots, jumpers, test points and other key components.

Most of the pots and jumpers are used in the interface circuits of the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use. The test data sheets supplied with each controller (in the drive door pocket) indicate the position of these adjustments as they were set in the factory. Adjustments not set in the factory can easily be made by the startup procedure in this book or in the Startup Guide and by use of Table 1.

NOTE: Only high impedance digital voltmeters or the optional "on-board" DVM (Test 3) should be used to make indicated measurements.

NOTE: Before changing any jumper positions, compare actual jumper positions with "as shipped" jumper positions through use of the Custom Software Description.



MISC ELEC 442 (R6/4/86)

FIGURE 18
CONTROL CARD

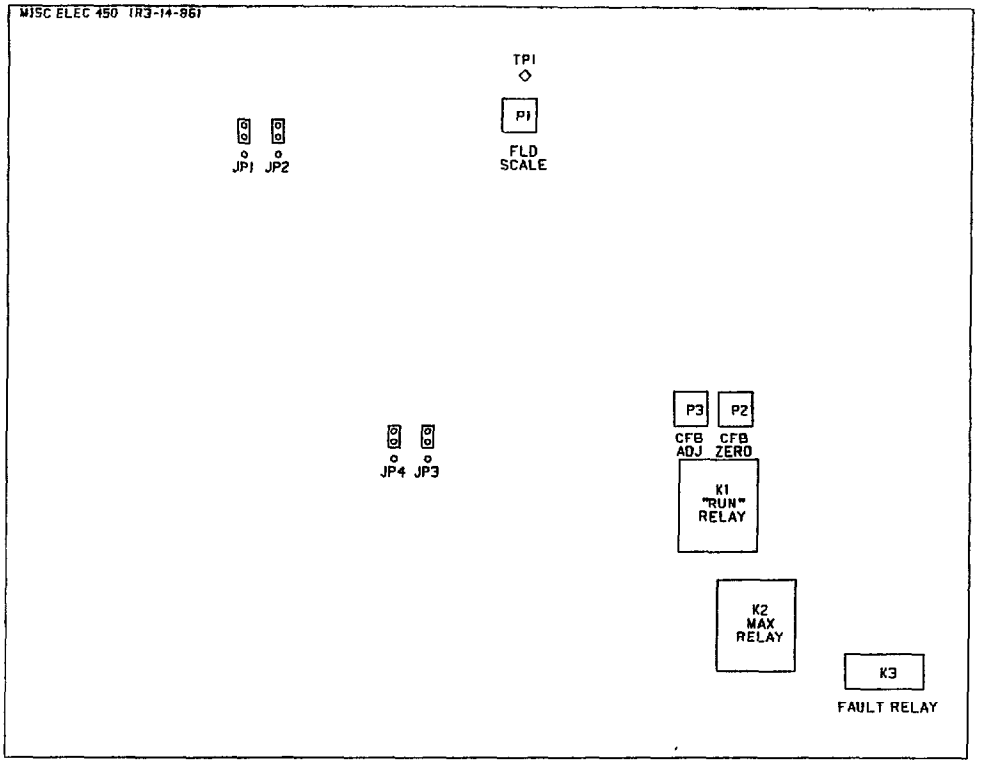


FIGURE 19
POWER SUPPLY CARD

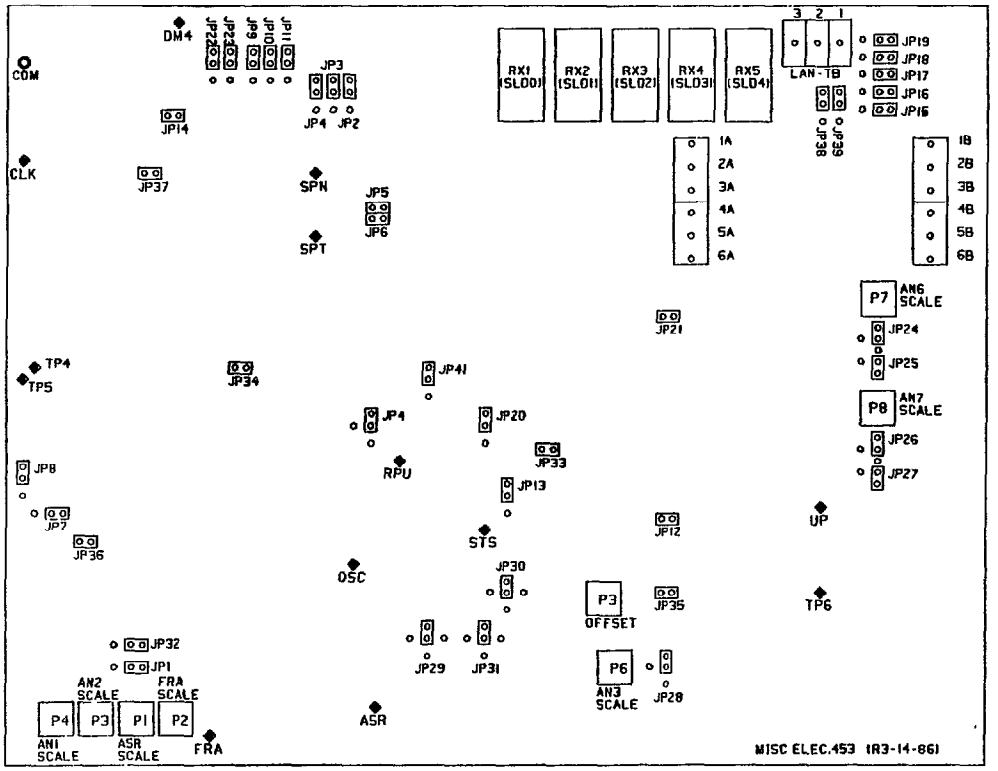


FIGURE 20
OPTIONAL APPLICATION CARD

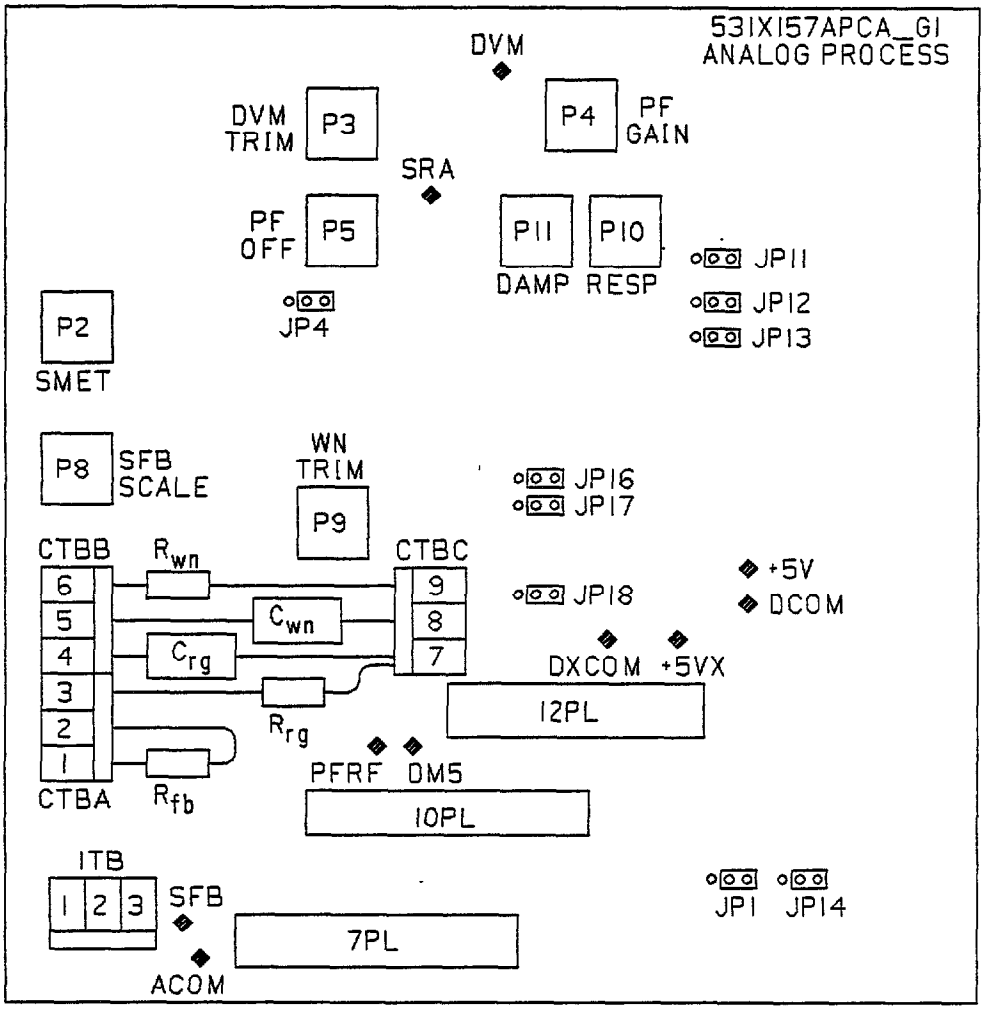


FIGURE 21
ANALOG PROCESS INTERFACE CARD

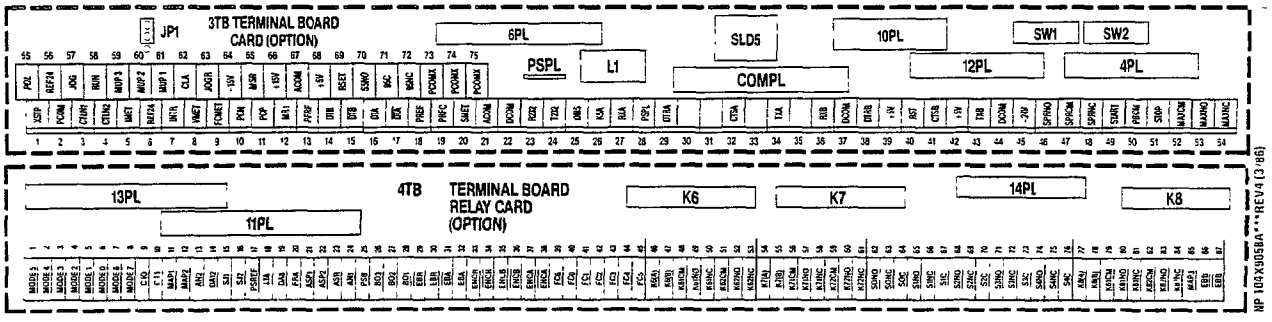


FIGURE 22
OPTIONAL TERMINAL BOARDS (3TB AND 4TB)

TABLE 1

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Pot	P1	MAN REF	To adjust test point SRS to four (4) volts when top speed reference is at MSR, 3TB65 (6PL-5; connector 6PL, pin number 5). Used in conjunction with jumper JP1.
		P2	IMET	This pot is used to adjust the output of the analog load meter circuit. Adjust the output voltage to be from 1 to 8 volts at rated load. Maximum current load on this signal is 1000 ohms/volt.
		P3	HP SEL	Adjusted by the manufacturer. To adjust in the field, set to make it visually the same on replacement cards.
		P4	OFFSET	Adjusted by the manufacturer, no need to be adjusted in the field unless directed to do so by fault codes. Refer to adjustment procedure in "Use of the Optional Programmer Module" section if a Programmer is available.
		P5	FCMET	This pot is used to adjust the output of the analog field current voltmeter circuit. Adjust the output from 1 to 5 volts at rated maximum field current. Maximum current load on this signal is 1000 ohms/volt.
		P6	DA0 SCL	This pot is used to adjust the gain of the 12-bit digital-to-analog converter (D A0) output from the Control card. CCW sets the gain at 0.25, full CW sets the gain at 1.25. The output saturates at 12 VDC.
		P7	DAMP	Not used.
		P8	VMET	This pot is used to adjust the output of the analog armature voltage meter circuit. Output is adjustable from 0 to 8 volts at rated voltage. Maximum current load on this signal is 10mA.
	Jumper	JP1	MAN REF	This jumper selects the maximum voltage range for the manual speed reference input. 1-2: 8 to 15 volts 2-3: 5 to 8 volts
		JP2 JP4 JP5 JP6 JP7		JP2 - JP7 Set as follows: 1-2: Should be in this position for proper operation.

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Jumper	JP8 JP9 JP10 JP11		These jumpers are set in accordance with the HP of the drive. Refer to the Custom Instruction Book for proper positions.
		JP12		Allows SCR test without Programmer 1-2: Disables SCR test (normal operation). 2-3: SCR test before power-up. An SCR test will be executed the first time a RUN is commanded. Subsequent RUN commands will execute normally. (Not used with elevators.)
		JP13		This jumper provides a hardware write protection for the software jumpers and adjustments. 1-2: Protected (should be in this position for maximum security and reliability of the drive). 2-3: Enables reprogramming. Do not perform a "hard" reset or turn power on or off in this position.
		JP14		This jumper selects proper line voltage. 1-2: 440 to 480 VAC 2-3: 380 to 415 VAC 2-5: 310 to 340 VAC 3-4: 220 to 240 VAC
		JP15		This jumper selects whether the load meter driver output is to be bipolar or absoluted. 1-2: Absoluted (always positive) 2-3: Bipolar (zero center)
		JP16 JP17 JP18		JP16 - JP18 set as follows: Set by factory - refer to jumper listing in Custom Instruction Book or drive door pocket for proper setting.

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Jumper	JP19 JP20		JP19 & JP20 set the polarity of RUN, JOG, POL and XSTOP. 1-2 & 3-4 Inputs are active when pulled down between -24 to 0.4 VDC. Inputs are inactive when open or tied to +3.5 to +24 VDC. 1-3 & 2-4 Inputs are active when tied between +3.5 to +24 VDC. Inputs are inactive when open or held at -24 to 1.5 VDC.
		JP21		Allows interactive card test. 1-2: Disables card test (normal operation). 2-3: Initiates card test if drive is not running. To exit card test, move JP21 back to the 1-2 position and hard reset drive (push SW2).
		JP24 JP25 JP26 JP27 JP28 JP30 JP31		JP22-JP28, 30 and 31 set as follows: Set by factory - refer to jumper listing in Custom Instruction Book or drive door pocket for proper setting.
		JP29		This jumper selects whether the DA0 output has a fixed gain or adjustable gain. 1-2: Fixed gain (see Table 3, RAM Variable Scaling). 2-3: Gain adjustable by P6-DA0 SCL.
	Switch	SW2	Hardware Reset	This switch causes a hardware reset ("hard" reset of the controller) when pressed. Used to clear some drive faults (refer to Table 5, Specific Fault Troubleshooting). The power supplies are toggled "OFF" and "ON" and the drive will return to the initial power up condition.

TABLE 1 (continued)

<table>
<tr><th>LOCATION</th><th>JUMPER OR POT</th><th>NUMBER</th><th>NOMEN-CLATURE</th><th>DESCRIPTION</th></tr>
<tr><td rowspan="4">MFC/ Power Supply card</td><td rowspan="3">Pot</td><td>P1</td><td>FIELD SCALE</td><td>P1 is used to calibrate the field current feedback using a known field current. This pot is set and sealed by the manufacturer.</td></tr>
<tr><td>P2</td><td>CFB ZERO</td><td>P2 is used to adjust offset of the current isolator circuit so that test point CFB (on Control card) is zero.</td></tr>
<tr><td>P3</td><td>CFB ADJUST</td><td>CAUTION
THE DRIVE MUST NOT BE RUNNING WHEN MAKING THIS ADJUSTMENT.

P3 is used to adjust the gain of the current isolator circuit so that CFB equals the value indicated on the label inside the right side of the card compartment (in front of the Power Supply card holder).
Pots P2 and P3 are interdependent and should be adjusted alternately.
WARNING
ELECTRICAL SHOCK HAZARD, HIGH VOLTAGE IS PRESENT ON THE POWER SUPPLY CARD. REMOVE POWER FROM THE DRIVE BEFORE HANDLING THE PLUG AND JUMPERS IN THIS PROCEDURE.

P3 can be set by disconnecting the current feedback signal (DRPL) and moving jumpers JP3 and JP4 to the 2-3 position. This applies a calibration voltage (approx. 0.1V) between pin IP and IN of DRPL. P2 should be adjusted prior to moving JP3 and JP4. Adjust P3 until test point CFB on the Control card measures the value indicated on the drive calibration label. After setting P3, P2 should be rechecked. Return JP3 and JP4 to the 1-2 positions and reconnect DRPL.
CAUTION
IF JP3 AND JP4 ARE LEFT IN THE 2-3 POSITION, ERRATIC OPERATION AND CARD FAILURES CAN OCCUR.</td></tr>
<tr><td>Jumper</td><td>JP1
JP2</td><td></td><td>JP1 and JP2 are used to select field current feedback signal (FC) gain.

Version G1 MFC/Power Supply card (Figure 10).
<table>
<tr><th>JP1</th><th>JP2</th><th>GAIN FACTOR</th><th>FIELD AMP RANGE</th></tr>
<tr><td>2-3</td><td>2-3</td><td>0.4</td><td>0.64 to 1.6 AMPS</td></tr>
<tr><td>1-2</td><td>2-3</td><td>1.0</td><td>1.6 to 4.0 AMPS</td></tr>
<tr><td>1-2</td><td>1-2</td><td>2.5</td><td>4.0 to 10.0 AMPS</td></tr>
</table>
Version G2 MFC/PSC (Figure 11) and optional Remote Field Module (Figure 12).
<table>
<tr><th>JP1</th><th>JP2</th><th>GAIN FACTOR</th><th>FIELD AMP RANGE</th></tr>
<tr><td>2-3</td><td>2-3</td><td>1.5</td><td>1.5 to 6.0 AMPS</td></tr>
<tr><td>1-2</td><td>2-3</td><td>3.0</td><td>6.0 to 12.0 AMPS</td></tr>
<tr><td>1-2</td><td>1-2</td><td>6.0</td><td>12.0 to 24.0 AMPS</td></tr>
</table></td></tr>
</table>

TABLE 1 (continued)

<table>
<tr><th>LOCATION</th><th>JUMPER OR POT</th><th>NUMBER</th><th>NOMEN-CLATURE</th><th>DESCRIPTION</th></tr>
<tr><td rowspan="2">MFC/ Power Supply card</td><td rowspan="2"></td><td>JP3
JP4</td><td></td><td>JP3 and JP4 are used for calibration of the armature current feedback signal.
JP3 JP4 Function
1-2 1-2 Normal Operating Mode
2-3 2-3 CFB calibration (See description at pot P3.)</td></tr>
<tr><td>JP5</td><td></td><td>K2 coil suppression capacitor selection jumper:
1-2: Capacitor in circuit (normal).
2-3: Capacitor out of circuit.</td></tr>
<tr><td>Power Connection card</td><td>Jumper</td><td></td><td>P1
P2
P3
P4
P5
P6
P7
P8
P9
P10</td><td>Select armature voltage feedback scaling resistors dependent on rated DC voltage output. Set by connecting wire jumpers with stab-on type connectors across the points indicated.
JUMPERS OUTPUT VOLTAGE
NONE 500V DC
P4-P8 and P5-P9 450V DC
P3-P7 and P6-P10 400V DC
P4-P7 and P5-P10 340V DC
P7-P8 and P9-P10 290V DC
P3-P4 and P5-P6 240V DC
P3-P8 and P6-P9 190V DC</td></tr>
<tr><td rowspan="2">3TB Terminal Board</td><td></td><td>JP1</td><td></td><td>JP1 selects whether optional SLD5 mounted on 3TB is controlled by the Control card or by an external contact closure.
1-2: Control card control of SLD5, programmed by ADR085, 086 and 087.
2-3: External control of SLD5 by connecting 3TB26 K5A to 3TB56, REF24.</td></tr>
<tr><td>Switch</td><td>SW1
SW2</td><td></td><td>SW1 and SW2 select the mode of serial link communication through COMPL on 3TB (depress switch as shown by dot to set for up or down position). For use with Data Terminal Equipment (DTE):
SWITCH SET
SW1-1 thru 3: Up Up
SW1-4: Down
SW2-1, 3, 4: Down Down
SW2-2: Up
For use with Data Communication Equipment (DCE), all switches set in down position.</td></tr>
</table>

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Analog Process Interface card	Pot	P2	SMET	Used to adjust the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA.
		P3	DVM TRIM	With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.
		P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the reference input circuit. PF GAIN adjusts for 6 to 10V at PCP to PCN.
		P8	SFB SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed. Also, see R_{fb}.
		P9	WNTRIM	Sets optimum action of sharp notch filter in speed regulator, if used. Also, see R_{wn} and C_{wn}.
		P10	RESP	Sets optimum response for best drive performance. Clockwise increases gain.
		P11	DAMP	Adjusts dampening of speed regulator with rate of change of armature voltage.
	Jumper	JP1		JP1 selects circuitry for the serial link communication mode. 1-2: Enables RS422 Reception. 2-3: Enables RS232C Reception.
		JP4		JP4 selects the speed meter (SMET) circuit mode. 1-2: Unipolar speed meter driver mode. 2-3: Bipolar meter drive mode (DC tach only).
		JP11 JP12 JP13 JP14		**JP11 JP12 JP13 JP14** 1-2 1-2 1-2 1-2 RS422 Serial Link - Standard 2-3 2-3 2-3 2-3 RS232C Serial Link - Standard (DTR low) 2-3 2-3 1-2 2-3 RS232C Serial Link - Standard (DTR hi) 1-2 2-3 2-3 2-3 RS422 Xmit/RS232C Handshake (Series 6 Programmable controller compatible)
		JP16		1-2: Activates sharp notch filter. 2-3: Filter inactive.

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Analog Process Interface Card	Jumper	JP17 JP18		Selects gain resistors in speed loop regulator circuit, in parallel with R_{rg} on CTBA3 to CTBC7. 1-2: Resistor in circuit (JP17=332K; JP18=182K) 2-3: Resistor out. (Do not set both JP17 and JP18 in 2-3 position unless R_{rg} is installed.
	Resistors & Capacitors		R or C R_{rg}* C_{rg}* R_{fb}* R_{wn} C_{wn}	Discrete resistors and capacitors are used on the Analog Process Interface card for some adjustments. Whenever replacing the Analog Process Interface card, assure that the proper value components are installed on the new card.. **Connects To — Function** CTBA3-CTBC7 — Speed Regulator Gain CTBB4-CTBC7 — Speed Regulator Gain CTBA1-CTBA2 — Tach Feedback Scaling CTBB6-CTBC9 — Notch Filter Response CTBB5-CTBC8 — Notch Filter Response * Specifications for the selection of these components are provided in the Custom Instruction Book for the drive.
Application card (optional)	Pot	P1	ASR SCALE	Not used.
		P2	FRA SCALE	P2 is used to scale the field reference adjust input at 13PL-20 (4TB20). One volt amplitude at test point FRA increases/decreases the FLDMX parameter setting by 204 counts.
		P7	AN6 SCALE	P7 is used to provide a variable input lead for general purpose amplifier #1. Refer to elementaries and operating notes for further information, if used.
		P8	AN7 SCALE	P8 is used to provide a variable input lead for general purpose amplifier #2. Refer to elementaries and operating notes for further information, if used.
	Jumper	JP1		Not used.
		JP5 JP6		JP5 and JP6 select polarity of the eight logic inputs, MODE 0-MODE 7, at 13PL-1 thru 8 (4TB1 thru 8). 1-2 & 3-4: The control is active low. Inputs are active when pulled below 0.8 volts. 1-3 & 2-4: The control is active high. Inputs are active when pulled above 3.5 volts or driven by TTL high output.

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Application card (optional)	Jumper	JP7		Not used.
		JP8		Not used.
		JP14		JP14 set as follows (if provided): 1-2: Should be in this position for proper operation. Open: Only for manufacturer's use.
		JP24 JP25 JP26 JP27 JP28 JP29 JP30 JP31		JP24 thru JP31 define functions for the on-board general purpose amplifiers (GP AMP #1 and GP AMP #2). Refer to drive elementaries, operating notes and Custom Software Description for more information (if used).
		JP32		JP32 selects the scaling range of P2-FRA SCALE. 1-2: Normal range. 2-3: Extended range.
		JP36 JP37		JP36 & JP37 set as follows: 1-2: Should be in this position for proper operation. Open: Only for manufacturer's use.
		JP40 JP41		Not used.
		JP45		JP45 selects whether P7-AN6 SCALE is in or out of the GMP AMP #1 circuit: 1-2: Connects P7 into amplifier circuit. 2-3: Disconnects P7 from amplifier circuit.
		JP46		JP46 selects whether P8-AN7 SCALE is in or out of the GMP AMP #1 circuit: 1-2: Connects P8 into amplifier circuit. 2-3: Disconnects P8 from amplifier circuit.
Programmer Card (optional)	Jumper	JP1 JP3 JP4 JP5		JP1, JP3, JP4 in 1-2 position. JP5 in 2-3 position for use with Control card 18PL.

TEST POINTS

As a further aid to troubleshooting, a series of test points have been provided to allow for specific signal observation and/or measurements. Test points not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
ACOM	Common for analog circuit power supplies.	
CFB	Armature current feedback signal - positive when forward SCRs are conducting. Maximum external loading is 5 mA.	Calibration label voltage level at rated drive current (approx. 1.0 VDC)
DA0	Scaled output from 12-bit digital-to-analog converter (DAC).	
DCOM	Common for digital circuit power supply.	
DFP	Delayed firing power for the SCR gate circuits. Maximum external loading is 25 mA. Enabled only when armature SCR firing is allowed.	+24V when enabled or 0V when disabled
MUP1 (DM1)	Multiuse point 1, connects to 6PL-9. Multiuse points may be used to jumper signals from other test points to the associated connector for custom applications. Refer to elementaries supplied with drive.	
MUP2 (DM2)	Multiuse point 2, connects to 6PL-10.	
MUP3 (DM3)	Multiuse point 3, connects to 6PL-11.	
FC	Field current feedback from Power Supply card. Maximum external loading is 1 mA. $FC = \frac{\text{MAX RATED MOTOR FIELD AMPS X 4 VDC}}{\text{MAX FIELD CURRENT SET BY JP1 and 2}}$ **NOTE:** See Table 1, Power Supply card for maximum field current set by JP1 and 2.	
JOGR	Jog reference input at 6PL-7.	
-15V	-15 VDC power supply.	-14.25 to -15.75 VDC
+15V	+15 VDC power supply.	+14.25 to +15.75 VDC
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
R/F	High when reverse SCRs being fired. Low when forward SCRs being fired.	High = 2 to 5 V Low = 0 to 2 V
SRS	Scaled speed reference signal.	4 volts at top speed reference
VFBB or VFB	Armature voltage feedback signal. Positive when forward motoring. Maximum external loading is 1 mA. Not for external use. **NOTE:** VFBB is affected by P8-VMET.	4 volts at rated armature voltage

POWER SUPPLY CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
TP1	Field current feedback signal. Maximum loading is 1 mA.	

PROCESS INTERFACE CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
DVM	Input probe connection for digital voltmeter. Requires optional Programmer.	
SFB	Analog speed feedback signal.	4 volts at top speed
PFRF	Output of the reference interface circuitry.	10 volts at top speed
DM5	Dummy point 5, connects to 10PL-16.	
SRA	Current reference (output from analog speed regulator).	±1V = 100% current

OPTIONAL APPLICATION CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
ASR	Auto System Reference signal.	4 volts at top speed reference
DA8	Output from the 8-bit D-A converter. Only on version APG cards.	±10 volts = maximum signal
DA12	Output from the 12-bit D-A converter. Only on version APG cards.	±10 volts = maximum signal
FRA	Field reference adjust signal.	±10 volts = full adjust
RPU	Encoder interface marker track, one pulse per revolution.	

SOFTWARE ADJUSTMENTS

Table 2 on the following pages is a listing of "standard" software adjustments (parameters) common in most **DC-300 E** drive applications. Included is the "**DC-300 E** Parameter Map" (equivalent analog representation of parameters), the "Key Parameter Adjustments", and the "RAM Variable Scaling" table, which defines the scaling of RAM addresses available for interrogation of the drive control "circuit". The parameters can be examined and adjusted with the optional Programmer (refer to "Use of Optional Programmer Module") or via the serial link.

Additionally, a Custom Software Description is shipped with every drive. The Custom Software Description describes the setting and/or function of various programmable jumpers and adjustments (hardware and software) within the drives that are identified with the specific model number of that drive. Included in that book is a listing of specific programmer options, their description and a set of software adjustment definitions pertinent to the specific drive.

PARAMETER MAP (Figure 23)

The Parameter Map is a diagnostic aid designed to reference specific software adjustments and correlate them to their specific regulating functions. For example, ADR031-BFLPG is represented as a potentiometer feedback across the field current regulating amplifier, giving an effective linear gain adjustment of input to output.

Software adjustments are shown next to their symbol with the mnemonic and the parameter address in a box beside it. Hardware adjustments are represented by a symbol, the adjustment name and the card it is located on (e.g.: FCMET is a potentiometer, P5 on the Control card). RAM addresses that can be used to examine outputs of certain stages of regulation through TEST 11, TEST 21, TEST 22 or D-A converters, are shown as a three-digit number in a circle, connected to the regulator section it is associated with (e.g.: Speed loop output can be examined by using RAM address 053 in Test 11, 21, 22 or the D-A converters).

The optional Programmer display scaling adjustments are shown in the lower left section of the map as well as their relation to feedback.

TABLE 2

KEY PARAMETER ADJUSTMENTS

The following parameter adjustments, listing Programmer mnemonic, address number, function name, and description, will prove helpful in setup and troubleshooting of the **DC-300 E** drive. This is a partial list and should be used in conjunction with the "Parameter Map". For a more complete list of parameters contained in a specific drive, refer to the Custom Software Description furnished with that drive.

The following is a description of column headers:

NAME	Mnemonic displayed by Programmer when interrogating parameter.
ADR	Parameter address.
UNITS	Unit designation of displayed number (example - RPM).
MIN	Minimum allowable value.
MAX	Maximum allowable value.
STEP	Amount by which value can be incremented or decremented.
DEFAULT	Standard value if no special setting is required.

NOTE: Parameters listed in Table 2 are grouped functionally per the parameter map (Figure 23).

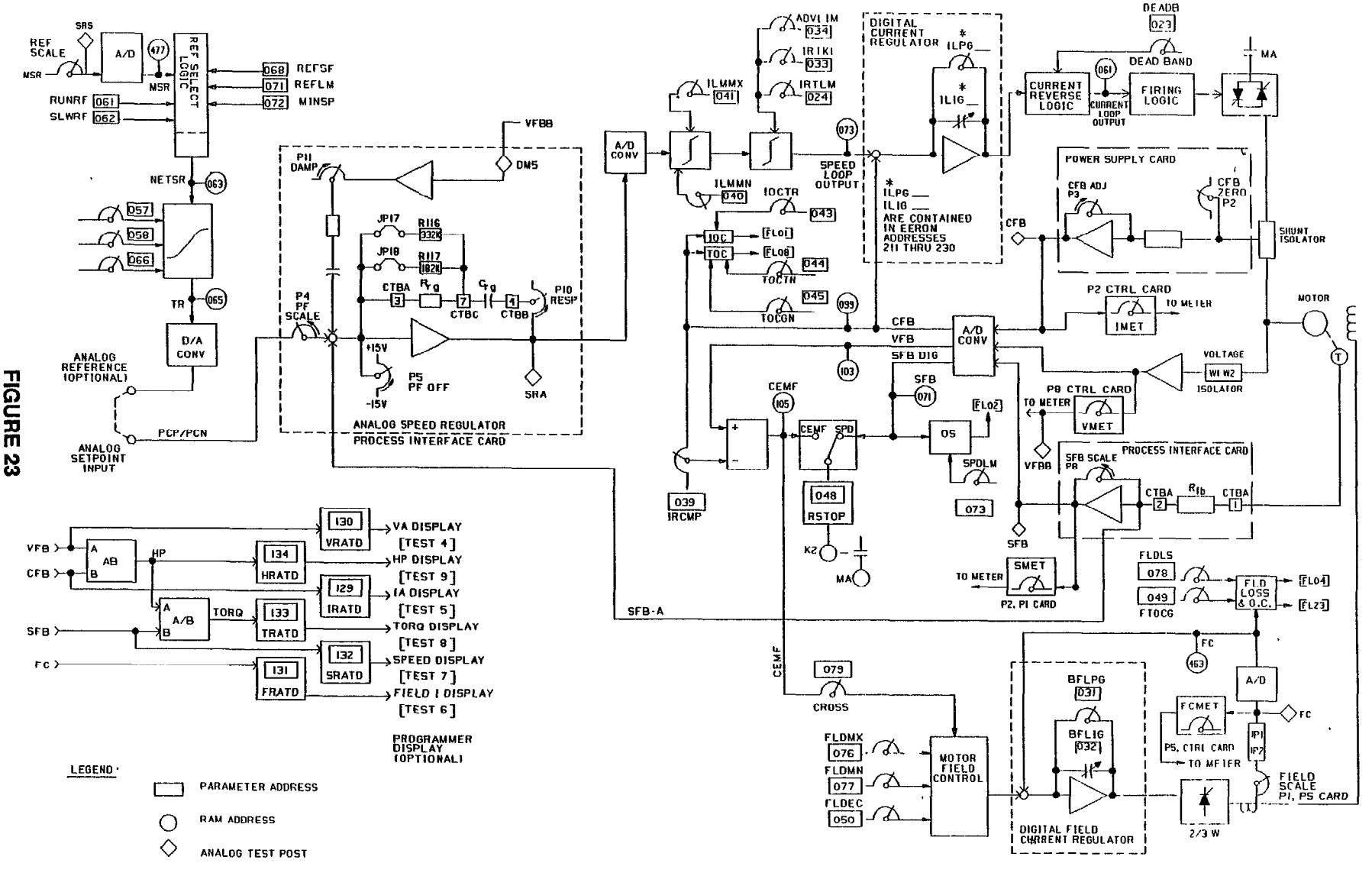


FIGURE 23
DC-300 E PARAMETER MAP

TABLE 2 (continued)
KEY PARAMETER ADJUSTMENTS

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

MOTOR FIELD

FLDMX 076 None 0 2047 1 800

Full Field Current – sets the motor field current below base speed. Also is the voltage reference in the open loop mode of field operation. In the current regulated mode, 1638 corresponds to 4 volts at test point FC. The field current scaling jumpers on the Power Supply card should be set to give the largest feedback at FC that does not exceed roughly 4.5 volts. In the open loop mode, 0 corresponds to fully retarded firing and 2047 corresponds to fully advanced firing.

$$FLDMX = \frac{2048 \times I_F\,(max)}{5 \times \text{Gain Factor}}$$

The gain factor is determined by the field current scaling jumpers on the Power Supply card (refer to Table 1).

FLDMN 077 None 0 2047 1 200

Minimum Field Current – sets the minimum field current level the drive will regulate. 1638 equals 4 volts at FC.

$$FLDMN = \frac{0.6 \times 2048 \times I_F\,(min)}{5 \times \text{Gain Factor}}$$

NOTE: This equation varies for material handling applications.

FLDLS 078 None 0 2047 1 100

Field Loss Trip Setpoint – sets the level of field current below which the drive will trip. 1638 equals 4 volts at FC.

$$FLDLS = \frac{2048 \times I_F\,(min)}{12.5 \times \text{Gain Factor}}$$

NOTE: A setting of 60,000 disables the field loss trip function. This equation varies for material handling applications.

CROSS 079 None 0 2047 1 1850

Field Crossover – sets the CEMF level at which the field will begin to weaken. A setting of 1638 corresponds to rated motor voltage (4 volts at VFB). Determine % of Rated Armature volts desired at Crossover. Multiply that % by 1638 to obtain CROSS parameter setting.

FLDEC 050 None 0 255 1 0

Field Economy – sets level of field current when drive is stopped. Determine % of FLDMX desired for field economy. Then FLDEC will equal that percent x 255.

Example: 70% of maximum field current desired for field economy.
0.70 x 255 = 178.5
Setting of FLDEC = 179.

NOTE: A setting of zero defaults FLDEC to 70% field current.

BFLPG 031 None 0 255 1 40

Base Field Loop Proportional Gain – sets the proportional gain of the field current regulator below base speed.

BFLIG 032 None 0 255 1 20

Base Field Loop Integral Gain – sets the integral gain of the field current regulator below base speed.

FTOCG 049 None 0 255 1 82

Field TOC Gain - - sets the time constant of the field timed overcurrent integrator. Set to 0 to inhibit this feature. For field forcing of FF per unit over a setting of FLDMX, the drive will trip in 6990/(FLDMX * FF * FTOCG) minutes. The default allows 0.1 P.U. forcing over a nominal FLDMX of 850 for 1 minute.

TABLE 2 (continued)
KEY PARAMETER ADJUSTMENTS

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

REFERENCES AND SCALING

REFSF 068 None 0 6553 1 6553

Reference Scale Factor – sets the factor by which references entered from the serial link and Programmer will be multiplied and by which speeds displayed on the Programmer and sent to the serial link will be divided. Enter the value of speed reference which corresponds to top speed (4 volts at SFB). Important - affects settings of RUNRF, SLWRF, JOGRF, R1RF, R2RF, SLKRF, parallel input scaling, and speed reference display. With Tach, REFSF = Top Speed RPM; for CEMF regulator, REFSF = Base Speed RPM at rated armature volts.

*** RUNRF** 061 RPM 0 6553 1 0

"Run" Key Reference – sets the preset reference for RUN commands from the serial link. Also may be used as the preset reference for manual start commands. Also serves as the top speed reference for motor operated rheostat (MRH) mode. May be a speed, torque or position depending on jumper setting in ADR006-RUNJP. Normally RUNRF = Top Speed (RPM).

*** SLWRF** 062 RPM 0 6553 1 0

"Slow" Key Reference – sets the preset reference for SLOW commands from the serial link. Normally SLWRF = Slow Speed (RPM).

*** JOGRF** 063 RPM 0 6553 1 10% of REFSF

"Jog" Key Reference – sets the preset reference for Jog commands from the serial link. Also may be used as the preset reference for manual jog commands. JOGRF = Jog Speed (RPM).

RPMSF 070 RPM 0 6553 1 6553

RPM Scale Factor – sets the speed in RPM which corresponds to 4 volts at SFB. NOTE: Change parameter only while the drive is not running, and toggle the reset switch afterwards to make the change fully effective. Important - affects setting of REFLM, MINSP, SPDLM, tapered current limit speed setting and digital tach scaling. Same as setting for REFSF.

*** REFLM** 071 RPM 0 6656 1 6560

Maximum Speed – sets the level at which speed reference will be clamped if the input reference magnitude exceeds the level. Maximum value should not exceed 1.015 x RPMSF. Typically, REFLM = RUNRF + 100.

*** MINSP** 072 RPM 0 6553 1 0

Minimum Speed – sets the level at which speed reference will be clamped if the input reference magnitude is less than the level. MINSP = desired Minimum Speed (RPM). Maximum value should not exceed RMPSF.

SPDLM 073 RPM 0 7936 1 7208

Speed Limit Trip Setpoint – sets the speed at which the drive will trip. SPDLM = RMP at drive overspeed trip. Should not exceed 1.2 x RPMSF.

***NOTE:** Parameters ADR061-RUNRF, ADR062-SLWRF, ADR063-JOGRF, ADR071-REFLM and ADR072-MINSP are only used if the optional "S-curve" is provided.

TABLE 2 (continued)
KEY PARAMETER ADJUSTMENTS

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

ARMATURE CURRENT REGULATOR

ILMMX 041 **% Rated** 10 510 2 150

Maximum Current Limit – sets the maximum DC current level allowed by the drive. One unit = 1% current.

ILMMN 040 **% Rated** 10 510 2 10

Minimum Current Limit – sets the minimum allowable current limit level; must be set **below** the desired **maximum** current limit setting. Also acts as the third current limit setting when tapered current limit is used. One unit = 1% current.

IOCTR 043 **% Rated** 10 510 2 400

Instantaneous Overcurrent Trip Setpoint – sets the current level at which the drive will trip. One unit = 1% current.

TOCTH 044 **% Rated** 10 510 2 120

Timed Overcurrent Sense Threshold – sets the current level at which the TOC integrator begins to time-up. One unit = 1% current. Refer to the Custom Software Description if multi-motor function is used.

TOCGN 045 **None** 0 255 1 24

Timed Overcurrent Response – sets the time constant of the TOC integrator. Refer to the Custom Software Description if multi-motor function is used.

$$TOCGN = \frac{66{,}666}{(\text{Trip TIme})\ (X\text{-}TOCTH)}$$

where X = % of armature current corresponding to desired trip time.

NOTE: The default values for TOCTH and TOCGN allow the drive to operate at 150% current for 90 seconds.

ILPG **ADR211 thru ADR230**
ILIG

Current loop proportional and integral gains for ten discrete current ranges. Odd addresses contain proportional gain parameters, ILPG0 through ILPG9. Even addresses contain integral gain parameters, ILIG0 through ILIG9.
Refer to the Custom Instruction Book for details on the values for these parameters.

ADVLM 034 **None** 1 255 1 255

Sets advance limit of SCR firing angle and thereby limits maximum armature voltage.

DEADB 023 **None** 0 255 1 0

Current Regulator Deadband – controls the sensitivity of the current loop reversal logic. A setting of 0 is maximum sensitivity (dither mode).

IRTLM 024 **PU/S** 1 255 1 50

Current Rate Limit – controls the maximum rate of change of current in response to commanded changes. Exact relation is current rate = 0.9 (IRTLM) per unit per second.

TABLE 2 (continued)
KEY PARAMETER ADJUSTMENTS

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

OUTPUT SIGNALS

DAC0 **090** None 0 65535 1 0

D-A #0 Output Control – controls what internal variable is sent to the 12-bit D-A converter (DAC0) on the Control card. Enter the desired RAM variable number into ADR090 to program DAC0.

RAM ADDR	VARIABLE NAME	RAM ADDR	VARIABLE NAME
053	Loop Output	071	SFB - Speed Feedback
055	CFB - Current Feedback	103	VFB - Voltage Feedback
061	Current Loop Output	105	CEMF
065	Timed Speed Reference	463	Field Current
069	Jog Reference	501	TOC Timer

Refer to the "RAM Variable Scaling", Table 3 at the end of this listing for relative value of RAM addresses through various outputs.

DAC8 **096** None 0 65535 1 0

D-A #1 Output Control – controls what internal variable is sent to the 8-bit D-A converter (DAC8) on the Application card. See DAC0 for programming information.

DAC12 **097** None 0 65535 1 0

D-A #2 Output Control – controls what internal variable is sent to the 12-bit D-A converter (DAC12) on the Application card. See DAC0 for programming information.

BIT #1 **098** None 0 65535 1 0

Bit#1 Output Control – controls what internal bit is sent to bit output #1 (B01) on the Application card. Possible outputs include:

RAM ADDR	VARIABLE NAME	RAM ADDR	VARIABLE NAME
061	Direction of Firing	148	Reference Enable
137	Fault	152	Precondition
140	Current Limit Flag	156	Zero Current Flag
141	CEMF Limit Flag		

In addition, the sign of all of the variables given in DAC8 output control may be used as outputs.

BIT #2 **099** None 0 65535 1 0

Bit#2 Output Control – controls what internal bit is sent to bit output #2 (B02) on the Application card. See Bit #1 for programming information.

BIT #3 **100** None 0 65535 1 0

Bit#3 Output Control – controls what internal bit is sent to bit output #3 (B03) on the Application card. See Bit #1 for programming information.

TABLE 2 (continued)
KEY PARAMETER ADJUSTMENTS

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
DIAGNOSTICS						
DIAG1	122	None	0	65535	1	0
		Diagnostic Variable #1 – selects 1 of 2 variables which can be displayed on the serial link or Programmer using "display diagnostics". See DAC8 output control for programming information.				
DIAG2	123	None	0	65535	1	0
		Diagnostic Variable #2 – selects second variable which can be displayed on the serial link or Programmer using "display diagnostics". See DAC8 output control for programming information.				
SCHRL	127	None	1	26	1	18
		Left Scaled Unit Character – determines which character is displayed as the left character of the scaled speed status message. 1 represents "A"; 2, "B"; 26, "Z".				
SCHRR	128	None	1	26	1	13
		Right Scaled Unit Character – see ADR127-SCHRL for programming information.				
IRATD	129	None	0	3276	1	100
		Displayed Arm Current Scaling – controls the factor by which per unit current is multiplied for display on the Programmer.				
VRATD	130	None	0	2621	1	100
		Displayed Arms Volts Scaling – controls the factor by which per unit armature voltage is multiplied for display on the Programmer.				
FRATD	131	None	0	43690	1	1000
		Displayed Field Current Scaling – controls the factor by which per unit field current is multiplied for display on the Programmer.				
SRATD	132	None	0	6200	1	100
		Displayed Speed Scaling – controls the factor by which per unit speed is multiplied for display on the Programmer.				
TRATD	133	None	0	3276	1	100
		Displayed Torque Scaling – controls the factor by which per unit torque is multiplied for display on the Programmer.				
HRATD	134	None	0	3276	1	100
		Displayed HP Scaling – controls the factor by which per unit horsepower is multiplied for display on the Programmer.				

TABLE 3
RAM VARIABLE SCALING FOR PROGRAMMER AND DAC OUTPUTS

RAM ADDR	VARIABLE NAME	VALUE AT 100% SIGNAL DECIMAL	VALUE AT 100% SIGNAL HEX[1]	VOLTS ON DAC[2] AT 100% SIGNAL
053	SLOPO-Speed Loop Output (10 Bit)	100	0064	2.0
055	CFB-Current Feedback (10 Bit)	100	0064	2.0
099	CFB-Current Feedback (14 Bit)	1600	0640	2.0
061	ILOPO-Current Loop Output	220	00DC	4.3
063	NETSR-Net Speed Reference (if used)	6553	1999	8.0
065	TR-Timed Reference (if used)	6553	1999	8.0
071	SFB-Speed Feedback	6553	1999	8.0
073	SLOPO-Speed Loop Output (14 Bit)	1600	0640	2.0
103	VFB-Voltage Feedback	1638	0666	8.0
105	CEMF-Counter Electromotive Force	1638	0666	8.0
463	FC-Field Current (rated current not scaled)	2047	07FF	10.0
477	MSR-Manual Speed Reference (if used)	6553	1999	8.0
501	TOCT-Timed Overcurrent Timer (trips at 1024)	1024	0400	0.3
067	SLOPE-Speed Loop Tracking Error (MSR versus SFB)	6553	1999	10.0

NOTE: (1) Negative numbers are represented in Two's complement.

(2) D/A converters DAC8 and DAC12 have an op-amp with a gain of two which is included in the listed values.

TROUBLESHOOTING

This section contains information on detecting and correcting problems which may be encountered when working on a **DC-300 E** drive controller. The **DC-300 E** controller contains a built-in fault detection system which indicates when a fault occurs and the nature of the fault. Tables 4 and 5 are intended to aid in the correction of problems both indicated by the drive fault system and those not indicated by the fault display. Following Table 5 is a "Stability" Troubleshooting section.

Table 4, the "BugBuster™ Troubleshooting Chart" is for aid in troubleshooting problems which have no associated fault or error indication or cannot be remedied through the standard fault system. Table 4 includes a "Quick Reference Guide", an outline of the possible problem descriptions included in the chart.

Table 5, "Specific Fault Troubleshooting" defines drive fault and error codes. Fault codes are displayed on the Diagnostic LEDs and Programmer module, and broadcast on the optional serial link. Error codes are only related to Programmer or serial link errors, and therefore, only displayed by the Programmer module or broadcast on the serial link.

Fault/Error Codes

The **DC-300 E** drive controller displays faults with a series of Diagnostic LEDs (interpreted as a binary number) or with a fault message (fault number and name) on the optional Programmer.

NOTE: In the normal condition the drive's LEDs will flash from right to left:

one at a time in the standby mode

two at a time in the RUN or JOG modes

The following table lists predefined fault codes in order by the assigned fault number. To determine the fault number from the Diagnostic LEDs, add the number value of the LEDs printed on the card above or below the LEDs (also add to this total the value of 64 if the LEDs are blinking). For example:

Types of Faults

The faults listed in Table 5 may be divided into four categories:

L - The fault is latched, i.e., once it occurs it will not clear itself even if the cause of the fault no longer exists. The controller must be reset to clear it.

N - The fault is non-latched, i.e., it will clear itself if the cause of the fault no longer exists.

T - The fault trips the drive, i.e., it causes the controller to remove power from the motor. The controller must be reset. All trip faults are latched.

A - The fault is annunciated only, i.e., if the drive is running it will not cause the drive to shut down. However, an annunciated fault will prevent the drive from starting if it is not running. Annunciated faults may or may not be latched.

There are several ways to reset the controller. A software reset may be executed from the Programmer card, or by serial command from a device connected to the serial link. A hardware reset may be executed by toggling the reset switch on the Control card, by momentarily connecting 6PL-1 to 6PL-4 (3TB69 to 3TB66), or by momentarily interrupting incoming power to the drive. Most latched faults may be adequately cleared by a software reset, however, some faults should be cleared by a hardware reset. These are indicated in Table 5 by the letter "H". If a software reset does not seem to restore the drive to normal operation, a hardware reset should be attempted.

FAULT CODE IDENTIFICATION

LED Indication	Binary Number						Decimal Number	Fault Number
	32	16	8	4	2	1		
Steady display	0	1	1	1	0	0	16+8+4 = 28	FL28
Blinking display	0	1	0	1	0	0	16+4+64 = 84	FL84

NOTE: 0 = LED OFF 1 = LED ON 1 = BLINKING LED

TABLE 4
BUGBUSTER TROUBLESHOOTING CHART
QUICK REFERENCE GUIDE

PROBLEM DESCRIPTION

DRIVE FAULTS, LED INDICATION

1. All six diagnostic LEDs off
2. All six diagnostic LEDs on
3. Fault messages displayed (repetitive)
4. Fault messages displayed (non-repetitive, random)

PROGRAMMER PROBLEMS

5. Programmer display off
6. Error messages displayed
7. Programmer ineffective - no errors displayed
8. Parameter changes ineffective
9. Diagnostics do not perform

FUSE BLOWING, SCR PROBLEMS, MOV FAILURES, CARD FAILURES

10. Fuse blowing, AC or DC power fuses
11. Field fuses blowing
12. Power supply fuses blowing
13. MOV failures
14. Repetitive card failures
15. Random SCR misfirings
16. Line voltage problems

OPERATIONAL PROBLEMS

17. 24 Volt logic not operating properly
18. Drive/Motor does not run
19. Field current or voltage wrong value
20. Drive runs in wrong direction
21. Drive does not reverse
22. Drive speed is incorrect
23. Serial link does not function properly
24. Erratic operation
25. Stability problems
26. Signal level detector chatter
27. Too much coast time on deceleration (regen. drives)

MOTOR PROBLEMS

28. Motor sparking
29. Motor runs hot
30. Motor audible noise
31. Motor vibration

NOISE, WIRING, POWER SUPPLIES, ETC.

32. EEROM memory rewrites
33. Noise or random nuisance faults
34. Wiring procedures
35. Checking power supplies

TABLE 4
BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
1. All Diagnostic LEDs OFF	1. Check Programmer display for drive fault (if used). 2. Check for 3-phase AC input. 3. Check for open AC line fuses (FU1-3) or breaker. 4. Check for open fuses on the Power Supply card (CFU1-8). 5. 2PL disconnected, open, loose or intermittent. 6. Power supplies loaded down or failed - refer to "Checking Power Supplies", this section. 7. If Hand-held Programmer is available, connect and attempt to enter the PARAMETER mode. Refer to Error Code List if any detected. A hard reset should clear this lock-up condition. 8. Replace Control card software (U1, U5 and U6). 9. Replace Control card.
2. All Diagnostic LEDs ON	1. Drive faulted - check fault code list. 2. U5 and U6 microprocessor interchanged on the Control card. 3. RSET-3TB69 connected to +15V. 4. Extended memory U1 installed - check JP24 per test data sheet in drive door pocket. 5. Replace Control card software (U1, U5 and U6).
3. Fault Messages Displayed (Repetitive)	1. See Specific Fault Troubleshooting, Table 5, for troubleshooting procedures related to specific drive fault number.
4. Fault Messages Displayed (Non-Repetitive, Random)	1. See problem description on "Noise or Random Nuisance Faults". 2. See "Wiring Procedure".
5. Programmer Display OFF	1. Is plug connected correctly? • Local Programmer - 10 conductors must be in **right** side of plug when connected. • Hand-held Programmer - red stripe must appear on the left-hand side of the cable leaving the Programmer on the left-hand side of the cable when inserted in 18PL. 2. Check power supplies - refer to "Checking Power Supplies", this section. 3. Replace Programmer card.
6. Error Messages	1. See Error Message Listing in Table 5..
7. Programmer Ineffective No Errors Displayed	1. Is plug inserted properly? • Local Programmer conductors must be in **right** side of plug when 10-cm is connected. • Hand-held Programmer - red stripe must appear on the left-hand side of cable leaving the Programmer and the left-hand side of plug when inserted in 18PL. 2. Check jumpers JP3 and JP4 on the Programmer card for proper position. 3. Check Programmer jumpers JP1 and JP2 - must be in 1-2 position. 4. Is the drive in Auto mode - this mode inhibits certain mode references. 5. Replace Programmer.

TABLE 4 (continued)
BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
8. Parameter Changes Ineffective	1. Check JP13 on write protect switch on the Control card, must be in 2-3 or "ENABLE" position for programming. 2. Check setting in ADR003-DGNJP - must be an odd number to allow reprogramming. 3. If the drive uses mult-motor operation for different gear ranges, spindle orient, etc., verify that the proper motor parameters are being adjusted. Check by setting the IOC trip setpoint of the motor parameters being adjusted to 10. Starting the drive will cause IOC fault if the proper motor parameters have been adjusted. 4. Check "Programmer Ineffective - No Errors Displayed" section. 5. Replace Control card software (U1, U5 and U6).
9. Diagnostics Do Not Perform	1. Verify that the particular function is available - refer to the Custom Software Description, software jumper descriptions. 2. Check JP13 on the Control card, must be in the 2-3 position for some Diagnostic tests to work. Refer to "Use of Optional Programmer Module". 3. Check the drive mode, Auto or Manual. The drive must be in Manual mode to run Diagnostics tests. 4. Check POL on 6PL-15 (3TB55) for reverse signal. 5. Refer to "Programmer Ineffective - No Errors Displayed", this section.
10. Fuse Blowing AC or DC Power Fuses	1. Compare the fuse size with the Renewal Parts List. 2. Check for failed SCRs. 3. Check the setting of ADR034-ADVLM. 4. Check for out of tolerance or sagging AC line voltage on regenerative drives - refer to "Line Voltage Problems" and "Noise or Random Nuisance Faults" this section. 5. Check the field power module for a short. 6. Using a megger, check for a ground in motor field, armature circuit or blower circuit. 7. Is the drive forced into Current Limit by an overhauling load? 8. Adjust the field control parameters to provide a lower armature voltage at top speed - lower ADR079-CROSS on field weakening drives or ADR076-FLDMX on base speed drives. 9. Refer to "Random SCR Misfirings" this section.
11. Field Fuses Blowing	1. Check for a ground in the motor field or conduit wiring. 2. Check the field power module for a short or failure. 3. Verify that all wires on the Power Supply card are connected to the correct stab-on. 4. Verify that FSPL is plugged into the proper position with the correct polarity. 5. Check for shorted MOVs on the Power Supply card. 6. Check for shorted Control Power Transformer (CPT) primary.

TABLE 4 (continued)
BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
12. Power Supply Fuses Blowing	1. Verify that the fuses are correct size. 2. Check for any shorted or over-loaded power supplies (Refer to "Checking Power Supplies", this section). 3. When using Hand-held Programmer, verify that it is properly connected to 18PL with red wire on the left-hand side of cable. 4. Replace the Power Supply card.
13. MOV Failures	1. Refer to "Line Voltage Problems", this section. 2. Motor fault causes AC fuses to open during regeneration. Check for grounded armature, field or wiring.
14. Repetitive Card Failures	1. Check all connections at 2TB, 3TB & 4TB. 2. Verify that all ribbon cables and plugs are connected properly (i.e., not offset by one row or pin). 3. Check the stab-on connection points on the Power Supply and Power Connection cards. 4. Verify that all voltages (incoming AC and internal DC) are correct. 5. Check for moisture. 6. Check for damage due to airborne chemical contamination.
15. Random SCR Misfirings	1. Check for inoperative SCRs. 2. Check for loose connections on the fuses, breaker, MA contactor, SCRs or any armature connections. 3. Check for loose connections on SCR gate leads, Power Connection card or 5PL. 4. Check for notching on the AC line - this could be caused by other drives on the same incoming AC line, of a horsepower equal to or greater than four to six times the horsepower of the problem drive. 5. Refer to "Line Voltage Problems", this section.
16. Line Voltage Problems	1. Verify that the line voltage is within drive specifications, -5% to +10%. 2. Verify that the AC line voltage is balanced line-to-line and line-to-ground. 3. Check for AC line notches or spikes. 4. Verify that the line frequency is within ±1 Hz of rated. 5. When AC line is supplied by a diesel driven alternator, check for changes in line frequency during drive acceleration or regeneration. 6. Check for more than one and one-half cycles of ringing following commutation notches in the AC line caused by the drive. Amplitude should be less than or equal to 200 volts. 7. Check for an ungrounded secondary on the power system transformer directly in front of the drive (if used). (The transformers are generally a delta/wye type with the neutral of the "wye" grounded through a high resistance or directly to ground.)

TABLE 4 (continued)
BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
17. 24 Volt Logic Not Operating Properly	1. Check the ±24 volt power supplies. 2. Check logic polarity control Jumpers per the Custom Software Description (ADR012-OPTJP), JP19 and 20 on the Control card, JP5 and JP6 on the Application card. 3. Check control logic connections at 3TB and 4TB. 4. 24 volt logic being switched too quickly to 4TB or Application card - there must be at least a 70 millisecond delay between two logic inputs changing state. 5. Check for proper polarity and common circuit if 24 volt supply is from a source other than drive. 6. Check that the drive is in the proper motor mode (0, 1, 2 or 3) for the particular motor or function being operated.
18. Drive/Motor Does Not Run	1. No connections to REF 24, ±15V or common should be made at 4TB. Move any such connections to 3TB. 2. FU1, 2, 3 or 4 open (or optional circuit breaker open). 3. CFU1, 2, 3, 4 or 5 open and/or MOVs shorted on Power Supply card. 4. Check all hardware and software jumpers per the Custom Software Description, particularly ADR012-OPTJP, JP19 and JP20 on the Control card and JP5 and JP6 on the Application card. 5. Check for the speed reference at 3TB or 4TB (PCN/PCP or PREF). 6. Check control logic voltage and state at 3TB and 4TB (RUN, JOG, SLOW, MODE 0 thru MODE 7, etc.) as well as power supply voltage levels. 7. Check that the drive is in the proper operate mode (AUTO or MANUAL). 8. Drive may be stalled in current limit.
19. Field Current or Voltage Wrong Value	1. Is the drive in field economy? 2. Check for proper motor field connections. 3. Check JP1 and JP2 on the Power Supply card for proper maximum field settings. 4. Check the P1-FIELD SCALE pot seal - factory adjusted for field current. 5. Check ADR076-FLDMX and ADR077-FLDMN per the Custom Software Description. 6. Check 4TB20 for connection to FRA if external field forcing is used. 7. Check for any external resistors or diodes in the field circuit. 8. Check the field power module for shorted elements. 9. Check the field current (Test point FC). Replace the Power Supply card if failed. 10. Replace the Control card.
20. Drive Runs in the Wrong Direction	1. Check for logic signal on 3TB15 (POL). 2. Armature connections reversed (correct armature and tach leads). 3. Field connections reversed (correct field and tach leads). 4. Check all software jumpers (ADR000 to ADR014) per the Custom Software Description. 5. Check the reference polarity. 6. Check logic polarity control jumpers per the Custom Software Description (ADR012-OPTJP), JP19 and JP20 on the Control card, JP5 and JP6 on the Application card. 7. Check the Custom Software Description to verify that the reverse function is enabled when operating from the serial link.

TABLE 4 (continued)
BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
21. Drive Does Not Reverse	1. Verify that the drive is regenerative. 2. Check the settings of the soft jumpers (ADR000 to ADR014) per the Custom Software Description. 3. Check the reference polarity. 4. Check control logic to POL (3TB55) if used for reversing. 5. Check logic polarity control jumpers per the Custom Software Description (ADR012-OPTJP), JP19 and JP20 on the Control card, JP5 and JP6 on the Application card. 6. Check the Custom Software Description to verify that the function is enabled when operating from the serial link.
22. Drive Speed is Incorrect	1. Verify that the motor design is suitable for the application. 2. Is the proper reference (PCP to PCN) enabled? Refer to Custom Software Description. 3. Check JP1 on the Control card for proper scaling of MSR input. 4. Adjust P1-REF SCALE on the Control card for MSR input. 5. ADR068-REFSF set incorrectly? 6. Check JP1 on the Application card for proper ASR scaling. 7. Adjust P8-SFB SCALE on Process Interface card. 8. Confirm the setting of P4-PFGAIN, on the Process Interface card. 9. ADR070-RPMSF may be misadjusted, check per the Custom Software Description. 10. Check for noisy tach signal. Refer to "Wiring Procedures", this section. 11. Check the setting of ADR041-ILMMX if the drive is suspected of being in current limit. 12. Check the setting of ADR034-ADVLM if the drive is suspected of being in voltage limit. 13. Check ADR002-REGJP and ADR079-CROSS per the Custom Software Description. 14. Check ADR076-FLDMX and ADR077-FLDMN per the Custom Software Description and by the calculations in the key parameter adjustment list. 15. Check resistor R_{fb} on the Analog Process Interface card (CTBA1 to CTBA2).
23. Serial Link Does Not Function Properly	1. Verify that JP11, JP12 and JP13 on the Process Interface card are set correctly. 2. Check the baud rate of the drive (ADR126-BAUDR) compared to serial link device (hard reset must be performed after BAUDR is changed). 3. Check serial link connections at 3TB22, 23 and 24 (DCOM, R232 and 3TB22). 4. If RS422 is used, verify that the handshaking lines are properly connected. Verify that sending device is responding to the drive's DTR signal (low level) when the device is allowed to transmit to the drive. 5. Replace the Process Interface card.
24. Erratic Operation	1. Check power supplies for magnitude and ripple (refer to "Checking Power Supplies"). 2. Verify that 1CPL on the MFC/Power Supply card is inserted correctly (not backwards or offset). 3. Check connections on all ribbon cables. 4. Tighten all connections on 2TB, 3TB, 4TB and all power connections. 5. Confirm correct hardware jumper setting per the Custom Software Description. 6. Check all parameters per the software test data sheet in the drive door or Custom Software Description. 7. Check for noisy ground connection - refer to "Wiring Procedures". 8. Examine CFB and VFB with an oscilloscope for stability - refer to "Stability".

TABLE 4 (continued)
BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
25. Stability Problems	1. Examine CFB with an oscilloscope to determine if any SCRs have failed. 2. Check settings of all parameters per the software test data sheet inside the drive door pocket. 3. Check the setting of the Process Interface card if tach is used. 4. Check Jumpers JP17 and JP18, resistor R_{rg} and capacitor C_{rg} on the Process Interface card per the Custom Instruction Book. 5. Verify that all contacts on the MA contactor are closing. 6. Check and adjust for any offset on CFB, P2-CFB ZERO, P3-CFB ADJUST (on the Power Supply card), JP3 and JP4 on the Power Supply card or the P4-OFFSET (on the Control card). 7. Refer to the "Troubleshooting Stability Problems" section in this manual.
26. Signal Level Detector Chatter	1. Is SLD looking at a "noisy" signal (e.g., CFB, tach feedback, etc.)? 2. Is enough hysteresis programmed into the appropriate parameter address?
27. Too Much Coast Time on Deceleration	Check settings in Custom Instruction Book and confirm regeneration is enabled.
28. Motor Sparking	1. Check motor commutator for contamination. 2. Check the setting in ADR024-IRTLM, may be too high. 3. Check the field current - refer to "Field Current or Voltage Wrong Value". 4. Check the field current stability - refer to "Stability Troubleshooting" section. 5. Check the commutator condition. 6. Confirm that all brushes are seated properly with proper brush tension. 7. Check the brush grade. 8. Check the brush neutral adjustment. 9. Check for sagging AC line voltage - refer to "Line Voltage Problems", this section. 10. If tapered current limit is required for the application, verify that ADR081-ADR084 reflect this.
29. Motor Runs Hot	1. Motor may be overloaded, check for mechanical problems. 2. Check the motor winding temperature. If the motor has Class F insulation and the winding temperature is less that 130°C, this is normal. 3. Check the rotation of the motor blowers. 4. Check the field current with respect to the motor nameplate rating. Adjust ADR076-FLDMX if necessary. 5. Check the field current when motor is not running. If the field current is full nameplate rating when the motor or blower is not running, check ADR050-FLDEC. 6. If motor runs hot at very low speeds, check motor application guide for full-torque speed range. 7. Confirm that all SCRs are firing properly by monitoring current feedback CFB.

TABLE 4 (continued)
BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
30. Motor Audible Noise	1. Loose pole bolts - tighten per manufacturer's torque specifications. 2. SCRs misfiring - check current feedback. 3. Cross-firing SCRs - increase deadband. 4. Unstable regulator - check "Troubleshooting Stability Problems" section.
31. Motor Vibration	1. Coast Stop Drive - if the vibration is electrically induced the vibration will disappear. If it does not disappear, check for mechanical causes. 2. Refer to "Stability" procedures. 3. Motor not mounted level. 4. Motor base vibrating.
32. EEROM Memory Rewrites	JP13 on the Control card must be in the 1-2 position during "Power UP" or hard reset.
33. Noise or Random Nuisance Faults	1. Verify that the 5-volt DC supply is not used externally. 2. Check if ACOM, DCOM or PCOM are used externally - PCOMX is the only point that should be used for common connections. 3. Verify that PCOM passes through a ferrite core (not required if PCOMX is on 3TB73-75). 4. Check for suppression on all relays, contactors and starters. 5. Check all power supplies - refer to "Checking Power Supplies", this section. 6. Refer to "Wiring Procedures", this section.
34. Wiring Procedures	1. Check the wire sizes on the armature, field and AC power connections. 2. Verify that the control wires (24-volt DC, 115-volt AC) do not run in the same conduit or run parallel closer than four inches to the power wires, causing noise pick-up. 3. Verify that all of the signal wires (i.e., tach, encoder, reference, load cell or RS-232C/422) are twisted pairs, shielded and shields grounded at the drive. 4. Verify that the signal wires are not grounded at more than one end or other than the drive end only. 5. Verify that the signal wires do not run in the same conduit or parallel closer than four inches to the power wires. 6. Verify that there is only one system ground. 7. Verify that the ground bus is large enough to prevent voltage drops or noise pick up.
35. Checking Power Supplies	1. Check the +15V test point on the Control card for +15VDC, ±5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 2. Check the -15V test point on the Control card for -15VDC, ±5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 3. Check the +5V test point on the Control card for +5VDC, ±1%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope.

TABLE 5
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000001	FL01	IOC	Instantaneous DC overcurrent (L,T) • Tighten armature connections. • Armature grounded. • Examine motor commutator for damage or evidence of flashover. • Verify proper brush seating and tension (refer to motor manual). • Mechanical Problems - binding shaft, unreleased brake, vibration, gearbox, or load problems. • IOC parameter set too low. Set at least 128 units above current limit settings. • Current feedback isolator failure - check 1PL & DRPL. • SCR misfiring - Check VFB or CFB waveforms. • Check for SCR failure. • Current regulator mistuned - adjust per the Custom Instruction Book.
000010	FL02	OVRSP	Overspeed (L,T) • ADR073-SPDLM set too close to ADR068-REFSF and ADR071-REFLM - set a minimum of 5% greater than REFSF and REFLM. • Speed overshoot occurs - speed regulator response too high, reduce gain. (Refer to Table 6, Troubleshooting "Stability" Problems.) • Loss of field current. • P8 on Process Interface card misadjusted. • Check R_{fb} on the Analog Process Interface card (CTBA1 to CTBA2).
000011	FL03	SPARE	

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000100	FL04	FLOSS	Loss of field current (L,T) • CFU1, 2 or 3 on Power Supply card open. • 1PL disconnected or damaged. • Field grounded. • Field lead connections made improperly. • Field regulator saturation; add "16" to ADR012-OPTJP. • Open field circuit components - resistors, contactors or DC fuses. • ADR078-FLDLS set too high - set at least 25% below ADR077-FLDMN and 50% below ADR076-FLDMX. • FC gain jumpers set wrong - JP1 and JP2 on Power supply card. • External field supply - set ADR078-FLDLS at "60000" to inhibit field loss. • Field regulator gain too low - adjust BFLPG, BFLIG (ADR031, 032).
000101	FL05	TPOL	Tach polarity incorrect (L,T) • Reversed tach connections. • Armature or field reversing contactors.
000110	FL06	JMPER	Software jumper illegally changed while running (L,T) • Drive must be stopped to change parameters in ADR000 to ADR014.
000111	FL07	RESET	Reset while running (L,T) • Drive powered up with RUN or JOG closed - check 6PL-12 and 6PL-13 (3TB58 and 57). • Momentary loss of incoming power while running. • Incorrect logic polarity, check ADR012-OPTJP.
001000	FL08	TOC	Timed overcurrent (L,T) • Motor mechanically overloaded - check for change in process load, worn bearings, gears or brakes. • CFU2 on Power Supply card open. • Field current is less than 85% of nameplate rated current-readjust field current parameters. • ADR044-TOCTH and ADR045-TOCGN set too low - check against the parameter list shipped with drive. • Current feedback isolator failure or 1PL disconnected or damaged. • P3-CFB ADJ on the Power Supply card misadjusted - readjust per procedure in Table 1. • Armature winding failed. • Dynamic braking failure (if used) - i.e., failure to close DB circuit.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
001001	FL09	LNKR	Serial Link trip (drive trip via serial link) (L,T) • Drive received external trip signal - check for external fault. • 12PL or 18PL incorrectly connected or damaged.
001010	FL10	LNKFL	Serial Link fault (communications failure) (L,T) • Programmer card jumpers incorrect - refer to Table 1. • Baud rate set incorrectly - check ADR126-BAUDR. • 12PL or 18PL incorrectly connected or damaged. • Intermittent connection between 3TB32 (CTSA) and 12PL-7 or 3TB41 (CTSB) and 12PL-14. • FL36 appears momentarily previous to FL10 latching - refer to FL36.
001011	FL11	DFPON	Delayed firing power on when it should be off (L,T) • Q2 failed on - check test point DFP. • Replace Control card.
001100	FL12	DFPOF	Delayed firing power off when it should be on (L,T) • Q2, Q3, R33 or U14 failed on Control card - replace Control card.
001101	FL13	EEXXX	Checksum error or parameter out of range (L, T, H) where XXX equals a 3-digit number, indicating an out-of-range parameter address or altered checksum. • Spontaneous parameter alteration - check all parameters and verify EEROM is not write "ENABLED" during power up or power down, or "hard" reset. • Value other than zero in ADR255-UNDFD - replace EEROM.
001110	FL14	ILOOP	Current loop regulator saturated (L,T) • Main contactor not picking up - check pilot relay operation. • DC fuse open (regenerative drives only). • 5PL is disconnected or damaged. • Loss or incorrect scaling of voltage feedback - check jumpers on the Power Connection card and Control card TP VFB or VFBB. • Loss of current feedback - check 1PL-20 and DRPL. • Armature not connected. • Regeneration enabled by soft jumper on single quadrant drive. • Brushes missing or unseated. • Controllers using the inner current loop with gain set too high - check against parameter list shipped with drive. • Current feedback miscalibrated - see Table 1, Power Supply card. • Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card per Table 1. • Power Supply card failed - replace. • Control card failed - replace.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
001111	FL15	VFBLS	Loss of armature voltage feedback (L,T) • Voltage feedback wires disconnected or damaged - check P1 and P2 between SCR conversion module and Power Connection card. • DC fuse (FU4) open - regenerative drives only. • 5PL disconnected or damaged. • CFB jumpers incorrect - check JP3 and JP4 on the Power Supply card per Table 1. • VFB circuit failure - check test point VFB. • Check SCRs.
010000	FL16	NOTRP	Trip from N.O. on Application card (L,T) • External input at 4TB8 (13PL-8). • Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card is present. If preceded by FL07 and soft reset, check logic polarity on ADR012-OPTJP. • If Microapplication card is used, check JP33 (should be in 1-2 position) and verify that U33 is installed correctly. • Replace Application card.
010001	FL17	XSTOP	Coast stop trip (L,T) • Check for input at XSTOP on 3TB1 (6PL-16). • Check ADR004-STPJP - a 4 causes coast stop when XSTOP is selected.
010010	FL18	STALL	Stalled overcurrent (L,T) • Mechanical problems - failed brake, frozen bearings, application problem, etc. • ADR046-STLTH and ADR047-STLGN set wrong - check against "as shipped" parameter list. • Drive running less than 1% of top speed and armature current greater than ADR046-STLTH. • CFU2 on Power Supply card open. • Field current is less than 85% of nameplate rated current while the drive is supplying armature current. • Current feedback isolator failure or 1PL disconnected or damaged. • P3-CFB ADJ on the Power Supply card misadjusted - readjust per procedure in Table 1. • Armature winding failed. • Dynamic braking failure (if used) - i.e., failure to open DB circuit.
010011	FL19	OVERV	Excessive armature voltage (L,T) • ADR034-ADVLM set too high - compare to parameter list. • Drive unstable - see Stability procedure in Troubleshooting section of Instruction Book. • VFB circuit failure - check test point VFB.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010100	FL20	TLOSS	Loss to tach feedback (L,T) • Tach leads disconnected. • Tach feedback circuit failure - 7PL disconnected or damaged. • Process Interface card scaling resistor (R_{fb}) incorrect. Refer to Custom Instruction Book. • P8-SFB SCALE on Process Interface card misadjusted. • Process Interface card failed - replace.
010101	FL21	NCTRP	Trip from N.C. interlock on Application card (L,T) • Loss of external input on 4TB7 (13PL-7). • Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card present. • If preceded by FL07 and soft reset, check logic polarity ADR012-OPTJP.
010110	FL22	PHLOS	AC phase loss (L,T) • JP14 on Control card in wrong position - see Table 1. • Fuse open in AC line - FU1-3. • 1PL disconnected or damaged. • Low or unbalanced AC line voltage - line-to-line or line-to-neutral more than 5% low. • Loose AC power connections. • CFU1 open. • Verify that ADR187-FUSLS contains a zero - change if necessary. • ±24-volt or ±15-volt power supplies failed - check CFU6 and 7 on Power Supply card. • ±15-volt power supplies outside limits (±5%) - replace Power Supply card.
010111	FL23	OVERF	Field overcurrent (L,T) • Field regulator unstable - refer to Stability procedure in Trouble-shooting section. • Field forcing at FRA (4TB20, 13PL-20) - check input scaling, see Table 1. • ADR076-FLDMX or ADR049-FTOCG incorrect - check parameter list. • Field SCR shorted - replace Motor Field Module. • Voltage regulated field - check for miswired or grounded field.
011000	FL24	U3 FL	Faulty RAM on U3 8155 (L,T) • Replace Control card.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
011001	FL25	COMTR	SCR commutation failure (L,T) • Low AC line voltage (less than 90% rated). • 1CPL disconnected. • 1CPL reversed - phase 1 CT should be wired to bottom of 1CPL at Power Supply card. • P3-HP SEL and JP8-11 gains set too high on Control card. • 1PL disconnected or damaged. • Gate plug(s) incorrectly connected. • SCR(s) failed.
011010	FL26	ACIOC	Circulating AC line fault (regenerative controllers only) (L,T) • Low or noisy AC line. • Incorrect setting of P3-HP SEL and JP8-11 on Control card. • Noise on SCR gate leads. • 1PL disconnected or damaged. • SCR(s) failed.
011011	FL27	LYSNC	Loss of AC line signal (L,T) • Fuse open in AC line - FU1-3. • 1PL disconnected or damaged. • Low or unbalanced AC line voltage - line-to-line or line-to-neutral more than 5% low. • Loose AC power connections. • CFU1 open. • 15-volt power supplies outside limits (±5%) - replace Power Supply card. • AC line frequency below 45Hz or above 63Hz (70Hz if called for by ADR001-FDBJP). • Check 50/60Hz square wave at test point SY on Control card - replace Control if no signal.
011100	FL28	RMODE	Illegal selection of RUN mode (N,A) • Do not command DIAGNOSTIC RUN or STATIC. • Attempted to run Test 15, S-TUNE when feature not supplied.
011101	FL29	CNLON	Control on input opened (L,T) • Open circuit between 3TB3 and 3TB4. • 6PL incorrectly connected or damaged. • Noise between 3TB3 and 3TB4.
011110	FL30	NOAPP	Application card failure (L,T) • Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card present. • 3PL incorrectly connected or damaged. • If Microapplication card is used (531X139APMA-M1 or M3), the Control card must be a 531X300CCHA-M2 or M3. • If Microapplication card is used, check JP33 (should be in 1-2 position) and verify that U33 is installed correctly. • Replace Application card.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
011111	FL31	NOSEQ	Phase sequence indeterminate (L,T) • CFU1 on Power Supply card open. • Excessively noisy or notched AC line. • Line unbalance, phase to ground by more than 15%. • 15-volt power supplies outside limits (±5%) - replace Power Supply card. • 1PL incorrectly installed or damaged.
100000	FL32	TOCAN	Timed overcurrent annunciation (N,A) • Motor mechanically overloaded - check for change in process load, worn bearings, gears or brakes. • CFU2 on Power Supply card open. • Field current is less than 85% of nameplate rated current. • ADR044-TOCTH and ADR045-TOCGN set too low - check against the parameter list shipped with drive. • Current feedback isolator failure or 1PL disconnected or damaged. • P3-CFB ADJ on the Power Supply card misadjusted - readjust per procedure in Table 1. • Armature winding failed. • Dynamic braking failure (if used) - i.e., failure to close DB circuit.
100001	FL33	PHSEQ	Phase sequence reversed (N,A) • Reverse any two AC input lines, recheck rotation of any three phase motors. • Excessively noisy or notched AC line. • Line unbalance, phase to ground by more than 15%. • 15-volt power supplies outside limits (±5%), noisy or excessive ripple, replace Power Supply card. • 1PL incorrectly installed or damaged. • ADR012-OPTJP not properly set for phase sequence insensitivity option.
100010	FL34	PLLSY	AC line PLL out of sync (N,A) • Excessively noisy or notched AC line. • CFU1 on Power Supply card open. • Line unbalance, phase to ground by more than 15%. • 15-volt power supplies outside limits (±5%) - replace Power Supply card. • On variable frequency AC lines - ADR029-PLLGN too low.
100011	FL35	O SPD	Motor rotating when RUN commanded (N,A) • RUN command initiated during motor rotation. • Test point SFB contains an offset at zero speed - adjust Process Interface card. • ADR004-STPJP not set properly for "flying restarts".

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
100100	FL36	LNKER	Undecipherable messages detected on serial link by drive or other device (N,A) • Excessive noise on 12PL-7 thru 14. • 12PL incorrectly connected or damaged. • 5V DC power supply noisy. • Baud rate incorrectly set - ADR126-BAUDR normally set at 4800. • Check Process Interface card jumpers JP1, JP11, JP12, JP13 and JP14. • Serial link message protocol incorrect, message too long, checksum byte incorrect or handshaking incorrect - refer to "Serial Link User's Guide". • Replace Process Interface card. • Replace control card.
100101	FL37	IHUNG	Armature current not zero when fully retarded (N,A) • JP3 and JP4 on Power Supply card in 2-3 position when running - should be in 1-2 position. • Offset voltage at test point CFB when zero armature current flowing - adjust P2-CFB ZERO on the Power Supply card. • DRPL disconnected or damaged. • 1PL disconnected or damaged. • Current isolator failed - replace Power Supply card.
100110	FL38	ELOCK	Attempt to change software adjustments when soft locked (N,A) • ADR003-DGNJP must be odd to program EEROM - add one to value in DGNJP if it's an even number.
100111	FL39	EFAIL	Software adjustment not stored in EEROM (N,A) • JP13 on Control card in hard lock position. • EEROM failed - replace.
101000	FL40	MXSPD	Speed reference greater than REFLM speed adjustment (N,A) • Manual Reference jumper JP1 on Control card improperly set - see Table 1. • P1-REF SCALE on the Control card misadjusted - see Table 1. • ADR071-REFLM set below ADR070-RPMSF. • P1-ASR SCALE, on the Application card misadjusted (if used) - see Table 1. • P4-PF GAIN or P5-PF OFF, misadjusted on Process Interface card - see Table 1. • Top speed reference for parallel input (if used) is set higher than ADR068-REFSF and ADR070-RPMSF. • Preset references (RUNRF, R1RF, R2RF, SLWRF) set too high (if used).
101001	FL41	KEYOF	Keypad function disabled (N,A) • Programmer key depressed for a function that is not enabled.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
101010	FL42	RLOGC	Run logic error (N,A) • JOG, RUN, or DIAGNOSTIC RUN called for with XSTOP open. • Keypad STOP called for during an external run cycle. • Incorrect logic polarity - check ADR012-OPTJP, JP19 and JP20 on the Control card and JP5 and JP6 on the Application card.
101011	FL43	U11FL	A-D converter failure (L,T) • Check for noise on ACOM, move wiring to PCOMX. • Control card failed - replace.
101100	FL44	U4LAT	U4 to U5 communication error (L,T) • Check PROM revision compatibility. • Check for noise on power supplies or analog inputs. • Replace Control card.
101101	FL45	SLOOP	Speed tracking fault (L,T) • Drive entered current limit or ADVLM limit when tracking enabled. • ADR036-SERLM set too low - should be between 1 and 7. • Speed regulator unstable - see "Stability Troubleshooting" section. • SRS and SFB should read same voltage at any speed.
101110	FL46	IOFST	Current feedback offset fault (L,T) • Current feedback signal at testpoint CFB offset by 35%. • P2-CFB ZERO on MFC/Power Supply card misadjusted.
101111	FL47	OADJ>	A-D converter offset too high (N,A) • P4-OFFSET on the Control card misadjusted - turn CW. • Check for noise on ACOM.
110000	FL48	OADJ<	A-D converter offset too high (N,A) • P4-OFFSET on the Control card misadjusted - turn CCW. • Check for noise on ACOM.
110001	FL49	NO U4	U4 to U6 communications error (L,T) • Check for PROM revision compatibility. • Check for noise on power supplies - or analog inputs. • Replace Control card.
110010	FL50	U4BAD	U4 RAM, timer or ALU is faulty (L,T) • Verify settings of ADR011, 014, 056, 089, 187, 209 and 210 per Custom Software Description. • Incorrect software installed for application check revision. • U4 failed - replace Control card.
110011	FL51	U1ROM	U1 ROM is faulty (L,T) • Replace U1 on the Control card. • Replace Control card.
110100	FL52	NCLSD	Annunciate N.C. interlock on Application card (N,A) • Check external input on 4TB7 (13PL-7). • Application card absent but selected by soft jumper - ADR010-APPJP should be zero if no Application card is present.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
110101	FL53	SCLER	SCALE ERROR (L,T) • Occurs if position Feedback is not enabled from the Process Interface card and the product of ADR070-RPMSF and ADR074-TCPPR is greater than 2,490,367 - check setting in ADR070 and 074.
110110	FL54	NO U5	U5 to U4 communication error (L,T) • Check for PROM revision compatibility. • Replace Control card or U5.
110111	FL55	U5BAD	U5 RAM, timer or ALU is faulty (L,T) • U5 failed - replace. • Replace Control card.
111000	FL56	U5ROM	U5 ROM is failed (L,T) • U5 failed - replace.
111001	FL57	NOPEN	Annunciate N.O. interlocks on Application card (N,A) • External input on 4TB8 (13PL-8). • Application card absent but selected by soft jumper - ADR010-APPJP should be zero if no Application card is present.
111010	FL58	U5COM	U4 not communicating correctly with U5 (L,T) • Check for PROM revision compatibility. • Noise on tach input. • Replace Control card.
111011	FL59	NO U6	U6 does not communicate properly with U4 (L,T) • Check for PROM revision compatibility. • Replace Control card.
111100	FL60	U6BAD	U6 RAM, timer or ALU is faulty (L,T) • U6 failed - replace. • Control card failed - replace.
111101	FL61	U6ROM	U6 ROM is faulty (L,T) • U6 failed - replace. • Control card failed - replace.
111110	FL62	SPARE	
111111	FL63	U6COM	U4 not communicating correctly with U6 (L,T) • Check for PROM revision compatibility. • Check for noise on tach input signal. • U1 extended memory - move JP24 to the 2-3 position. • U5 and U6 (TUP) in wrong sockets. • Replace Control card.

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000000	FL64	SPARE	
000001	FL65	A1FOP	SCR A1F is not operating (L,T) • Gate circuit 1F disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000010	FL66	A2FOP	SCR A2F is not operating (L,T) • Gate circuit 2F disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000011	FL67	A3FOP	SCR A3F is not operating (L,T) • Gate circuit 3F disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000100	FL68	A4FOP	SCR A4F is not operating (L,T) • Gate circuit 4F disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000101	FL69	A5FOP	SCR A5F is not operating (L,T) • Gate circuit 5F disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000110	FL70	A6FOP	SCR A6F is not operating (L,T) • Gate circuit 6F disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.

*(see door label for module location)

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000111	FL71	FWDOP	All forward SCRs are not operating (L,T) • 5PL disconnected or damaged. • Replace Power Connection card. • Replace Control card.
001000	FL72	ARMOP	Armature circuit open (L,T) • DC fuse open (regenerative drives only). • Armature circuit open. • Brushes missing or disconnected. • 5PL disconnected or damaged. • Main contactor not picking up - check pilot relay operation. • Loss of current feedback - check 1PL-20 and DRPL. • Armature not connected. • Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card. • Power Supply card failed - replace. • Control card failed - replace.
001001	FL73	A1ROP	SCR A1R is not operating (L,T) • Gate circuit 1R disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001010	FL74	A2ROP	SCR A2R is not operating (L,T) • Gate circuit 2R disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001011	FL75	A3ROP	SCR A3R is not operating (L,T) • Gate circuit 3R disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001100	FL76	A4ROP	SCR A4R is not operating (L,T) • Gate circuit 4R disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.

*(see door label for module location)

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
001101	FL77	A5ROP	SCR A5R is not operating (L,T) • Gate circuit 5R disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001110	FL78	A6ROP	SCR A6R is not operating (L,T) • Gate circuit 6R disconnected or open. • 5PL disconnected or damaged. • Failed SCR - check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001111	FL79	REVOP	All reverse SCRs are not operating (L,T) • 5PL disconnected or damaged. • ADR002-REGJP not set correctly for non-regenerative drives. • Replace Power Connection card. • Replace Control card.
010000	FL80	SCTFL	SCR short circuit test failed (L,T) • Check for shorted SCRs - replace.*
010001	FL81	A1FSH	SCR A1F or A4R is shorted (L,T) • Check for shorted SCRs - replace.*
010010	FL82	A2FSH	SCR A2F or A5R is shorted (L,T) • Check for shorted SCRs - replace.*
010011	FL83	A3FSH	SCR A3F or A6R is shorted (L,T) • Check for shorted SCRs - replace.*
010100	FL84	A4FSH	SCR A4F or A1R is shorted (L,T) • Check for shorted SCRs - replace.*

*(see door label for module location)

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010101	FL85	A5FSH	SCR A5F or A2R is shorted (L,T) • Check for shorted SCRs - replace.*
010110	FL86	A6FSH	SCR A6F or A3R is shorted (L,T) • Check for shorted SCRs - replace.*
010111	FL87	NOUAP	Microapplication card failure (L,T) • Microapplication card absent but selected by soft jumper - ADR014-SPCJP should be set to disable Microapplication card interface (refer to Custom Software Description). • 3PL incorrectly connected or damaged. • U30 or U33 on Microapplication card failed - replace. • Replace Microapplication card.
011000	FL88	UAPWD	MUP watchdog misoperated (L,T) • Application card (G2) installed when Microapplication card (G1) is called for - check setting in ADR014-SPCJP per Custom Software Description. • U30 on Microapplication card failed or jumpers set incorrectly - replace card if necessary. • Refer to BugBuster, "Noise or Random Nuisance Faults" item.
011001	FL89	MUPTO	RUP to MUP communication failure (L,T) • U30 on Microapplication card failed - replace. • Replace Microapplication card.
011010	FL90	NOMSG	RUP to MUP watchdog detected no message received within time period expected (L,T) • Serial Link or Local Area Network transmission lines disconnected. • 12PL damaged or disconnected. • External device failed to send message in time period set by ADR005-SERJP.
011011	FL91	SPARE	
011100	FL92	SPARE	
011101	FL93	SPARE	
011110	FL94	SPARE	
011111	FL95	WATCH	FUP watchdog detected RUP has misoperated (L,T) • Same as FL58.
100000	FL96	PASSED	Annunciates when SCR test has passed.

*(see door label for module location)

TABLE 5 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
100001	FL97	OCTFL	SCR open circuit test failed (L,T) • DC fuse open (regenerative drive only). • Armature circuit open. • Brushes missing or disconnected. • 5PL disconnected or damaged. • Main contactor not picking up - check pilot relay operation. • Loss of current feedback - check 1PL-20 and DRPL. • Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card. • Power Connection card failed - replace. • Power Supply card failed - replace. • Control card failed - replace.
100010	FL98	OCTMF	Multiple open SCRs (L,T) • Check all gate plugs. • 5PL disconnected or damaged. • Check SCRs for failure - replace. • Replace Power Connection card. • Replace Control card.
100011	FL99	ALLOP	All SCRs tested open (L,T) • DC fuse open (regenerative drive only). • Armature circuit open. • Brushes missing or disconnected. • 5PL disconnected or damaged. • Main contactor not picking up - check pilot relay operation. • Loss of current feedback - check 1PL-20 and DRPL. • Armature not connected. • Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card. • Power Supply card failed - replace. • Control card failed - replace.

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

TABLE 5 (continued)
SPECIFIC ERROR TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
—	ER01	BUART	Faulty UART on the 8031 U.C. • This is a normal response to Test 50. • Check JP3 and JP4 on Programmer card. • Programmer card failed - replace.
—	ER02	BKPAD	The keypad has shorted leads. • Replace ribbon cable. • A key pressed during power up or reset. • Programmer card failed - replace.
—	ER03	BDMEM	The random access memory is faulty. • Programmer card failed - replace.
—	ER04	BPRGM	The program memory checksum is incorrect. • Programmer card failed - replace.
—	ER05	BTIMR	The timers in 8031 U.C. are faulty. • Programmer card failed - replace.
—	ER06	NADDR	An address is not specified. • "ENTER" key pressed without specifying a parameter address.
—	ER07	2 KEYS	Two keys were pressed together. • Press keys - one at a time. • Keypad failed - replace. • Programmer card failed - replace.
—	ER08	ILBRD	Illegal broadcast. • Programmer card failed - replace.
—	ER11	NOACK	No acknowledgement received. • Device connected to the serial link not responding to the acknowledged requests from the Programmer card. • Wires to the serial link disconnected. • Driver or receiver chip failed on Programmer, Process Interface card, or external device. • Processors locked up, use hard reset. • If Hand-held Programmer, may be failed U6 port - replace. • Noise on transmission wires. • RS232C on hand-held cable.
—	ER12	N-CTS	No "clear-to-send". • Serial link transmission lines disconnected. • Check JP3 and JP4 on Programmer card. • Noise on 5-volt power supply. • Ground loops or unsuppressed relays. • Programmer card failed - replace.

TABLE 5 (continued)
SPECIFIC ERROR TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
—	ER13	ORBUF	Receive buffer has overflowed. • Message sent to the drive which exceeds the maximum number of bytes (11 data bytes). • Baud rate set incorrectly. • Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. • Message protocol used incorrectly - refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • External serial link device failed.
—	ER14	CKSUM	Incorrect checksum received. • The serial link message checksum has been calculated improperly by external Serial Link device - refer to the Serial Link User's Guide. • Baud rates set incorrectly. • Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. • Message protocol used incorrectly - refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • External serial link device failed.
—	ER15	BDMSG	Something other than a "start of transmission" was received when "S.O.T." was expected. • "Start of transmission" byte of the message sent over the serial link has been determined incorrectly - refer to the Serial Link User's Guide. • Baud rates set incorrectly. • Byte protocol set incorrectly on external device (1 start bit, 8 data bits, and 1 stop bit). • Message protocol used incorrectly - refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • External serial link device failed.
—	ER21	MTRNG	The motor is rotating at start of "self-tune" tune-up program. • Test point SFB contains an offset to zero speed - adjust P7-SFB ZERO on the Process Interface card. • A/D offset misadjusted - adjust P4-OFFSET on the Control card. • Noise on tach leads. • 7PL damaged or disconnected. • Failed Process Interface card. • Programmer card failed - replace.
—	ER22	N-ACL	Max current is reached without detecting motor acceleration. • Check switches and jumpers for tach on Process Interface card. • Drive and Programmer card not communicating properly. • Software jumpers misprogrammed.

TABLE 5 (continued)
SPECIFIC ERROR TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
—	ER23	TPSPD	Motor has reached 50% of program's speed. • Check jumpers and switches on Process Interface card. • Motor and inertia too large for test.
—	ER24	MTRNG	Motor accelerated for more than 10 seconds while finding the friction current. • Test point SFB contains an offset at zero speed - adjust the Process Interface card. • A/D offset misadjusted - adjust P4-OFFSET on the Control card. • Motor being driven by external source.
—	ER25	MTRNG	Motor was still turning 20 seconds after finding the friction current. • Test point SFB contains an offset at zero speed - adjust the Process Interface card. • A/D offset misadjusted - adjust P4-OFFSET on the Control card. • Check jumpers or switches on the Process Interface card. • Pulse rate set incorrectly for digital tach. • Soft jumpers set incorrectly.
—	ER26	N_IFB	Reached 25° retarded firing and no current feedback measured. • DC fuse open. • Armature loop open. • Check SCR. • CFB circuit failed, replace Power Supply card.
—	ER27	N_IFB	Reached 37° retarded firing and no current feedback measured. • DC fuse open. • Armature loop open. • Check SCR. • CFB circuit failed, replace Power Supply card.

TROUBLESHOOTING
STABILITY PROBLEMS

There are three broad categories of drive symptoms broadly classified as "Stability" problems. Technically, two of these are not actually stability problems but, due to their behavior, are included in this troubleshooting procedure. These categories are as follows:

- **"Stability" problems:** "Hunting", oscillations in speed or current, etc. In general all problems in this category are cyclical in nature, with well-defined sinusoidal periods of oscillation. Usually these problems are electrically induced and can be tuned out with proportional and integral gain adjustments.

  There are at least five regulating loops in most DC-300E drives, any one which can be mistuned causing stability problems under the right conditions. They are as follows:

  (1) Current regulator - continuous current
  (2) Current regulator - discontinuous current
  (3) Speed regulator
  (4) Field current regulator - base speed
  (5) Field current regulator - weak field speed

NOTE: Various forms of position regulators are also available, which, due to their variety are not addressed here.

- **Mechanical Oscillations** - Problems which are a direct result of mechanical deficiencies but under certain conditions can be aggravated or excited by electrical frequencies. Examples of this are vibration, audible motor noise, gear chatter, etc.

- **Erratic Operation** - Intermittent speed or current fluctuations, mechanical "bumping", or other disturbances non-sinusoidal or non-periodic in nature. These may be induced by misfiring SCRs, intermittent grounds, "noise", or mechanical problems.

I. Before troubleshooting the drive, first verify the following:

NOTE: A programmer is required to make the following parameter checks in this part and adjustments as required in Part II.

- All software jumpers, ADR000 through ADR014 are in accordance with drive parameter list.
- All hardware jumpers are in accordance with the jumper selection chart.
- All software parameters are in accordance with the parameter list.
- Resistor (R_{rg}) and capacitor (C_{rg}) on Process Interface card are correct.
- All wiring is correct and in agreement with the drive elementaries.

II. Next, use Table 6 to decide which type of problem actually exists, keeping the following pointers in mind.

- DC-300 E drive current loops are almost always stable.
- Most stability problems, but not all, are corrected by adjusting speed loop parameters.
- It is always worthwhile to verify field current stability first.
- In general, when troubleshooting cascading or series regulating loops, the correct procedure is to stabilize the inner regulator first and then move outward; i.e., current regulator, then speed regulator, then position regulator (if furnished).
- If at any time the drive becomes stable as a result of a step, record any parameter changes, and do not continue with the remaining procedures.

TABLE 6

SYMPTOM	PROCEDURES
A. Unstable when unloaded but stable when loaded, drive is stable but exhibits gearbox or motor noise or runs erratically.	1. Check for failed SCRs - monitor test point CFB on the Control card with an oscilloscope. 2. Current reversal too sensitive - increase ADR023-DEADB. 3. See Custom Instruction Book for gain changes in current loop regulator. 4. If instability still exists, return adjusted parameters to their original setting and skip to Symptom E.
B. Unstable above and /or below base speed and over a wide speed range (Stabilizing the Field Current Regulator).	1. Confirm Field Current Regulator stability - monitor test point FC on the Control card with an oscilloscope or VOM. If stable, skip to the next appropriate symptom. 2. Field current is unstable - reduce ADR031-BFLPG until FC is stable. Do not reduce below the setting in ADR032-BFLIG (NOTE: ADR031 may already be lower than ADR032 in some applications, maintain this relationship). Lower ADR032 if necessary. 3. Document any change and proceed to Symptom C or D as necessary.
C. Speed instability below base speed.	1. See Custom Instruction Book for gain changes in speed loop. 2. If no improvement, return the adjusted parameters to their original settings. 3. If there is some improvement but instability still exists - document changes and skip to Symptom E.
D. Speed stability problems above base speed.	1. See Custom Instruction Book for gain changes in field parameters, ADR031-BFLPG and ADR032-BFLIG. 2. Check setting of ADR079-CROSS. 3. If no improvement, return the adjusted parameters to their original settings. 4. Check setting of P11-DAMP on Process Interface card. 5. If there is some improvement but instability still exists - document changes and skip to Symptom E.

TABLE 6

SYMPTOM	PROCEDURES
E. Drive is unstable: - over a small part of the speed range, or - when loaded but stable when unloaded, or - such that the oscillating frequency is proportional to speed.	1. Confirm that the problem is not tach related, correct tach feedback problem (misaligned or loose coupling, belt slippage, wiring or excessive ripple). 2. Verify that the power supplies are clean with an oscilloscope (refer to "Checking Power Supplies" in the Troubleshooting section). 3. Run drive to top speed and push the "RESET" button on the Control card. The drive will **coast** stop. Mechanically induced problems will still be present as the motor decelerates. If problem disappears, continue to next step. 4. Recalculate resistor R_{rg} and capacitor C_{rg} per the Custom Instruction Book. 5. Reduce ARD037-SLEAD and double ADR015 and ADR016. 6. Determine resonant frequency "Fr" in Hz with an oscilloscope, if possible. Convert to radians per second as follows: "Wr" = 6.28 x "Fr". 7. If no improvement, return the parameters to their original settings and skip to Symptom G.
F. Undirectional Stability Problems.	1. Try running the motor in the opposite direction to confirm that stubbing brushes or brush position is not the problem. 2. Inspect the load to make sure that nothing is wrong mechanically. 3. Make certain that all SCRs are firing in both directions by monitoring test point CFB on the Control card with an oscilloscope.
G. Additional Procedures.	If the above procedures do not result in stable operation, additional steps may be required including: 1. Checkout of the current loop. 2. Running the field regulator open loop to verify stability. If such procedures are required, contact GE DRIVES, Product Service Engineering, for instructions. **(814-875-3851)**

TYPICAL WAVEFORMS

The following are examples of typical waveforms that can be seen on an oscilloscope. Each figure is accompanied by the name of the signal, the test point which it is derived from (in parenthesis), and the mode of operation if applicable. Also attached is the proper time setting for the waveform.

(For information on using an oscilloscope, consult the owner's manual.)

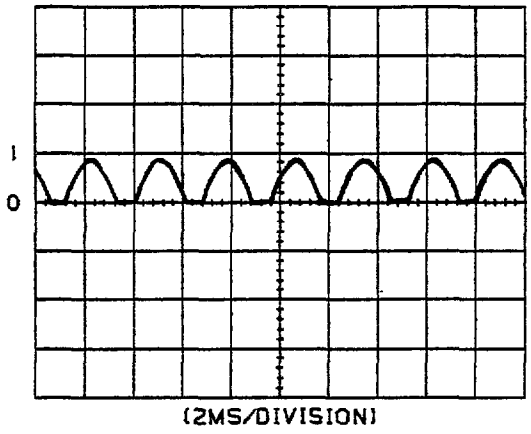


FIGURE 24
CURRENT FEEDBACK (CFB) IN
DISCONTINUOUS CURRENT

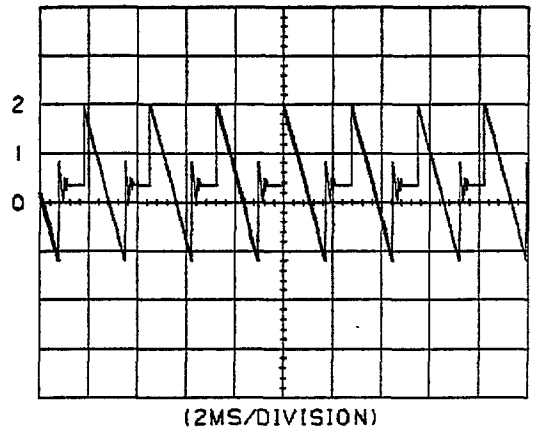


FIGURE 25
VOLTAGE FEEDBACK (VFB) IN
DISCONTINUOUS CURRENT

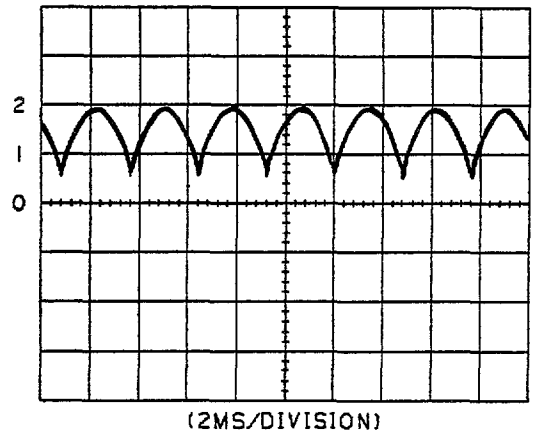


FIGURE 26
CURRENT FEEDBACK (CFB) IN
CONTINUOUS CURRENT

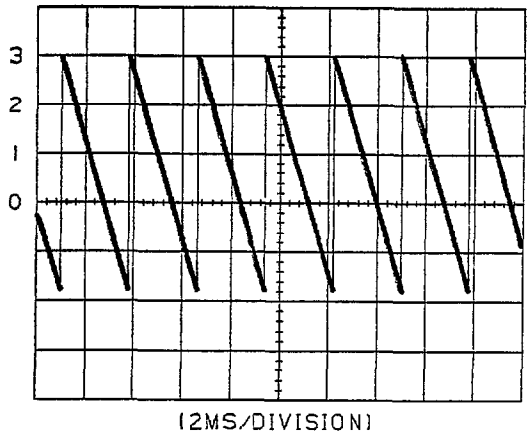


FIGURE 27
VOLTAGE FEEDBACK (VFB) IN
CONTINUOUS CURRENT

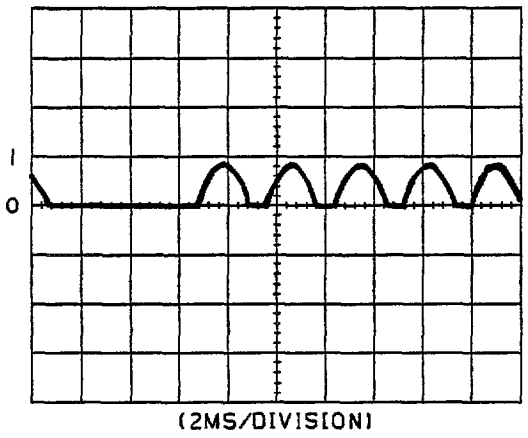


FIGURE 28
CURRENT FEEDBACK (CFB) WITH
ONE SCR OPEN OR DISCONNECTED

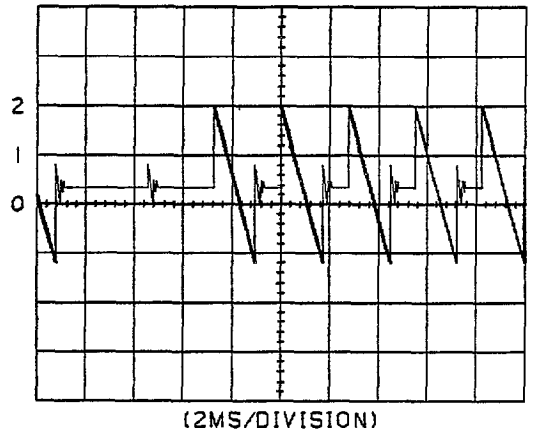


FIGURE 29
VOLTAGE FEEDBACK (VFB) WITH
ONE SCR OPEN OR DISCONNECTED

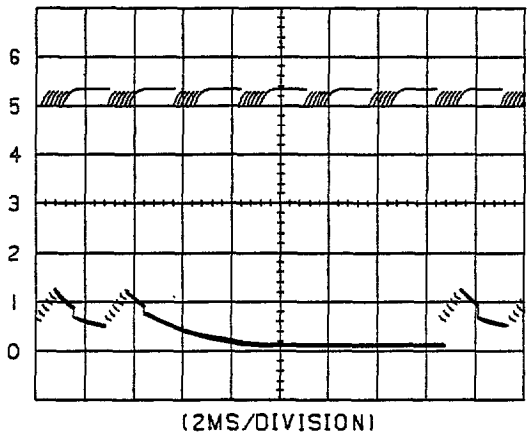


FIGURE 30
TOP – DELAYED FIRING POWER (DFP)
BOTTOM – GATE PULSES FOR A1 SCR,
DISCONTINUOUS CURRENT

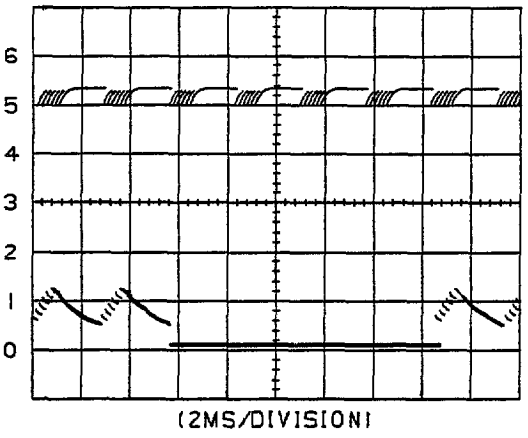


FIGURE 31
TOP – DELAYED FIRING POWER (DFP)
BOTTOM – GATE PULSES FOR A1 SCR,
CONTINUOUS CURRENT

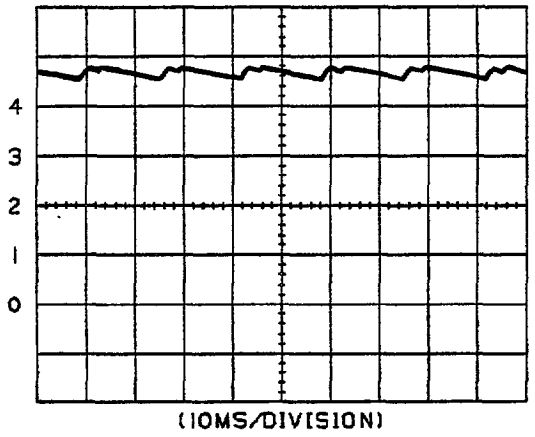


FIGURE 32
FIELD CURRENT (FC) AT FULL FIELD

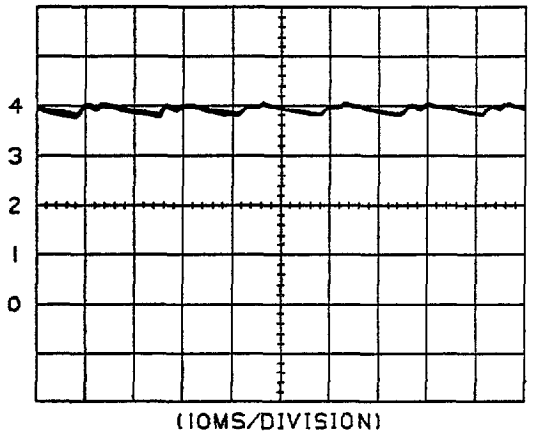


FIGURE 33
FIELD CURRENT (FC) WEAK FIELD

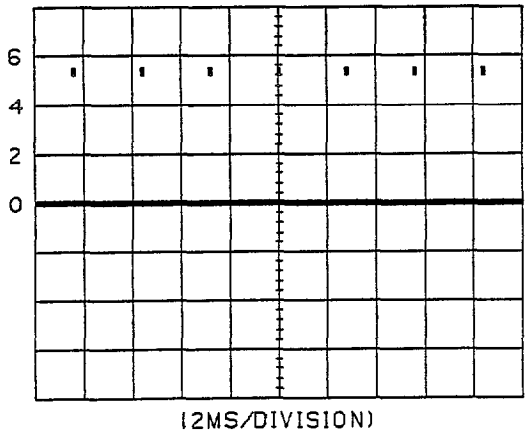


FIGURE 34
INITIAL PULSE UNIT (IPU)
INDICATED INITIAL PULSE OF EACH
ARMATURE SCR FIRING PULSE BURST

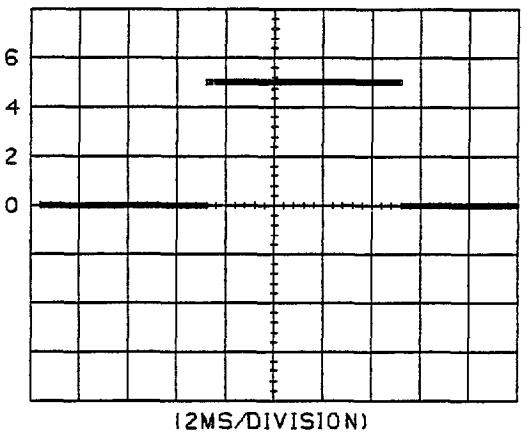


FIGURE 35
SYNC SIGNAL (SY) - SQUARE
WAVE SYNCHRONIZED TO
AC LINE FREQUENCY

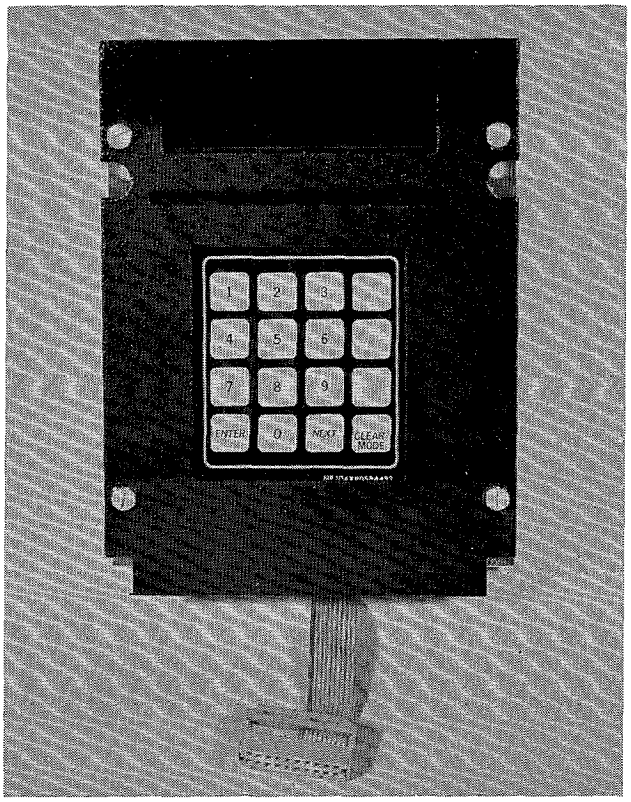


FIGURE 36
OPTIONAL "LOCAL" PROGRAMMER

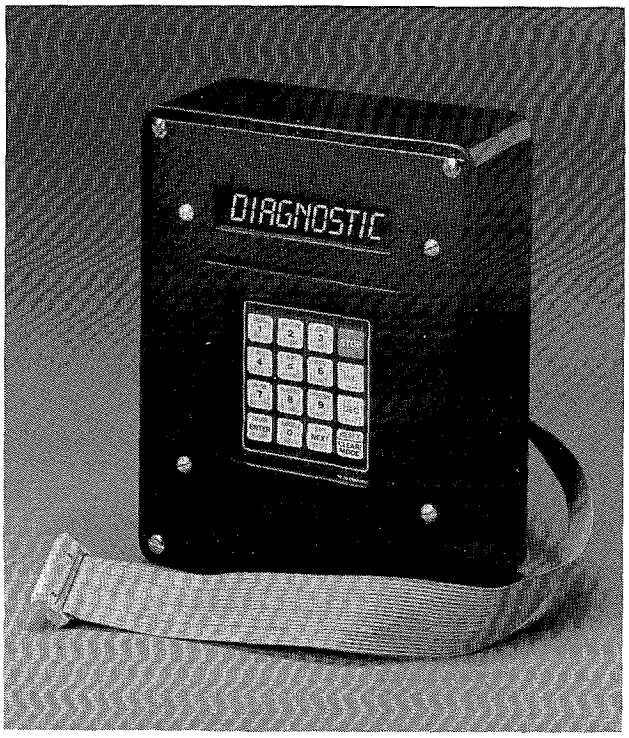


FIGURE 37
OPTIONAL "HAND-HELD" PROGRAMMER

USE OF THE OPTIONAL PROGRAMMER MODULE

The optional Programmer assembly is available in two configurations, as pictured in Figures 36 and 37. The Local configuration is mounted on the Control card and plugged into 18PL. The Hand-held configuration is contained in a box and plugged directly into the Control card 18PL.

CAUTION

THE HAND-HELD PROGRAMMER CABLE MUST BE ORIENTED PROPERLY OR THE CONTROL AND PROGRAMMER CARDS WILL BE DAMAGED.

The Hand-held Programmer uses a different cable than the Local Programmer and care must be exercised when the Hand-held Programmer is plugged in.

To properly orient the cable, hold the unit in front of the drive and extend the ribbon cable back to 18PL, so that the cable is not twisted, and plug into 18PL. The red stripe in the ribbon cable should be on the left when looking at the front of the Hand-held Programmer and should be on the left when plugged into 18PL.

The local Programmer cable is oriented with the red wire on the left side of the cable when the operator is facing the drive.

The optional Programmer provides an alphanumeric readout of fault codes (instead of the diagnostic LEDs) and operation/adjustment of the drive through three modes --OPERATE, PARAMETER, and DIAGNOSTIC modes.

THREE MODES OF OPERATION

A. **OPERATE** Mode - The "normal" mode, allows the red and black keypad functions to be active (the **RED** keys are dominant). This is the mode the drive will enter when power is applied.

B. **PARAMETER** Mode - Allows the user to examine and change as necessary any of 254 EEROM (E^2) Parameter addresses to "customize" the drive for a given application. In this mode all **BLACK** keys are active plus [INC], [DEC] and [STOP].

C. **DIAGNOSTIC** Mode - Allows the user to utilize any of the internal digital displays, plus run a series of diagnostic checks and self-tuning procedures. In this mode, both the **BLUE** and **BLACK** key functions are active (with **BLUE** keys dominant), plus [INC], [DEC] and [STOP].

CHANGING BETWEEN MODES

NOTE: Throughout this section, a number or function encased in brackets [ ] indicates a programmer key pressed.

The drive will normally enter the OPERATE mode as soon as power is applied.

A. To enter the PARAMETER mode, first confirm that the display reads "OPERATE", then press [SET], [DRV#], [7], [7], and [ENTER]. The display should now show "PARAMETER".

B. To enter the DIAGNOSTIC mode, the drive must first be in the PARAMETER mode. If the display shows "PARAMETER", simply press [CLEAR/MODE] and the display will immediately show "DIAGNOSTIC". If a particular parameter is being displayed, press [CLEAR/MODE] causing "PARAMETER" to be displayed then press [CLEAR/MODE] again; "DIAGNOSTIC" will appear.

C. To return to the OPERATE mode, simply press [CLEAR/MODE] once in the DIAGNOSTIC mode, or if the drive is in the PARAMETER mode, press [CLEAR/MODE] either two or three times, as required until the display returns to OPERATE.

NOTE: The Programmer may appear to "hang up" due to programming errors. In such cases it may be necessary to initiate a "hardware reset" using the RESET switch (SW2) on the Control card.

WARNING

PERTAINING TO THE REMAINDER OF THIS SECTION, DO NOT USE DIAGNOSTIC RUN, SELF-TUNE (I-TUNE OR S-TUNE), SQUARE-WAVE REFERENCE TEST OR ANY PROGRAMMER "RUN" FUNCTIONS ON ANY CRANE OR ELEVATOR DRIVE.

OPERATE MODE - (RED Keys dominant)

In the OPERATE mode, any of the **RED** key functions, if activated (if key is not activated, the Programmer will display "*FL41* KEYOF" momentarily), can be used as follows:

WARNING

OPERATE MODE IS TO BE USED ONLY AS A DIAGNOSTIC TOOL; MOTOR WILL ROTATE, IGNORING SAFETY CIRCUITS. DO NOT ATTEMPT TO USE [RUN], [SLOW], [JOG], [R1], [R2], [REV], [A/M], [RATIO] OR [PRGSV].

[STOP] - Controlled stop.

[REFR] - Run at speed reference at which drive was last run, (includes preprogrammed keypad references or changes caused by using the [INC] and [DEC] keys, MSR or ASR).

[DRV#] - Used to examine the drive number (in most cases "00"). Also part of access code to PARAMETER mode.

[SET] - Used to adjust the settings of various keypad functions; activated by pressing [SET], the [RUN], [SLOW], [JOG], [R1], [R2], or [RATIO], then the desired setting in decimal numbers, then [ENTER]. Also, used for entering Parameter mode. The EEROM must be write "ENABLED" to make these adjustments.

[INC] & [DEC] - Increases or decreases the speed of any of the keypad Run or Jog functions, or the Ratio function; simply enter the desired function and then press [INC] or [DEC] as desired.

[RESET] - Resets the drive after fault or error messages appear. The reset key is referred to as a "Software Reset".

The normal display in the OPERATE Mode will be "M XX% YY%" where "M" indicates Manual mode using the Manual System Reference, MSR (alternatively an "A" will indicate Auto mode using the Auto System Reference, ASR on the optional Application card), XX% indicates percent of rated speed, and YY% indicates percent of rated load.

PARAMETER MODE - (BLACK Keys, plus INC, DEC & STOP active)

A. To enter the PARAMETER mode press [SET], [DRV#], [7], [7], and [ENTER].

B. To **examine** the present value of any parameter simply key in any three-digit parameter address (do not press [ENTER]). The name for that address will immediately be displayed along with the present contents of that address. Press [CLEAR/MODE] to clear present parameter (displays "PARAMETER") then the address number for the next parameter desired.

Occasionally, it may become necessary to check a series of parameters. Simply call up the lowest parameter address and press [NEXT] as often as required until the highest parameter number is reached.

C. To change parameters, two conditions must be present:

1. The EEROM (E^2) enable device must be in the "ENABLE" position.

JP13: 1-2; position; E^2 write "PROTECT"
2-3; E^2 write "ENABLE"

CAUTION

DO NOT APPLY OR REMOVE POWER FROM THE DRIVE OR PERFORM A "HARDWARE RESET" WITH THE E^2 ENABLE DEVICE IN THE "ENABLE" POSITION.

2. Parameter address #3 (designated as ADR003-DGNJP) must contain an **odd** number (1, 3, 5, etc.). If an even number exists, add "1" to the present value and enter the new number.

Example: Examine and change the drive acceleration and deceleration rates. **Assume** the drive acceleration rate is set at 2 seconds and the deceleration rate is set at 10 seconds. Change the acceleration rate (ADR057-ACELT) from 2 seconds to 7.5 seconds. Reduce the drive deceleration rate (ADR058-DCELT) from 10 to 5.2 seconds.

1. Confirm that the E^2 enable device is in the "ENABLE" position and that ADR003-DGNJP contains an odd number.

2. After examining ADR003, press [CLEAR/MODE] the [0], [5], [7]. The display shows "ACELT. 20", corresponding to a 2-second acceleration rate.

3. Either press [INC] until the display value increases to "75" or press [7], [5]. In either case the display should now show "ACELT. 75".

4. Press [ENTER], placing the new value into EEROM memory.

5. Press [NEXT], commanding the drive to go to the next parameter. ADR058-DCELT; the display shows "DCELT. 100"; corresponding to a 10-second deceleration rate.

6. Press [DEC], until the display shows "DCELT. 52" or press [5], [2].

7. Press [ENTER] placing the new value into memory.

8. Press [CLEAR/MODE]; the display will now show "PARAMETER".

9. To examine other addresses simply key in the appropriate three-digit parameter address number.

NOTE: This next step is **extremely important** to prevent accidental reprogramming of the drive.

10. To **seal** the new parameter values, **remove** the "1" from ADR003-DGNJP returning it to the "as shipped" **even** number, and return E^2 enable device to the "PROTECT" position.

D. As in the OPERATE mode, the display may return to "'M. XX% YY%" after a period of inactivity, but the drive is still in the PARAMETER mode (indicated by the apostrophe before the M) awaiting further keypad entries. Simply enter the next parameter address to be examined.

E. To **exit** the PARAMETER mode, press [CLEAR/MODE] either once or twice to go to the DIAGNOSTIC mode. To re-enter the OPERATE mode, press [CLEAR/MODE] again.

DIAGNOSTIC MODE

The Diagnostic Tests listed below will aid in the start-up, tune-up and troubleshooting of the drive. To use these diagnostic features either a Local or Hand-held Programmer is required. Availability of certain tests may be dependent on hardware or software configurations provided.

WARNING

DO NOT USE DIAGNOSTIC RUN [DGR], DIAGNOSTIC STATIC [DGS], SELF-TUNE (I-TUNE OR S-TUNE), TEST 25, OR ANY PROGRAMMER RUN FUNCTIONS ON ANY CRANE OR ELEVATOR APPLICATIONS.

STANDARD DIAGNOSTICS TESTS:

Test 1	[DGS] - DO NOT USE
Test 2	[DGR] - DO NOT USE
Test 3	[DVM] - Displays Digital Voltmeter
Test 4	[ARM V] - Displays armature voltage.
Test 5	[ARM I] - Displays armature current in amps.
Test 6	[FIELD I] - Displays field current in amps.
Test 7	[RPM] - Displays motor speed.
Test 8	[TORQ] - Displays motor torque in 1 lb-ft.
Test 9	[HP] - Displays motor horsepower.
Test 10	Displays software revision of Control card microprocessors.
Test 11	Displays diagnostic variables (selectable).
Test 51	Revision date of Programmer software.
Test 52	Revision level of Programmer software.
RAM TEST	Displays single diagnostic variable (selectable).
DAC0, DAC8 & DAC12	Digital to Analog converter outputs.
BITS 1, 2 & 3	Status bit outputs.

ADVANCED DIAGNOSTICS TESTS:

Test 20	Electronic Wire-Check.
Test 21	Diagnostic Variable Pairs.
Test 22	Diagnostic Variables through DAC Outputs.
Test 25	Square-Wave Reference.

A. STANDARD DIAGNOSTIC TESTS

NOTE: For Tests 3 through 9 simply enter the DIAGNOSTIC mode and press the appropriate key. To enter all other diagnostic tests, press [TEST], [X], where "X" is the number of the particular test to be performed, and then press [ENTER].

TEST 3 - DVM MODE
(Requires Optional Process Interface Card)
(Also see Diagnostic Mode for Drive Setup)

Test 3 provides the capability of utilizing the Local or Hand-held Programmer and Process Interface card as a digital voltmeter (DVM) for checking various analog test points (TP) on the drive cards. DC voltage in the +/-40 volt range can be measured. This requires a test probe wire (supplied in the miscellaneous parts packet shipped with the drive) from test point DVM on the Process Interface card to the particular test point to be monitored. The DVM may be calibrated by monitoring the +5 volt test point and adjusting P3-DVM TRIM on the Process Interface card.

Pressing [TEST] discontinues DVM mode and displays "CLR TEST".

TESTS 4 thru 9 VARIOUS DISPLAYS
(ARM V, ARM I, FIELD I, SPEED, TORQ, HP)

Tests 4 thru 9 are accomplished by setting the drive in DIAGNOSTIC mode and then entering the desired test number. To change to a different test simply push another number from 4 thru 9. To cancel the last test push [TEST] once, the Programmer will display "CLR TEST".

TEST 10 - SOFTWARE REVISION

Test 10 will display the revision level of the microprocessors on the Control card. For example, "R20F20T20" says the regulator microprocessor (RUP) is version 2.0 software, the firing microprocessor (FUP) is version 2.0 and the tachometer microprocessor (TUP) is version 2.0. This can be verified by comparing the information with white labels on the individual microprocessor chips on the Control card.

TEST 11 - DIAGNOSTIC VARIABLE DISPLAY
(also see Tests 21 and 22)

Test 11 can be used to display, in hexadecimal numbers, certain drive variables for the purpose of comparison. For example, speed reference and speed feedback may be compared. To accomplish this, several things must be done. First, place the drive in PARAMETER mode as described previously. Next determine, from the list which follows, the RAM address of the regulator variables to be monitored. Then put

the first variable RAM address number into parameter ADR122-DIAG1 and the second variable RAM address into parameter ADR123-DIAG2. The last step is to put the drive in DIAGNOSTIC mode and select Test 11 as described above. The programmer will then display "DGXXXX. YYYY", where "XXXX" is the hexadecimal value of the RAM address entered in ADR122, and "YYYY" is the hexadecimal value of the RAM address entered in ADR123.

NOTE: The value displayed is in 2's complement. Values from 0000 to 7FFF are positive magnitude (0 to +32767). Values from FFFF to 8000 are negative magnitude (FFFF=-1, 8000=-32768).

RAM ADDRESSES FOR "ANALOG" SIGNALS

RAM ADR	DESCRIPTION	RAM ADR	DESCRIPTION
073	Speed Loop Output	103	Voltage Feedback
099	Current Feedback	105	CEMF
061	Current Loop Output	463	Field Current
065	Timer Speed Reference*	494	ACOM
069	Jog Reference*	501	TOC Timer
071	Speed Feedback		

*If used.

NOTE: These addresses will contain a variable number, representing an "analog" signal. Refer to "RAM Variable Scaling" Table 3 in the Adjustment Section for the relative hexadecimal values of the displayed variables.

RAM ADDRESSES FOR LOGIC SIGNALS

RAM ADR	DESCRIPTION	RAM ADR	DESCRIPTION
140	Current Limit Flag	156	Zero Current Flag
148	Reference Enable	137	Fault
152	Precondition		

NOTE: This list contains addresses of signals which will be either "high" (display "8080") or "low" (display "0000"). If the signal is high, then the statement is true. For example, if "137" is in ADR122, Test 11 is initiated, and a fault condition exists, then the display will show "DG8080.YYYY", (YYYY value will depend on the number in parameter ADR123). If the drive is not faulted then the display will be "DG0000.YYYY".

TEST 51 - PROGRAMMER CARD SOFTWARE REVISION DATE

Test 51 allows a check of the software date of the Programmer microprocessor (PUP).

TEST 52 - PROGRAMMER CARD SOFTWARE REVISION

Test 52 provides a check on the software level of the "PUP".

RAM TEST - DISPLAYS RAM LOCATION VALUE

RAM test allows a "one-shot" look at a RAM address (refer to TEST 11 for variables). The value displayed is in decimal numbers. To display a RAM address, press [RAM], the Programmer will display "***RAM***". Then press the keys corresponding to the desired RAM variable. The drive will display the following: "XXX YYYY", where "XXX" is the RAM address and "YYYY" is the RAM value. Press [CLEAR/MODE] to clear the display.

DAC0, DAC8 and DAC12 (DAC8 and DAC12 Require an Optional Application card)

WARNING

THESE OUTPUTS MAY ALREADY BE UTILIZED BY THE DRIVE. REFER TO THE DRIVE ELEMENTARIES AND CUSTOM SOFTWARE DESCRIPTION.

Three selectable analog outputs, DAC0, DAC8 and DAC12, are available for test purposes and are programmed similar to Test 11. To program DAC0, enter the Parameter mode and call up ADR090-DAC0. Enter the signal to be monitored from Table 3, "RAM Variable Scaling". Putting that RAM Address in ADR090 will cause that signal to appear at the DAC0 test point. DAC8 and DAC12 perform in a similar manner, but the signal RAM address is loaded into ADR096-DAC8 or ADR097-DAC12 and the output appears at the DAC8 or DAC12 terminal on 4TB.

STATUS BITS 1, 2 and 3 (Requires Optional Application card)

WARNING

THESE OUTPUTS MAY ALREADY BE UTILIZED BY THE DRIVE. REFER TO THE DRIVE ELEMENTARIES AND CUSTOM SOFTWARE DESCRIPTION.

ADR098-BIT#1, ADR099-BIT#2 and ADR100-BIT#3 are programmed in the same manner as DAC8 and DAC12. They are different in that they are logic outputs rather than analog outputs.

Signals listed in Test 11, Logic Signals, can be used to select addresses to enter in ADR098, ADR099 and ADR100. Variables listed under Test 11, "Analog Signals", may also be used in the bit outputs to indicate the polarity of the signal (positive or negative). These outputs also appear on 4TB as "B01", "B02", and "B03" respectively.

B. ADVANCED DIAGNOSTIC TESTS

TEST 20 - ELECTRONIC WIRE-CHECK

WARNING

SIGNAL LEVEL DETECTORS AND BIT OUTPUTS MAY CONTROL EXTERNAL DEVICES. CONSULT THE SYSTEM ELEMENTARIES BEFORE USING THIS FEATURE.

Test 20 allows five different types of wire-checking and output forcing test as follows:

1) Signal Level Detector (SLD) forcing.
2) Status Bit monitoring.
3) Fault Code testing.
4) Logic input monitoring (Application card).
5) Logic input monitoring (Control card).
6) Microapplication card input monitoring (future).

To perform all tests, it is first necessary to be in Diagnostic mode with the EEROM write enabled (write protect the EEROM after this test is completed). Then key in [TEST], [2], [0], and [ENTER]. This causes Test 20 to be active and the display to show "'1.'.00111111". The left-most digit indicates the test level in this case, the "'1.'." indicates level 1 or Signal Level Detector (SLD) forcing.

To change the type of test, press the [INC] or [DEC] key until the correct test number is displayed in the left-most digit. For example, if Test 20 is called for and the initial display is shown above, press [INC] once to run the next level (20-2), Status Bit monitoring. The programmer display will be similar to "'2.'.11111100". Each test will be described below.

NOTE: Test 20-1 temporarily changes the value of Parameter ADR009-SLDJP. If the AC control power is removed while keys 0 thru 5 are depressed, ADR009-SLDJP may be changed and will not contain the correct value when AC power is reapplied. The value of SLDJP should be noted prior to initiating this test.

TEST 20-1 - SIGNAL LEVEL DETECTOR (SLD) FORCING (Requires Application card)

This display will show "'1.'.00111111". SLD0 is energized if the right most digit is 1. and dropped out if it is a zero. Scanning from right to left, the status of all six SLDs (SLD0 through SLD5) can be seen. That is, the second digit from the right is SLD1, the third digit from the right is SLD2, etc. To change the state of SLD0, depress [0] key.

NOTE: SLDs cannot be toggled while the drive is running or the drive will fault. While the [0] key is depressed, SLD0 will be maintained in the opposite state it was originally. When the [0] key is released, SLD0 returns to its original state. Keys 1-5 operate in the same manner for SLD1 thru SLD5.

TEST 20-2 - STATUS BIT OUTPUTS (Requires Application card)

Status bits B01, B02 and B03 (ADR098, ADR099 and ADR100 respectively) are programmed as previously described. If test 20-2 is called for, B01 will appear as the right most character in the display, B02 will be one to the left of B01, and B03 is one to the left of B02. If Bit#1 is "high" the right most character will be "1"; if it is "low", it will be displayed as [0].

TEST 20-3 - FAULT CODE OUTPUTS (Requires Application card)

To test remote fault code lines and their effects on external equipment, place the drive in Test 20-3. Next enter the desired fault code by keying in [0]. The programmer then displays "FAULT". Next key in number of fault code desired and press [ENTER]. (NOTE: Fault 63 will cause FC5 to be high.)

This test does **not** cause the drive to fault, unless a number greater than 127 is entered. This test is a way of forcing the fault code output lines on the Application card to check interfaces with external equipment using the fault output lines. The Diagnostic LEDs will also display the appropriate fault code.

TEST 20-4 - LOGIC INPUT MONITORING (Requires Application card)

If Test 20-4 is selected the display will show "'4.'.XXXXXX" where Xs will be either "1" or "0". A "1" will be displayed if a particular input is tied to REF24 and the drive is programmed for **negative** logic (JP19 and JP20 on the Control card are in 1-2 and 3-4 position respectively; and JP5 and JP6 on the Application card are in the 1-2 and 3-4 position respectively). From right to left the digits are MODE 0 thru MODE 7.

TEST 20-5 - LOGIC INPUT MONITORING (Control card)

Text 20-5 is similar to Test 20-4 except that the left four digits do not apply. Starting at the first digit from the right, the status of four logic inputs will be displayed as:

"5.XXXXDCBA".

A = Status of RUN command inverted ("0" indicates RUN is initiated).
B = Status of JOG command inverted ("0" indicates JOG is initiated).
C = Status of POL command inverted ("0" indicates POL is initiated).
D = Status of XSTOP command ("1" indicates XSTOP initiated).

TEST 21 - DIAGNOSTIC VARIABLE PAIRS

NOTE: Test 21 will cause ADR122-DIAG1 and ADR123-DIAG2 to change.

Test 21 performs in a similar manner to Test 11 except that several pairs of RAM addresses are already pre-programmed in memory. Test 21 is entered from the Diagnostic mode by keying in [TEST], [2], [1], and [ENTER]. The programmer will then display "TST21". Next key in [0], [1], [2], [3], or [1], [0] (corresponding to Tests 21-0, 21-1, 21-2, 21-3, or 21-10 respectively), and [ENTER], to display the pairs of RAM addresses shown in the table below.

As in Test 11, the display will show "DGXXXX.XXXX". The left and right four characters are as selected from the listing. Test 21-3 will display the variables programmed into ADR096-DAC8 and ADR097-DAC12.

If [1], [0] is entered, calling for Test 21-10, then the display will prompt with "ADDR1". At this time, key in the first RAM address you wish to monitor and press [ENTER] (typical RAM addresses are listed under Test 11). The display will then prompt with "ADDR2". Key in the second RAM address and press [ENTER]. The display will now be DGXXXX.XXXX", where the left four digits are the value of the signal address stored in ADDR1, and the right four digits are the value of the signal address stored in ADDR2.

RAM VARIABLE PAIRS FOR TESTS 21 AND 22

TEST NUMBER	LEFT 4 DIGITS (RAM-Addr)		RIGHT 4 DIGITS (RAM-Addr)	
0	Speed loop output	(053)	Current loop output	(061)
1	Speed Reference	(065)	Speed feedback	(071)
2	Position Error	(471)	In position flag	(154)
3	DAC8	(ADR096)	DAC12	(ADR097)
10	Diag. Var. #1	(ADR122)	Diag. Var. #2	(ADR123)

DIAGNOSTIC MODE FOR DRIVE SETUP

NOTE: Step A of the following procedure utilizes Diagnostic Test 21-10. All other steps utilize the DVM mode. These procedures allow accurate tuning of all analog pots as listed below.

Control card	OFFSET	P4
Control card	REF SCALE	P1
Power Supply card	CFB ZERO	P2
Power Supply card	CFB ADJUST	P3
Application card	ASR SCALE	P1
Application card	FRA SCALE	P2

A. Place the drive in DIAGNOSTIC mode. Enable EEROM and key in [TEST], [2], [1], [ENTER], then [1], [0] and [ENTER]. The drive will display "ADDR1". Key in [4], [9], [4] and [ENTER]. The drive will display "ADDR2". Key in [0], [7], [1] and [ENTER]. The drive will display "DG.XXXX.XXXX". Use the left four characters and P4-OFFSET on the Control card to adjust the drive offset to zero.

B. Place the drive in the DVM mode by pressing [DVM].

C. To adjust the current feedback signal, which is generated on the Power Supply card, the following steps are necessary. With power on the drive and the MA contactor open, adjust P2-CFB ZERO on the Power Supply card such that TP CFB on the Control card reads 0 volts.

   Remove AC power from drive. Disconnect connector DRPL from the Power Supply card and move JP3 and JP4 to the 2-3 (calibrate) position. Reapply AC power to the drive and adjust P3-CFB ADJ on the Power Supply card such that TP CFB on the Control card reads the value of the calibration label on the inside of the card rack.

   Turn off AC power to the drive, return JP3 and JP4 to the 1-2 position, and reconnect DRPL to the Power Supply card. The armature current shunt isolator is now calibrated to provide approximately 1.0 VDC at TP CFB for rated drive current.

D. Calculate the tachometer voltage at top speed from the tachometer nameplate and motor top speed. Refer to the Custom Instruction Book to calculate resistors R_{rg} and R_{fb}. Start the drive and run at half speed (2.0 VDC at TP SRS on Control card if running in manual mode or 2.0 volts at TP ASR on Application card if running in auto mode). Monitor motor speed, either with a direct reading mechanical tach or with a voltmeter between 1TB-1 and 1TB-3 on the Process Interface card. Adjust P8-SFB SCALE on the Process Interface card to 50% of motor speed.

   Run the drive at top speed and recheck the motor speed. Readjust P8-SFB SCALE as necessary for the correct top speed. TP SFB on the PI card should now read 4.0 VDC and be equal to TP SRS voltage. This completes the calibration of the speed feedback circuit and setup of the drive's analog pots.

E. If the FRA input to the Application card is used it will be necessary to adjust P2-FRA SCALE on the Application card for proper amount of field forcing. With the proper voltage applied to the FRA terminal 4TB (see elementary drawing for your specific drive) adjust P2-FRA SCALE on the Application card, for the correct field current with field forcing applied (also shown on elementary drawings). Use Test 6 in DIAGNOSTIC mode to monitor field current while adjusting P2-FRA SCALE.

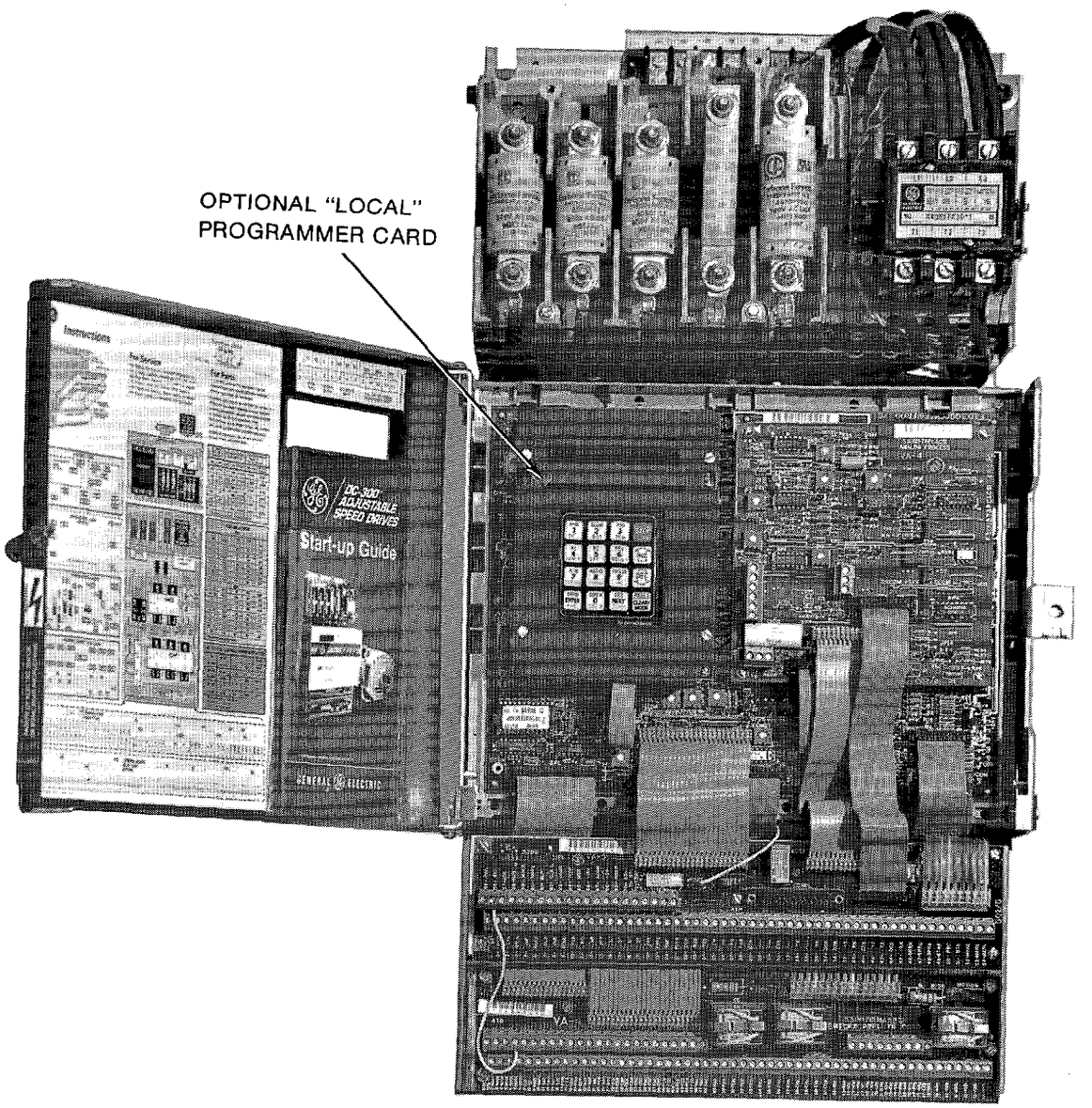


FIGURE 38
DC-300 E CONTROLLER WITH OPTIONAL PROGRAMMER CARD

PARTS REPLACEMENT

REPLACEMENT OF SCR MODULES (Figures 4 and 39)

1. Lower the card carriers to expose the rear of the unit.
2. Disconnect the wires from the damaged module and remove the bus bar and mounting screws.
3. Clean the mounting surface of old lubricant and dirt.
4. Apply a thin film of General Electric GL322 Versilube® Plus lubricant (Part No. 104X214AB059) to both contacting surfaces and install the module. Tighten screws to the torque specifications below.

CARD REPLACEMENT PROCEDURE (Figure 40)

CAUTION

TREAT ALL CARDS WITH STATIC SENSITIVE HANDLING TECHNIQUES. USE "GROUNDING" STRAP WHEN CHANGING CARDS AND STORE CARDS IN THE ANTI-STATIC BAGS THEY ARE SHIPPED IN.

1. Carefully disconnect all cables.
2. Release all hold-down tabs and remove board.
3. Set all jumpers on the replacement board exactly as the jumpers on the failed board.
4. Install the new board verifying that all tabs snap tightly into position.
5. Reconnect all cables verifying that they are properly seated at both ends.
6. Set all pots in the same physical position as the pots on the failed card and fine tune as required per instructions in Table 1 of the Adjustments section.
7. If the Control card is changed, remove the EEROM from the failed card and install in the new card. If the EEROM from the failed card is suspected to be damaged, it will be necessary to reprogram a new EEROM to the custom parameter list. See Figure 41 for typical EEROM removal tool and grounding techniques.
8. If the Analog Process Interface card is changed, transfer resistors and capacitors from CTBA, CTBB and CTBC (R_{fb}, R_{rg}, C_{rg}, R_{wn}, C_{wn}).

SCR MODULE PART NUMBER	HEATSINK MOUNTING SCREWS	*BUS BAR MOUNTING SCREW
104X125DC020 104X125DC021	Min. 40 inch pounds Max. 53 inch pounds	Min. 106 inch pounds Max. 133 inch pounds
104X125DC022 104X125DC023	Min. 28 inch pounds Max. 30 inch pounds	Hand Tighten
104X125DC026 104X125DC027	Min. 40 inch pounds Max. 49 inch pounds	Min. 22 inch pounds Max. 31 inch pounds

* Support the bus bar while tightening screws to avoid excess torque on connected modules.

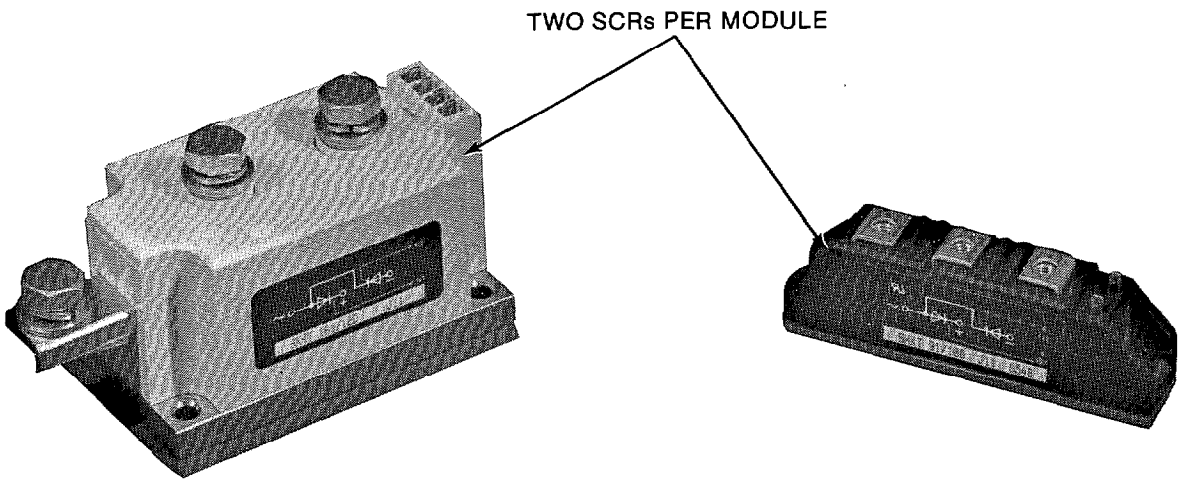


FIGURE 39
TYPICAL ARMATURE SCR MODULES

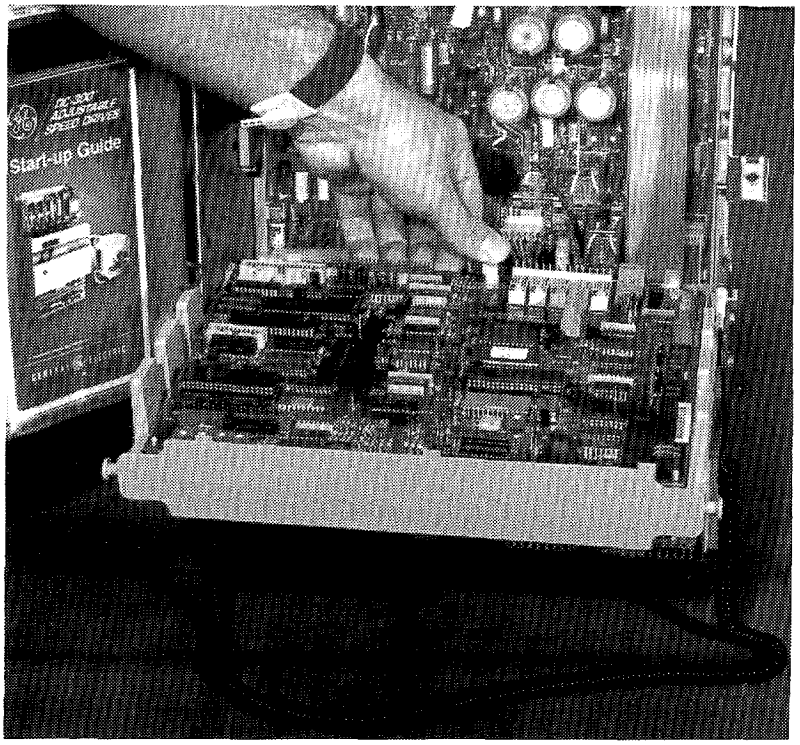


FIGURE 40
DRIVE CARD WITH SIX CARD RELEASE TABS
PUSH TABS AWAY FROM CARD TO REMOVE

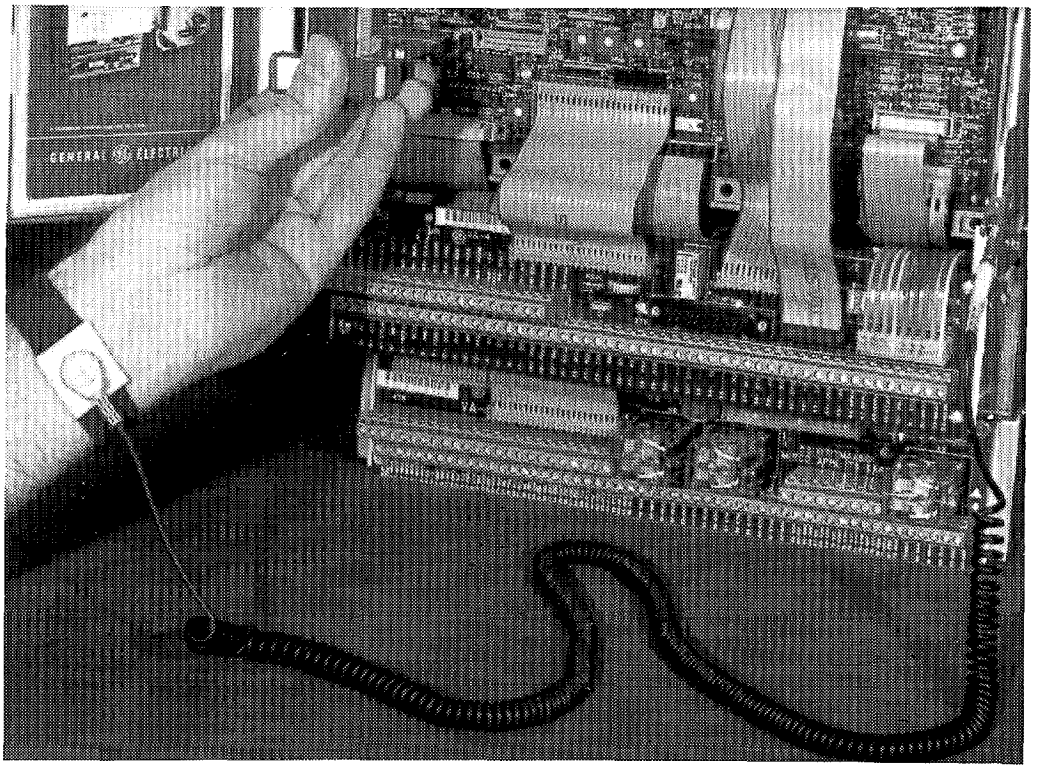


FIGURE 41
TYPICAL EEROM REMOVAL AND/OR REPLACEMENT EQUIPMENT
EXERCISE CARE WHEN REMOVING OR INSERTING EEROM
TO PREVENT BENDING PINS

SPARE AND RENEWAL PARTS

Renewal parts information for this equipment is included in the **CUSTOM INSTRUCTION BOOK** provided for the drive. The renewal parts list is generated specifically for the equipment furnished on each order and provides information on parts relative to the particular rating and options incorporated in the drive at the time of shipment. If the renewal parts information is missing, a replacement list may be obtained by contacting your local **GE Sales Office** or **GE DRIVES Authorized Distributor** with the drive **MODEL and REQUISITION numbers.** Parts lists may also be obtained direct from **GE DRIVES Technical Services, (814) 875-2075.**

It is recommended that the parts list be reviewed and a set of spare parts be maintained to minimize down time should repairs be required. Renewal parts may be ordered by contacting your local **GE Sales Office** or an **Authorized GE Parts Distributor in your area.**

PART NUMBER IDENTIFICATION

The parts listed are divided into four categories, each with its own part number structure:

Order-Specific Assemblies
Common Assemblies
Component Parts
Circuit Cards

Order-Specific Assemblies are generally major assemblies consisting of a group of common assemblies that make up the specific controller provided. Other items which are obtained specifically for the order may also be provided with a similar part number structure.

Example:

7VXYZ999CD01

- 7V -- assembly provided by GE DRIVES DEPARTMENT
- XYZ999 -- identification number (original order)
- CD -- assembly type (CD = Core Drive)
- 01 -- unit identification number

Common Assemblies are subassemblies, consisting of a group of component parts, which are in common use in GE DRIVES products and are not specific to a particular order. The part number consists of the number "36" followed by an alpha character and may be up to 14 characters in total length.

Example:

36C77345AAG02 is the part number for an external Motor Field Control Assembly for a DC-300 controller.

Component Parts are the basic components used in the assemblies and generally represent the lowest discrete level. Component part numbers consist of a combination of alphanumeric characters, defining the class of component and the specific item.

Example:

104X109AD038
- 104X – component part
- 109 – component type (fuse)
- AD038 – specific item (500 V, 3.2A FNQ)

Circuit Cards are identified by a catalog number giving the product class, card function and revision levels. It should be noted that some cards contain software which is designated by the suffix "Mn" where "n" is the software number. These cards are also available less software with suffix "Gn" designation. The factory may substitute later versions of cards based on availability and design enhancements.

Example:

531X300CCHADM4
- 531X – card for GE DRIVES DEPARTMENT microprocessor based controller
- 300CCH – Control card for DC-300 controller
- ADM4 – revision level, including software

As an aid in expediting card identification, a partial listing of the cards, functions and associated root part numbers follows:

PRINTED CIRCUIT CARDS	CATALOG NUMBER (Root)
Control card (with microprocessors)	531X300CCH_
Power Connection card (4 Quadrant)	531X121PCR_
Power Connection card (1 Quadrant)	531X122PCN_
Power Supply card	531X111PSH_
Analog Process Interface card	531X157APC_
PRINTED CIRCUIT CARD -- OPTIONAL	
Programmer card	531X135PRG_
Microapplication (1)	531X139APM_
Terminal Board card - 3TB	531X170TBS_
Terminal Board Relay card - 4TB	531X137TRG_
Microapplication TB card - 4TB	531X171TMA_
Programmer card Assembly	36B605548A_G01

INTERCONNECTION CABLES

CONNECTOR	CATALOG
1PL	36A358208AAG02
2PL	104X154CB111
3PL	36A358208AAG04
4PL-3	36A358208AAG07
5PL	36A358208AAG01
6PL	36A358208AAG08
7PL	36A358208AAG05
8PL-10	36A358208ABG01
9PL	36A358208CXG01
10PL-3	36A358208AAG09
11PL	36A358208AAG11
12PL	36A358208AAG10
13PL	36A358208AAG12
14PL	36A358208EYG01
16PL (6 foot)	36A358208AAG18
16PL (8 foot)	36A358208AAG19
16PL (10 foot)	36A358208ARG01
16PL (15 foot)	36A358208ARG02
18PL	36A358208ADG01

NOTE: Connectors used may vary with options provided.

WARRANTY PARTS AND SERVICE

1. Consult the **GE DRIVES Terms and Conditions** for details of the product warranty, including warranty period and parts or service coverage.

2. Warranty replacement parts or service assistance may be obtained direct from the factory during regular working **hours (8 a.m. - 5 p.m.; M-F)** by calling:

GE DRIVES
PRODUCT SERVICE DEPARTMENT
ERIE, PA
(814) 875-3851

3. An **EMERGENCY HOT-LINE** has been established by GE DRIVES to provide a 24 hour telephone answering service to our customers.

GE DRIVES
EMERGENCY HOT-LINE
(814) 455-3219

For a **PROMPT RESPONSE THE NEXT WORKING DAY** the answering service operator will record the required information and your **NAME, COMPANY and PHONE NUMBER.** In cases of emergency, where assistance must be obtained immediately, the operator may be requested to contact a Customer Service Representative (for parts) or Product Service Engineer (for technical assistance) at home.

4. So that we may serve you more expediently, please have the following information ready whenever contacting the factory:

DRIVE SERIAL NUMBER & MODEL NUMBER
PART NUMBER & DESCRIPTION

5. GE DRIVES may request that failed parts, replaced under warranty, be returned to the factory. A **RETURN AUTHORIZATION** tag or number is to be obtained and included with any parts being returned. When returning parts, **pack them carefully** to prevent damage in transit and ship to:

GENERAL ELECTRIC COMPANY
DRIVES DEPARTMENT
1100 LAWRENCE PARKWAY
ERIE, PA 16531

ATTENTION: PARTS REPAIR & EXCHANGE

RECOMMENDED POWER STUD WIRING, TERMINALS & CRIMPING TOOLS

ITEM	CRIMPING TOOL			FOR WIRE SIZE	TERMINAL TYPE
	AMP. CAT. NO.	TYPE			
1	49592	Hand		14--12 AWG	Solistrand*
2	49935	Hand		12--10 AWG	Solistrand
3	69062	Hand-hydraulic with self-contained dies		8--2 AWG	Solistrand
	ELECTRO-HYDRAULIC POWER UNIT 115VAC AMP CAT. NO.	HEAD CAT. NO.	DIE CAT. NO.		
4	69120-1	69065	46323-2	1--10 AWG	Solistrand
5	69120-1	69065	46323-2	2/0 AWG	Solistrand
6	69120-1	69065	46324-2	3/0 AWG	Solistrand
7	69120-1	69065	46325-2	4/0 AWG	Solistrand
8	69120-1	69065	46326-2	250 MCM	AMPower
9	69120-1	69065	46327-2	300 MCM	AMPower

*Trademark of AMP Incorporated.

GLOSSARY OF TERMS

Base Speed	–	Maximum speed of a motor at rated armature volts and field amps.
DFP	–	Delayed Firing Power to armature SCRs.
EEROM	–	Electrically Erasable Read-Only Memory, also called EE or E^2 (E squared). Contains the drive Parameters.
EE Enable Device	–	Device used to "Enable" or "Protect" EEROM programming. A hardware jumper (JP13) located on the Control card.
Field Weakening	–	Lowering of field current at rated armature volts to allow higher speeds. Actual armature voltage where field weakening begins is called the crossover point (adjusted by Parameter ADR079-CROSS).
FUP	–	Firing Microprocessor, U5 on the Control card.
"Hard" Reset	–	Hardware Reset, resets the drive cards. Used to clear some faults and allow certain parameter changes. Activated by depressing SW2 on the Control card or connecting 3TB69 to 3TB66.
K1	–	"RUN" Relay located on Power Supply card.
K2	–	MA Auxiliary Relay, pilot relay for MA contactor. K2 is located on the Power Supply card.
K3	–	Fault Relay located on the Power Supply card.
LED	–	Light Emitting Diode.
MA	–	Main AC Contactor (see General Description).
MFC	–	Motor Field Control (see General Description).
MOV	–	Metal Oxide Varistor.
Parameters	–	Adjustable software settings used to program and tune the drive regulators.
PI	–	Process Interface card.
PL	–	Plug (may be plug or receptacle).
PSC	–	Power Supply card.
PUP	–	Programmer Microprocessor, U1 (and U3 PROM) on Programmer card.
RAM	–	Random Access Memory
Regenerative	–	Ability to return power from the armature to the AC line. Creates breaking effect on motor. (Regeneration, regen., etc.)
RS232C	–	Serial Link Communication Standard.
RS422	–	Serial Link Communication Standard.
RUP	–	Regulator Microprocessor, U4 (and U1 PROM) on Control card.
SCR	–	Silicon Controlled Rectifier (Thyristor).
SLD	–	Signal Level Detector, SLD0 thru SLD5, (RX1 or K1 thru RX5 or K5) on optional Application card and SLD5 on 3TB.
"Soft" Reset	–	Software Reset initiated by pressing the RESET key on the optional Programmer.
TUP	–	Tachometer Microprocessor, U6 on the Control card.

NOTES

NOTES

GE Drives

GENERAL ELECTRIC COMPANY
DRIVES DEPARTMENT
ERIE, PENNSYLVANIA 16531

GEK-85774 (1/89) 100 (P)